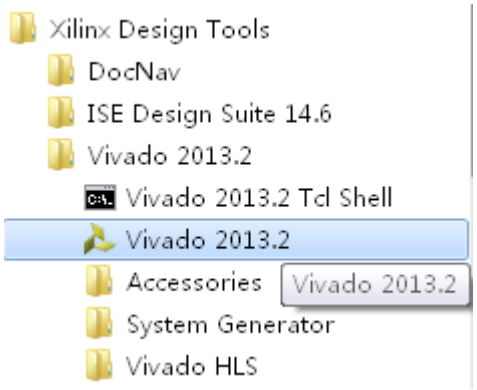
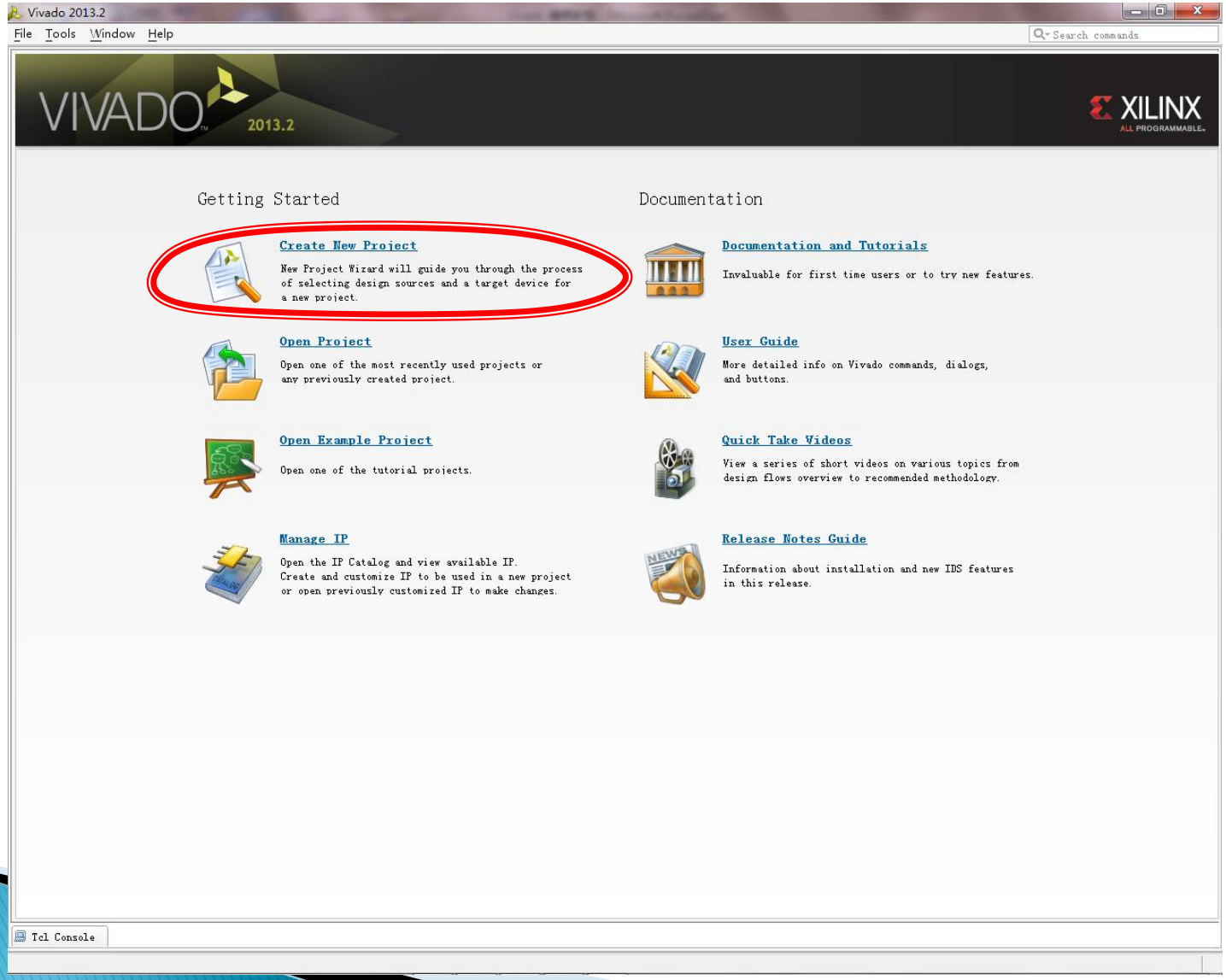
# Vivado 简明教程

AVNET 南京办
蔡键龙

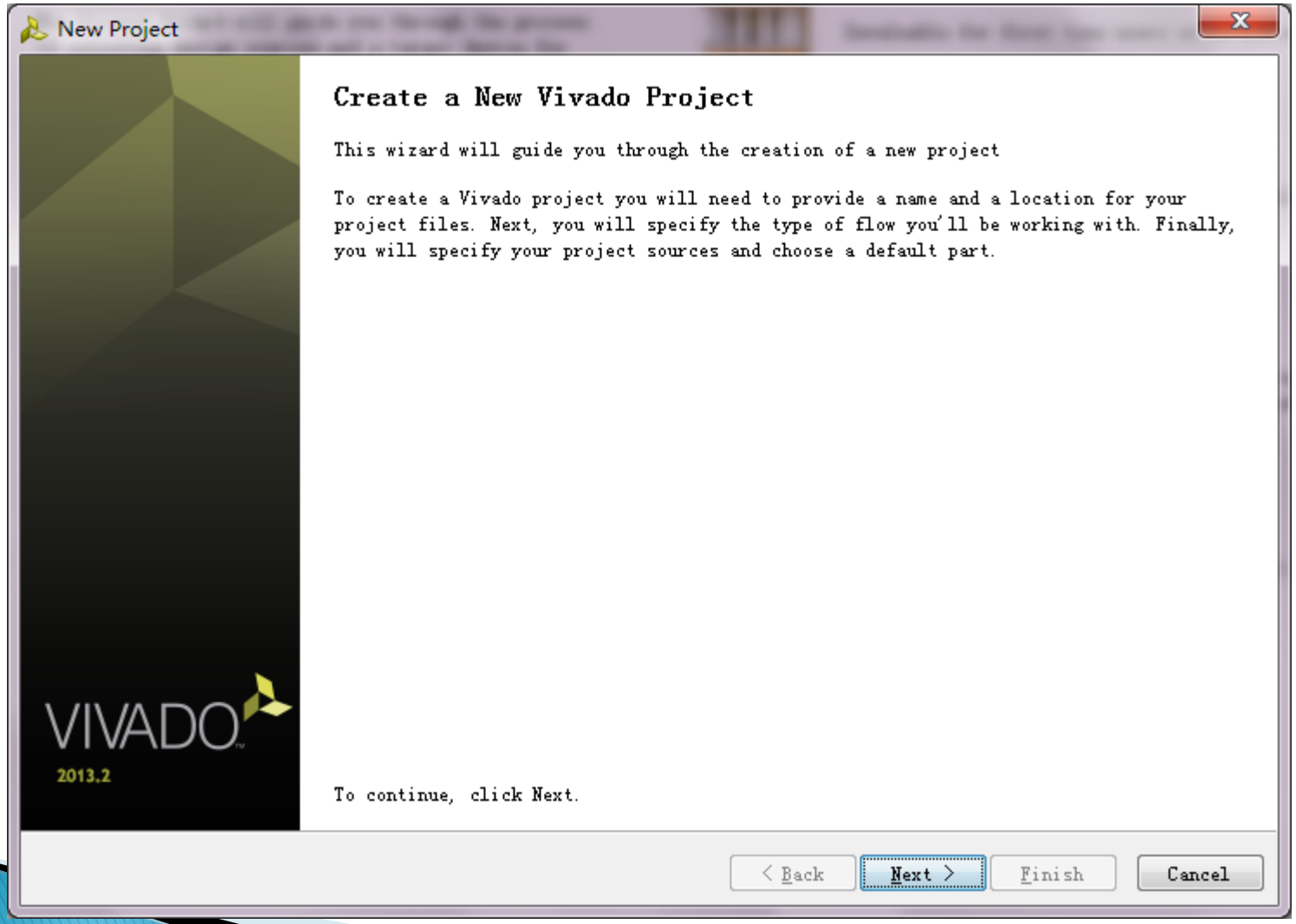
# 启动Vivado


Xilinx Design Tools
DocNav
ISE Design Suite 14.6
Vivado 2013.2
Vivado 2013.2 Tcl Shell
Vivado 2013.2
Accessories
Vivado 2013.2
System Generator
Vivado HLS

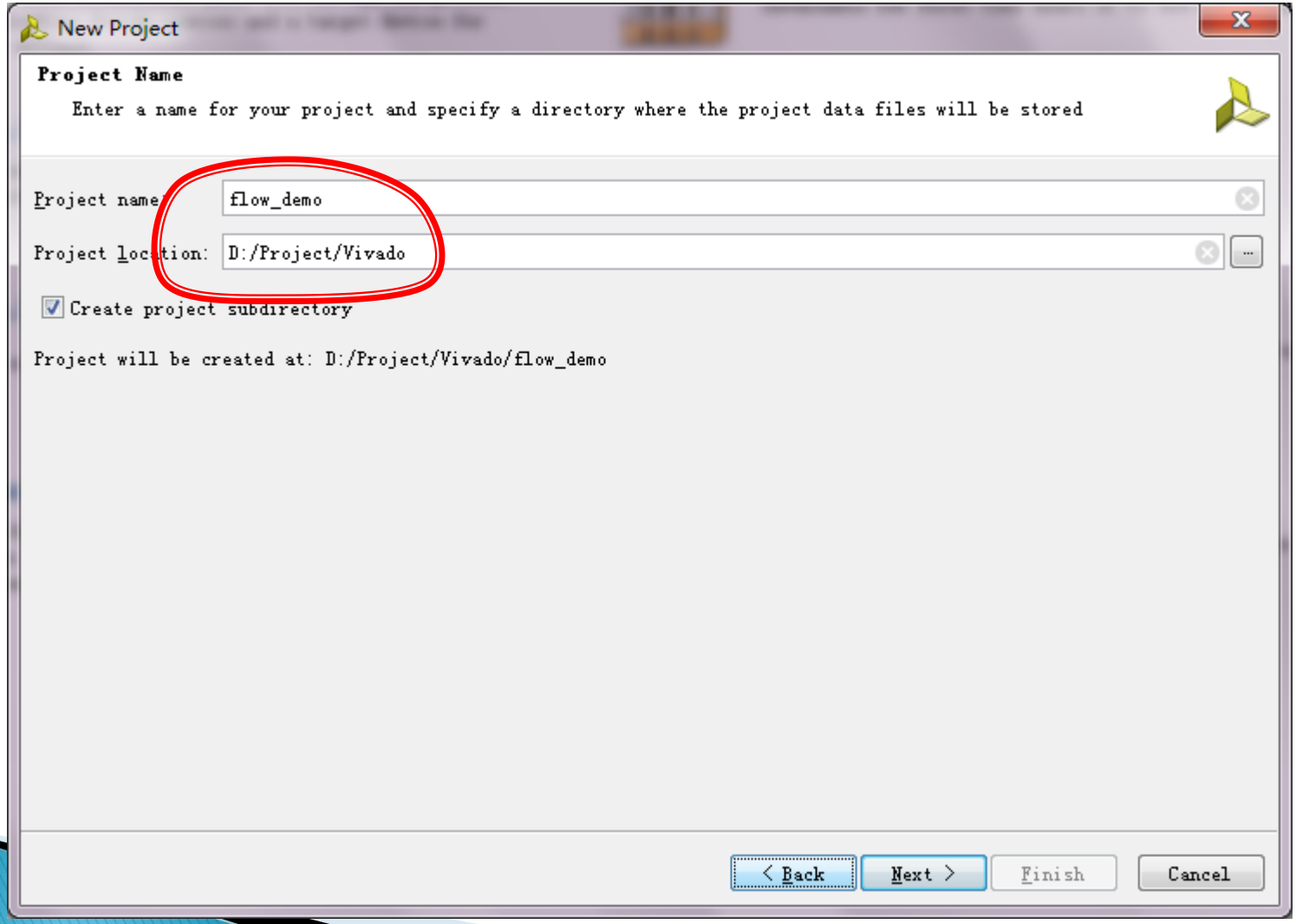
# 新建工程


Vivado 2013.2
File Tools Window Help
Search commands
VIVADO 2013.2
XILINX
ALL PROGRAMMABLE
Getting Started
Create New Project
New Project Wizard will guide you through the process of selecting design sources and a target device for a new project.
Open Project
Open one of the most recently used projects or any previously created project.
Open Example Project
Open one of the tutorial projects.
Manage IP
Open the IP Catalog and view available IP. Create and customize IP to be used in a new project or open previously customized IP to make changes.
Documentation
Documentation and Tutorials
Invaluable for first time users or to try new features.
User Guide
More detailed info on Vivado commands, dialogs, and buttons.
Quick Take Videos
View a series of short videos on various topics from design flows overview to recommended methodology.
Release Notes Guide
Information about installation and new IDS features in this release.
Tcl Console

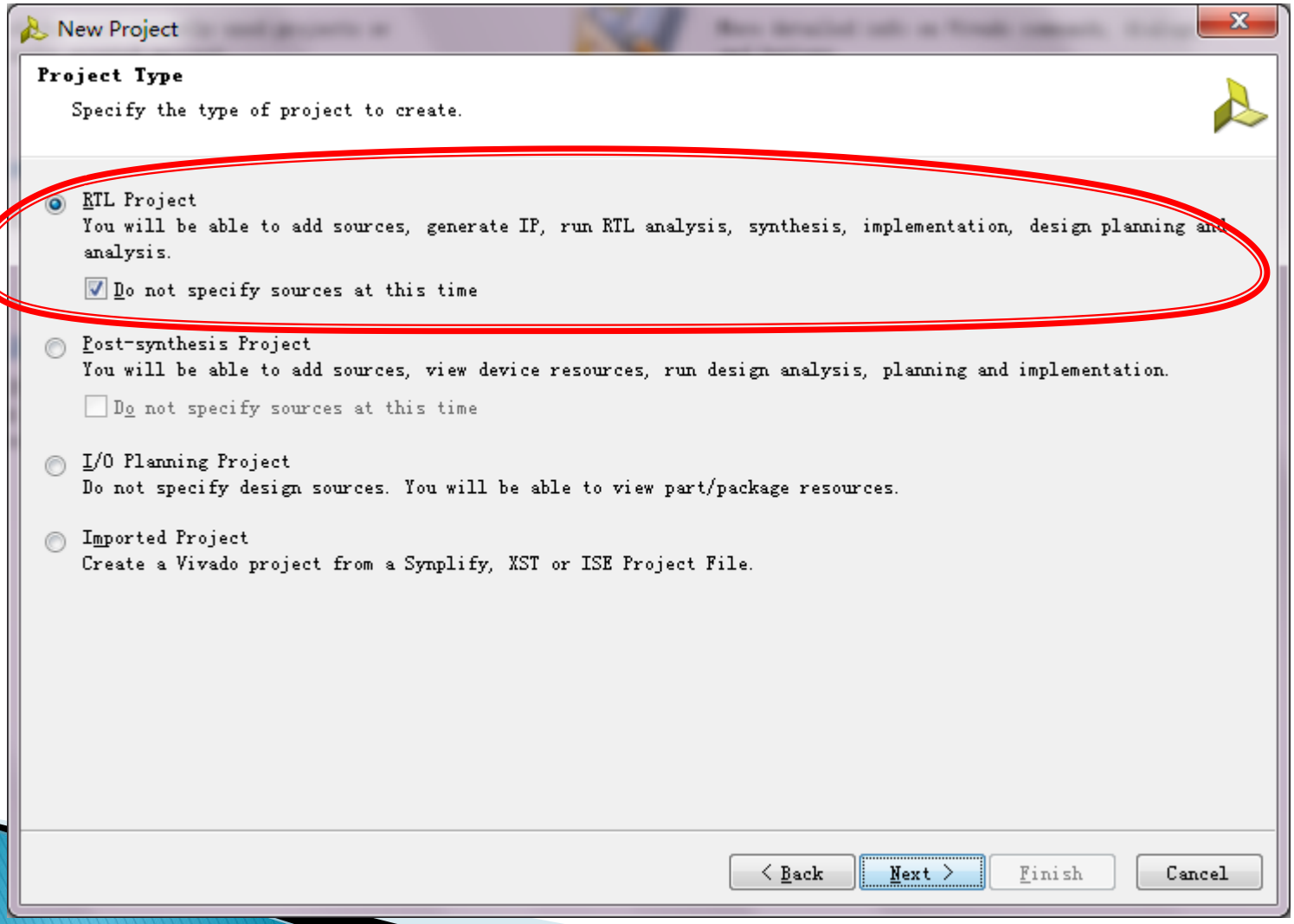
# 新建工程


New Project
Create a New Vivado Project
This wizard will guide you through the creation of a new project
To create a Vivado project you will need to provide a name and a location for your project files. Next, you will specify the type of flow you'll be working with. Finally, you will specify your project sources and choose a default part.
VIVADO
2013.2
To continue, click Next.
< Back
Next >
Finish
Cancel

# 新建工程

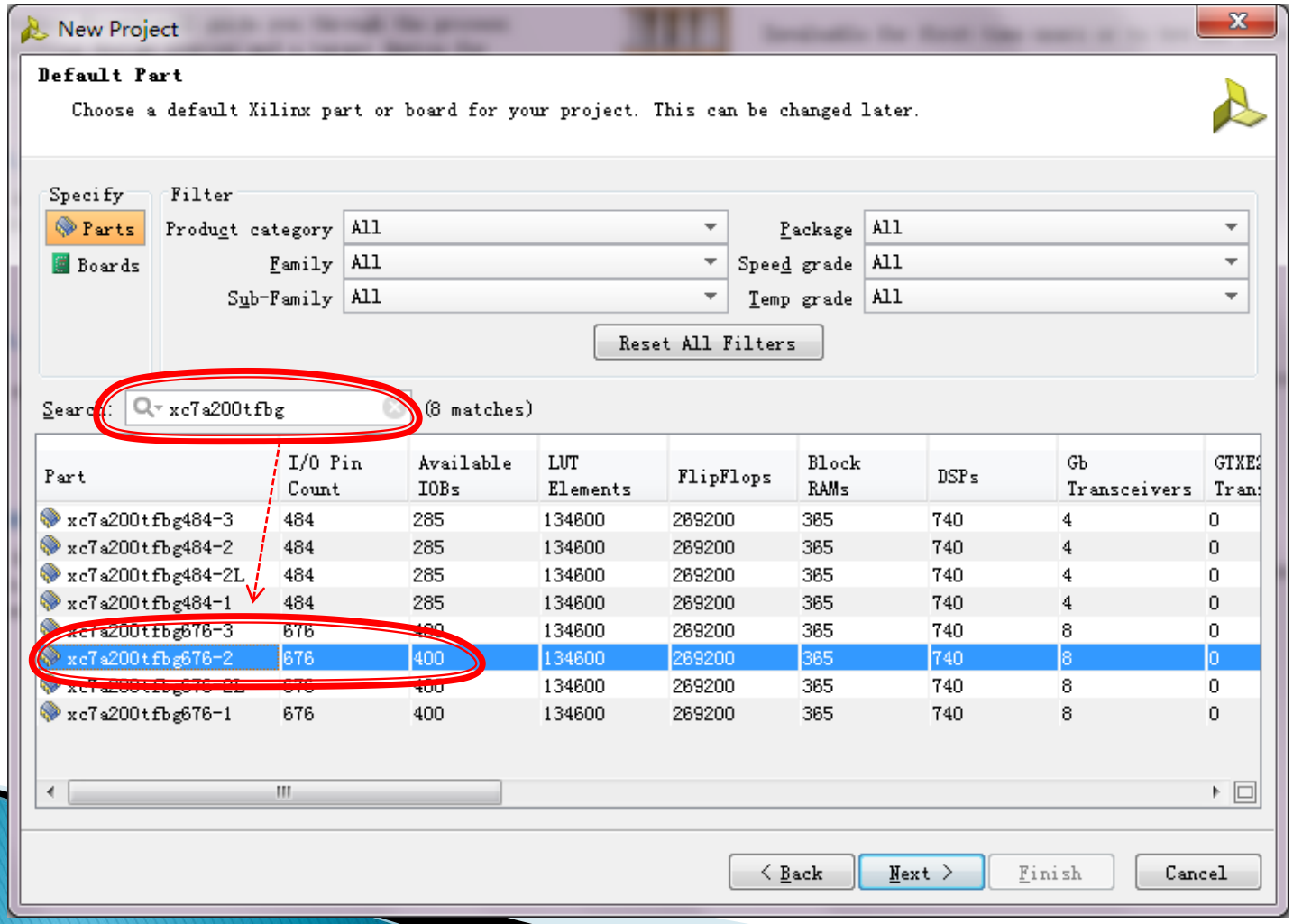

New Project
Project Name
Enter a name for your project and specify a directory where the project data files will be stored
Project name:
flow_demo
Project location:
D:/Project/Vivado
Create project subdirectory
Project will be created at: D:/Project/Vivado/flow_demo
< Back
Next >
Finish
Cancel

# 新建工程

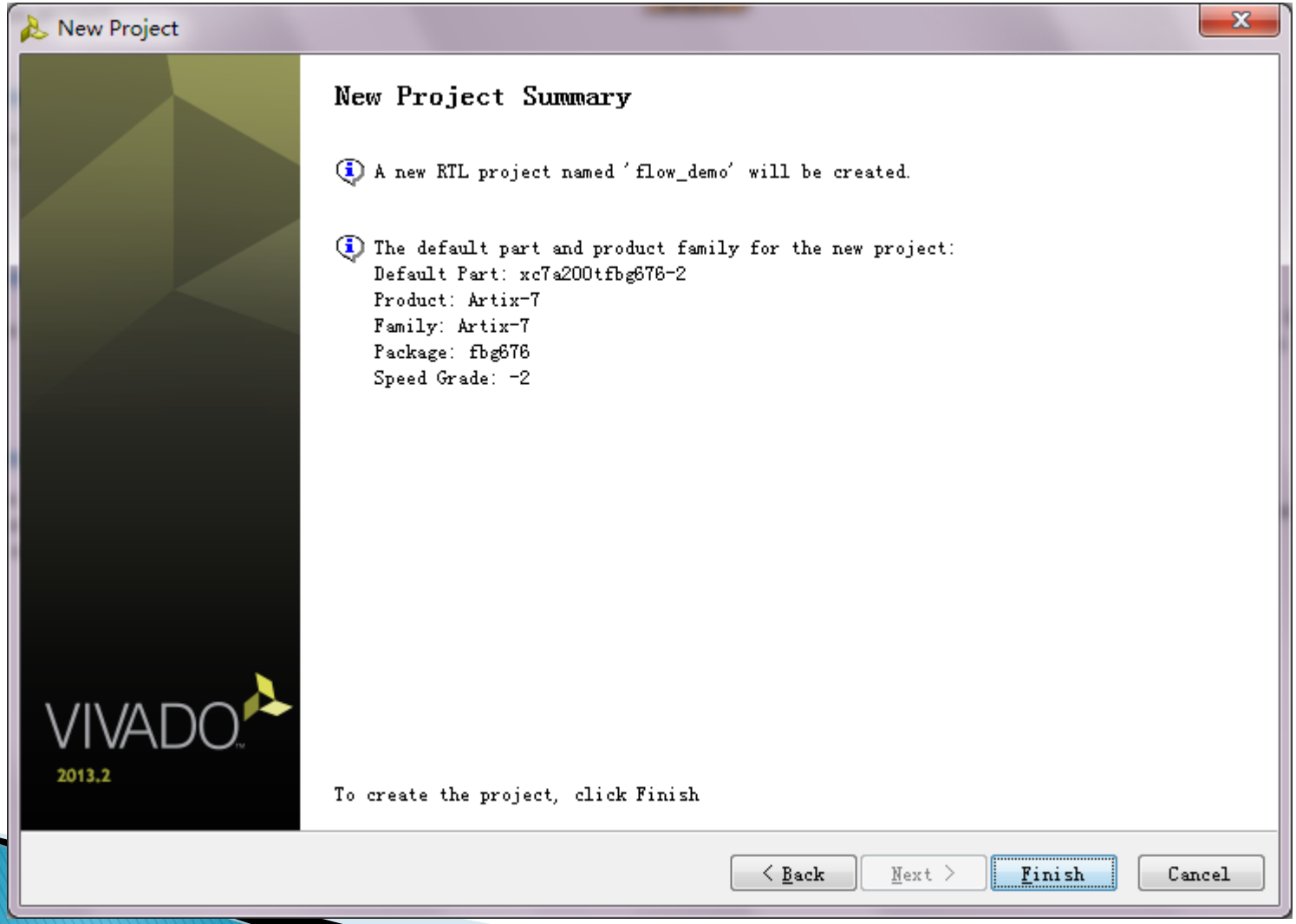

New Project
Project Type
Specify the type of project to create.
RTL Project
You will be able to add sources, generate IP, run RTL analysis, synthesis, implementation, design planning and analysis.
Do not specify sources at this time
Post-synthesis Project
You will be able to add sources, view device resources, run design analysis, planning and implementation.
Do not specify sources at this time
I/O Planning Project
Do not specify design sources. You will be able to view part/package resources.
Imported Project
Create a Vivado project from a Synplify, XST or ISE Project File.
< Back
Next >
Finish
Cancel

# 新建工程

- 选定所用的器件



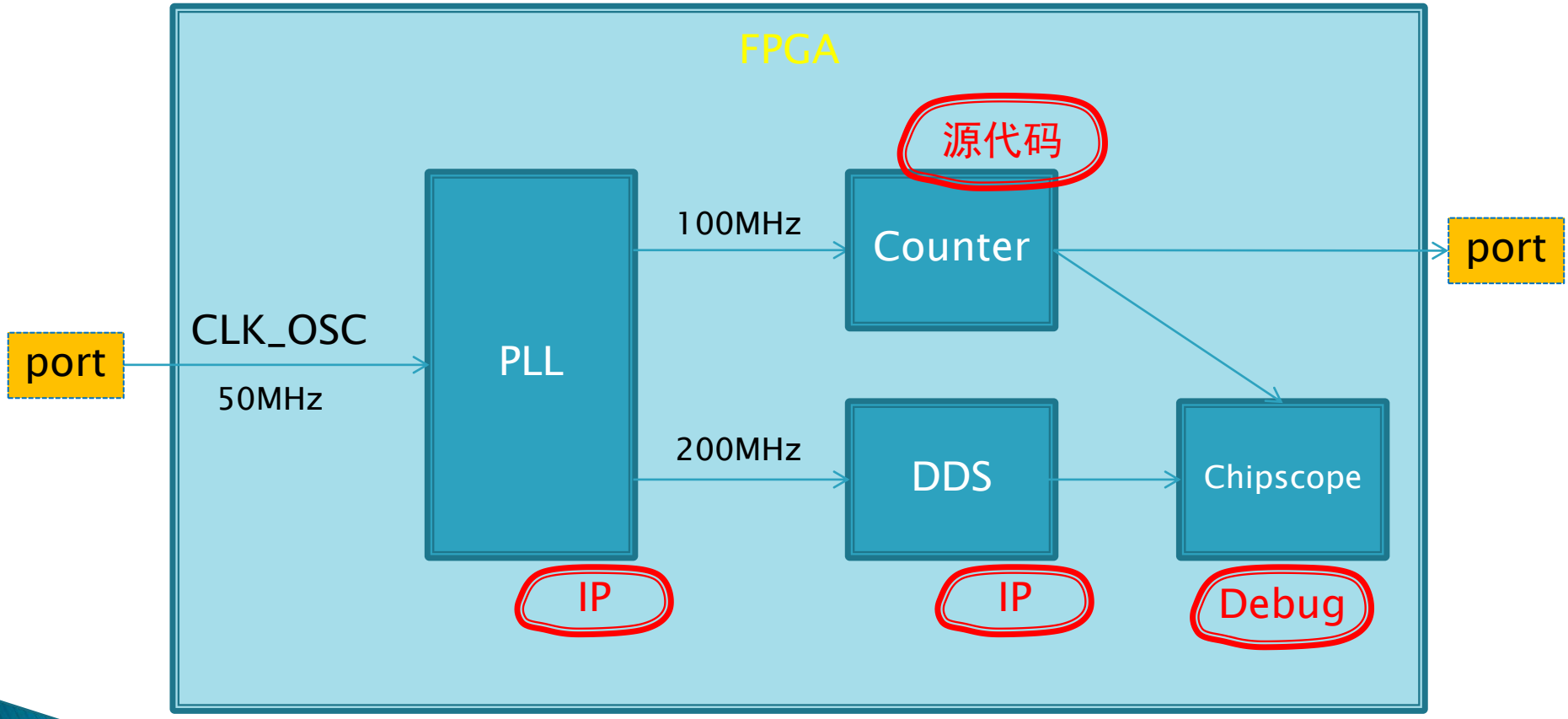
# 新建工程


New Project
New Project Summary
A new RTL project named 'flow_demo' will be created.
The default part and product family for the new project:
Default Part: xc7a200tfbg676-2
Product: Artix-7
Family: Artix-7
Package: fbg676
Speed Grade: -2
VIVADO
2013.2
To create the project, click Finish
< Back
Next >
Finish
Cancel

# 示例工程的内容

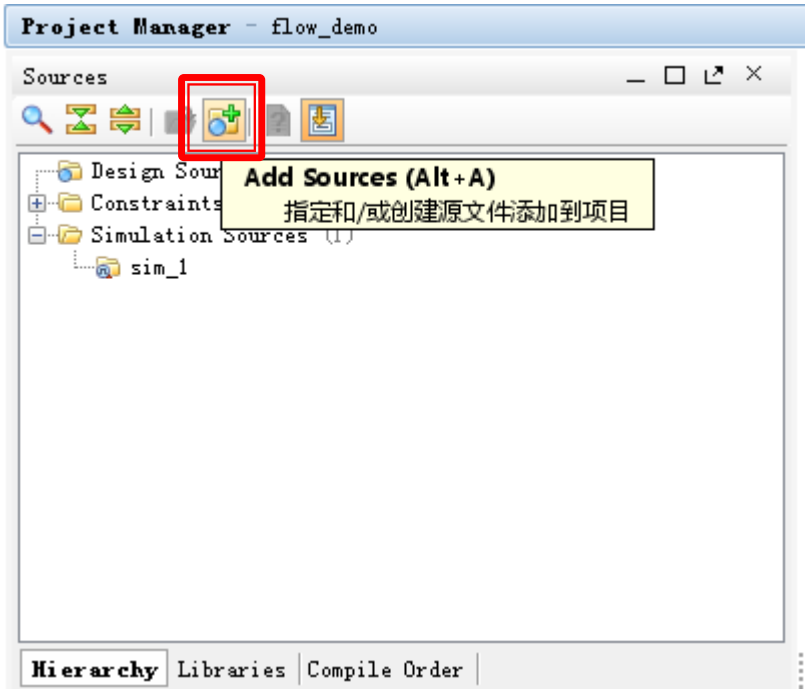
- 源代码输入
- 调用及例化IP
- 功能仿真
- Chipscope例化
- 时钟约束
- 管脚锁定
- 工程实现
- 生成bit文件

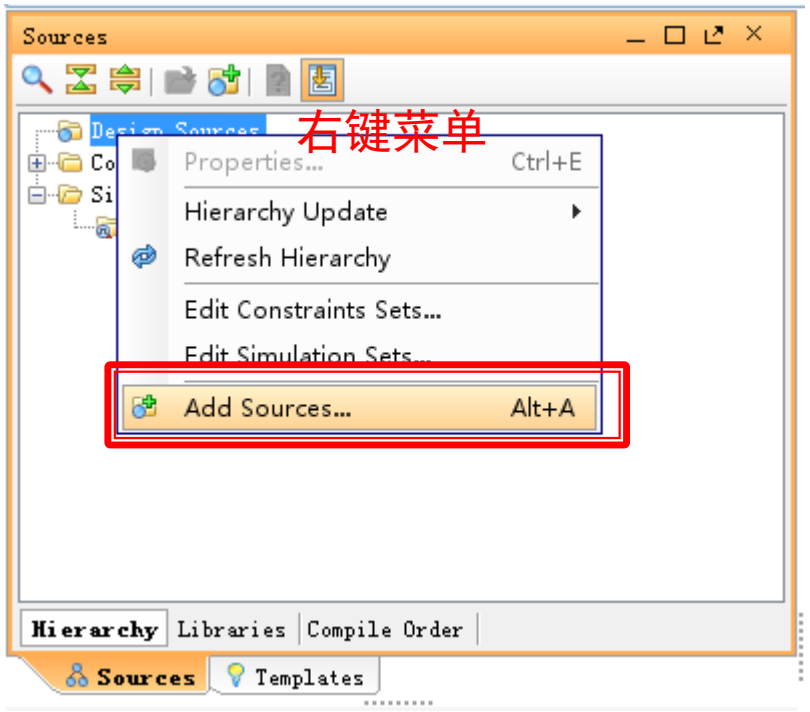
# 示例工程的架构


FPGA
源代码
port
CLK_OSC
50MHz
PLL
100MHz
Counter
port
200MHz
DDS
Chipscope
IP
IP
Debug

# 示例工程

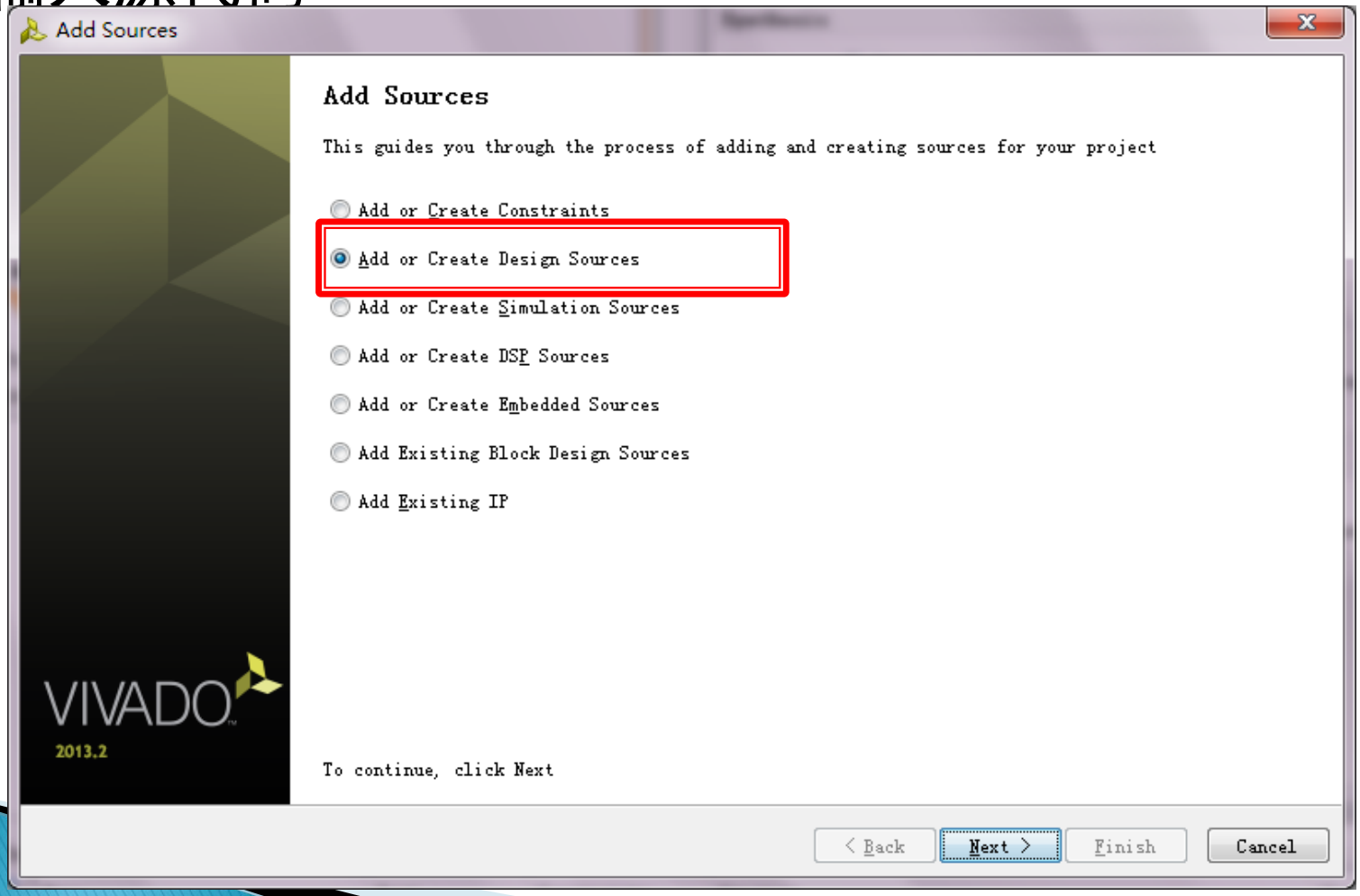
- 输入源代码


Project Manager - flow_demo
Sources
Add Sources (Alt+A)
指定和/或创建源文件添加到项目
Design Sour
Constraints
Simulation Sources (1)
sim_1
Hierarchy
Libraries
Compile Order


Sources
右键菜单
Properties... Ctrl+E
Hierarchy Update
Refresh Hierarchy
Edit Constraints Sets...
Edit Simulation Sets...
Add Sources... Alt+A
Hierarchy
Libraries
Compile Order
Sources
Templates

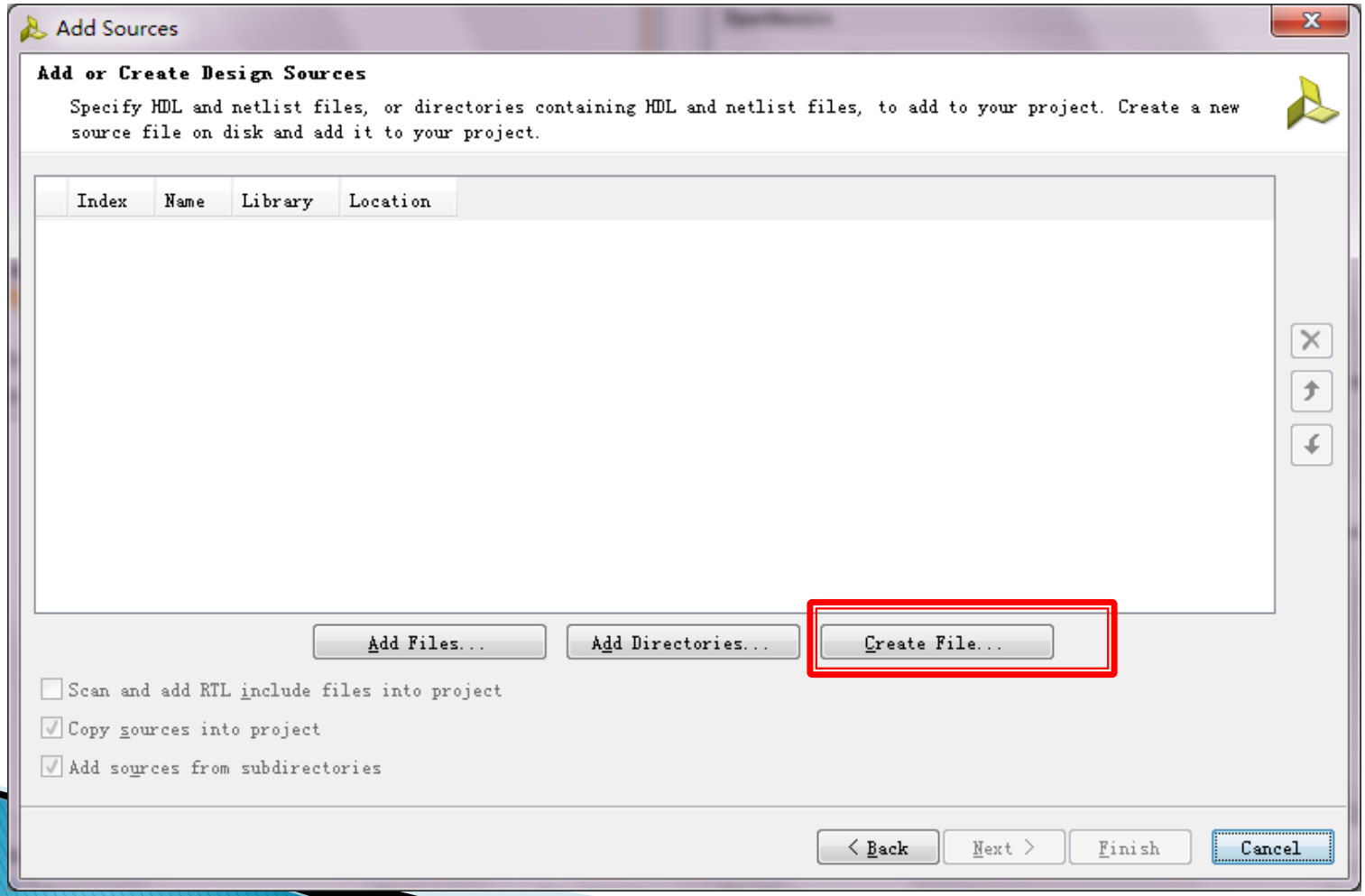
# 示例工程

- 输入源代码


Add Sources
Add Sources
This guides you through the process of adding and creating sources for your project
Add or Create Constraints
Add or Create Design Sources
Add or Create Simulation Sources
Add or Create DSP Sources
Add or Create Embedded Sources
Add Existing Block Design Sources
Add Existing IP
VIVADO
2013.2
To continue, click Next
< Back
Next >
Finish
Cancel

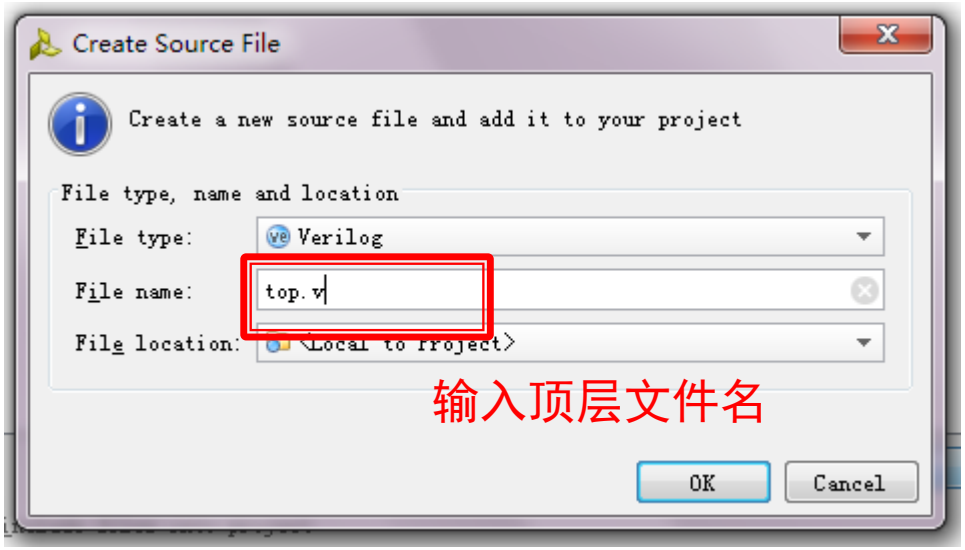
# 示例工程

- 输入源代码


Add Sources
Add or Create Design Sources
Specify HDL and netlist files, or directories containing HDL and netlist files, to add to your project. Create a new source file on disk and add it to your project.
Index
Name
Library
Location
Add Files...
Add Directories...
Create File...
Scan and add RTL include files into project
Copy sources into project
Add sources from subdirectories
< Back
Next >
Finish
Cancel

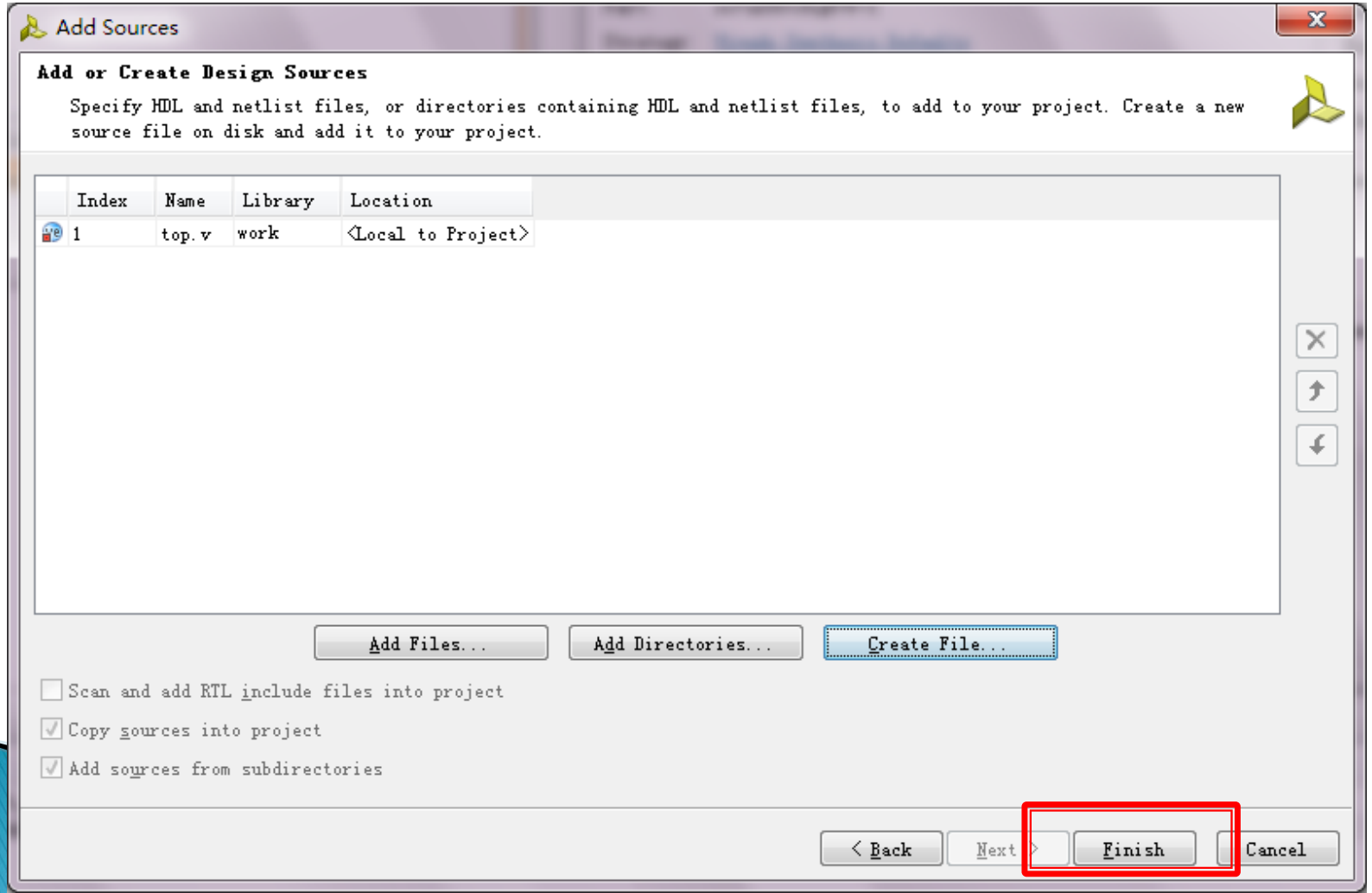
# 示例工程

- 输入源代码


Create Source File
Create a new source file and add it to your project
File type, name and location
File type:
Verilog
File name:
top.v
File location:
<Local to Project>
输入顶层文件名
OK
Cancel

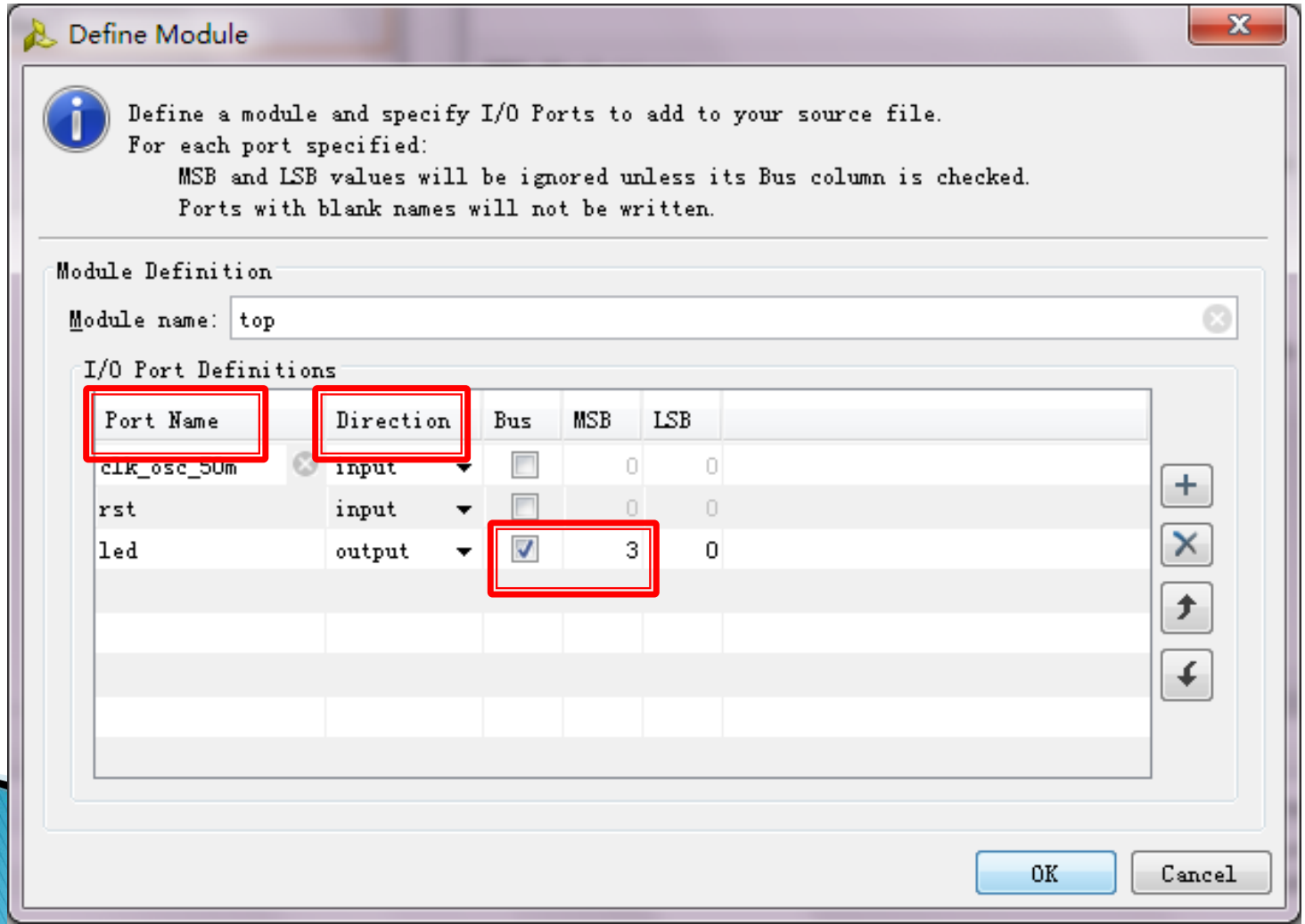
# 示例工程

- 输入源代码


Add Sources
Add or Create Design Sources
Specify HDL and netlist files, or directories containing HDL and netlist files, to add to your project. Create a new source file on disk and add it to your project.
Index
Name
Library
Location
1
top.v
work
<Local to Project>
Add Files...
Add Directories...
Create File...
Scan and add RTL include files into project
Copy sources into project
Add sources from subdirectories
< Back
Next >
Finish
Cancel

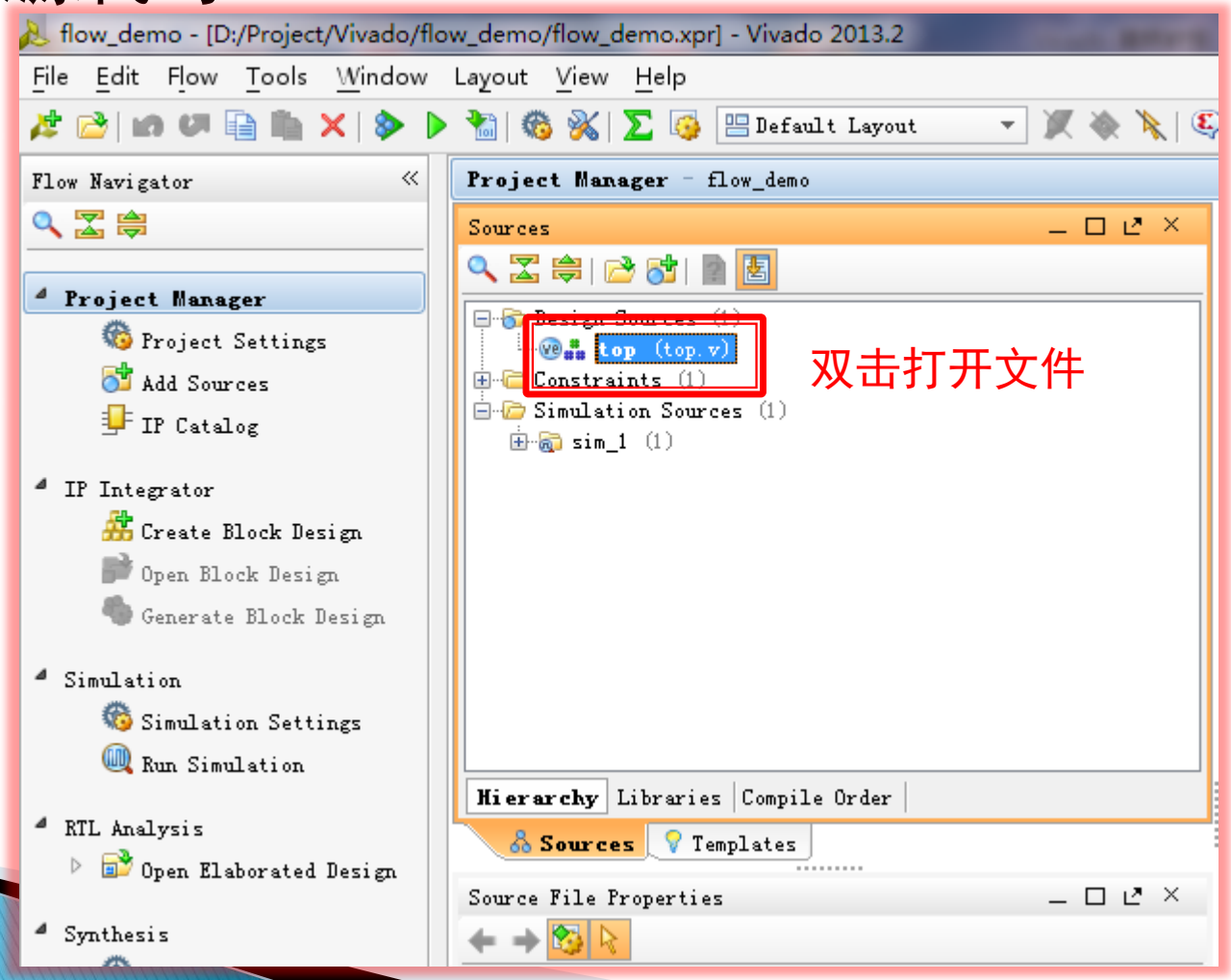
# 示例工程

- 输入源代码


Define Module
Define a module and specify I/O Ports to add to your source file.
For each port specified:
MSB and LSB values will be ignored unless its Bus column is checked.
Ports with blank names will not be written.
Module Definition
Module name: top
I/O Port Definitions
Port Name
Direction
Bus
MSB
LSB
clk_osc_50m
input
0
0
rst
input
0
0
led
output
3
0
OK
Cancel

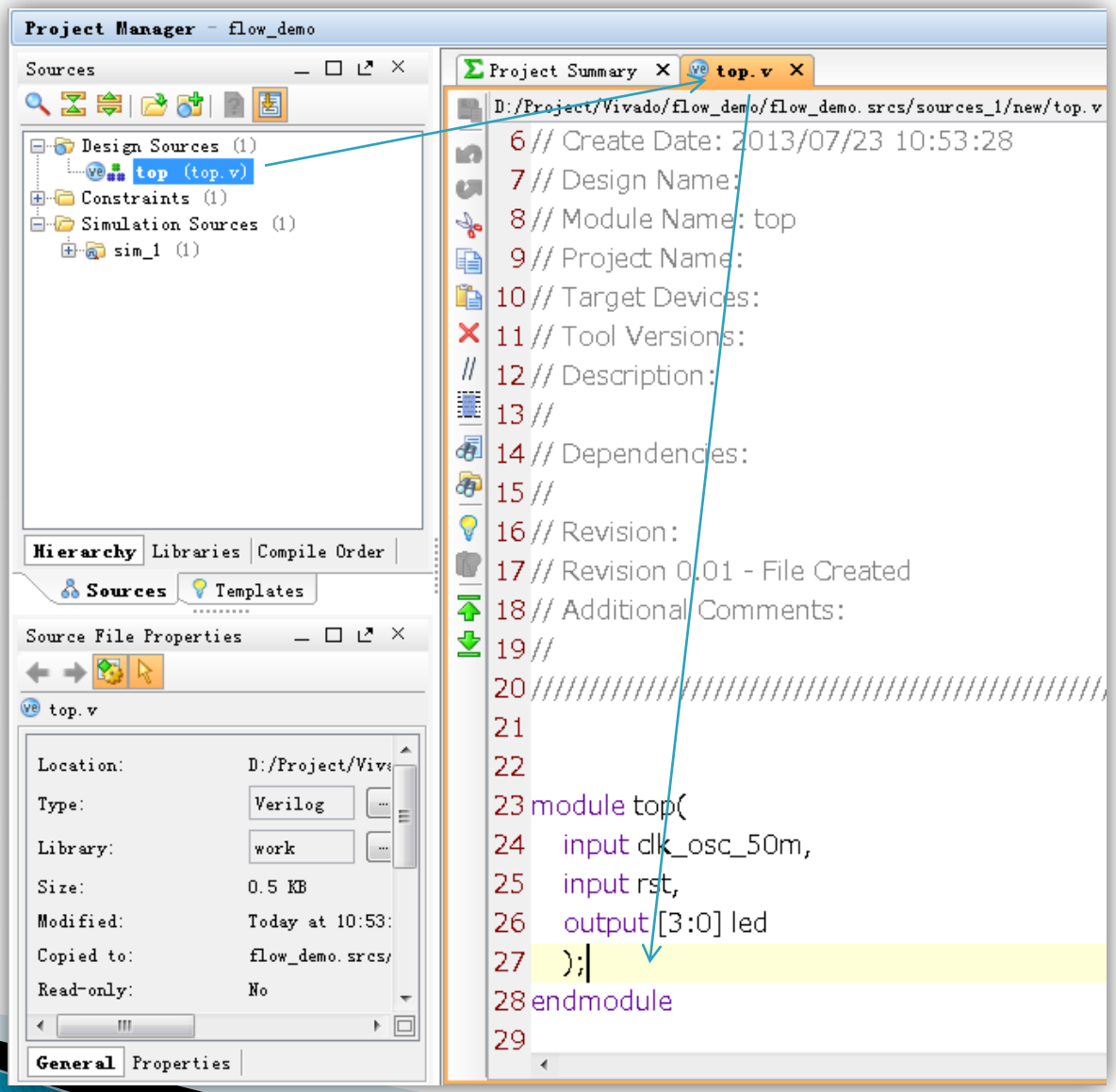
# 示例工程

- 输入源代码


flow_demo - [D:/Project/Vivado/flow_demo/flow_demo.xpr] - Vivado 2013.2
File Edit Flow Tools Window Layout View Help
Default Layout
Flow Navigator
Project Manager
Project Settings
Add Sources
IP Catalog
IP Integrator
Create Block Design
Open Block Design
Generate Block Design
Simulation
Simulation Settings
Run Simulation
RTL Analysis
Open Elaborated Design
Synthesis
Project Manager - flow_demo
Sources
top (top.v)
Constraints (1)
Simulation Sources (1)
sim_1 (1)
双击打开文件
Hierarchy Libraries Compile Order
Sources Templates
Source File Properties

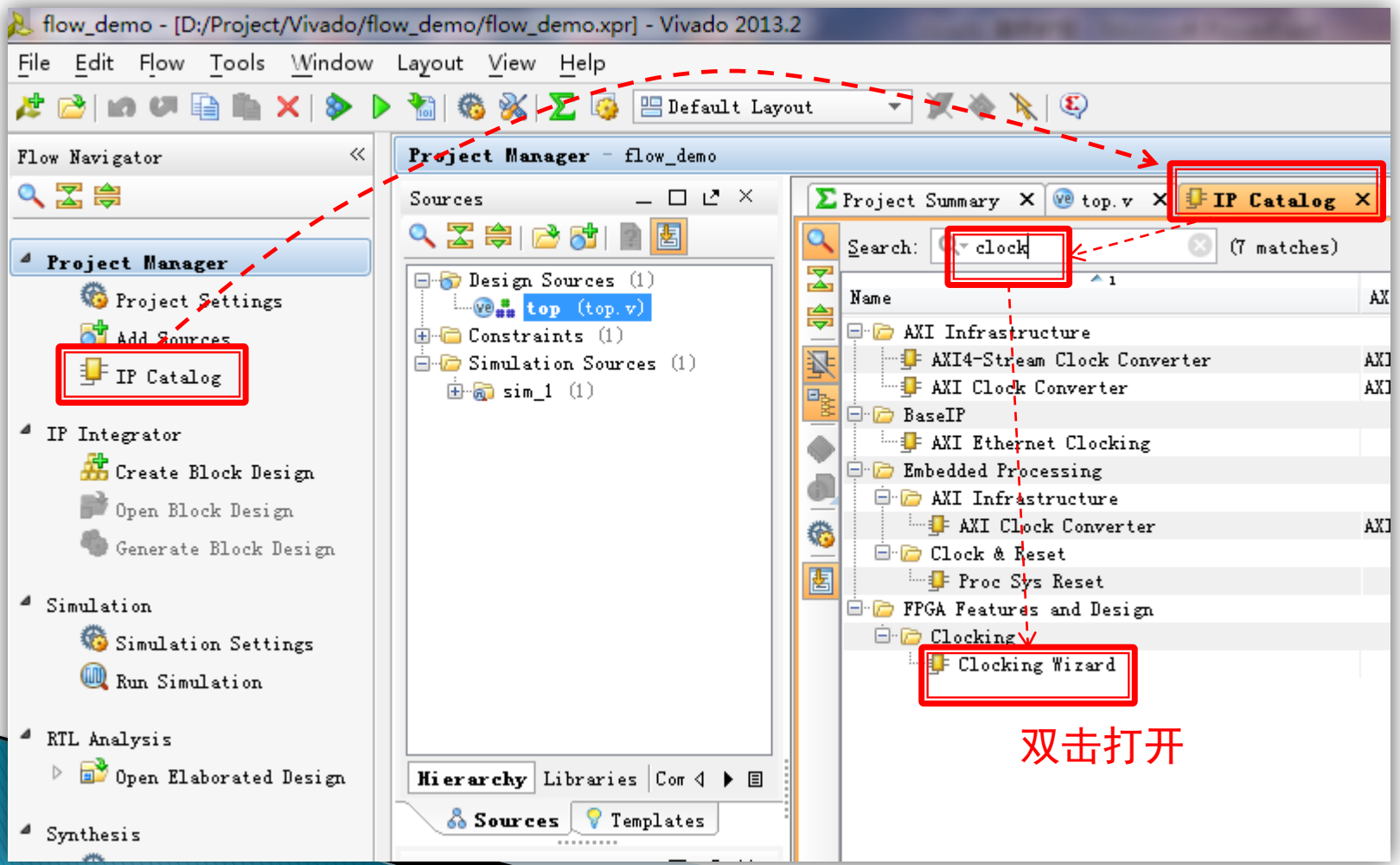
# 示例工程

- 输入源代码


Project Manager - flow_demo
Sources
Design Sources (1)
top (top.v)
Constraints (1)
Simulation Sources (1)
sim_1 (1)
Hierarchy
Libraries
Compile Order
Sources
Templates
Source File Properties
top.v
Location: D:/Project/Viv
Type: Verilog
Library: work
Size: 0.5 KB
Modified: Today at 10:53:
Copied to: flow_demo.srcs/
Read-only: No
General
Properties
Project Summary
top.v
D:/Project/Vivado/flow_demo/flow_demo.srcs/sources_1/new/top.v
6 // Create Date: 2013/07/23 10:53:28
7 // Design Name:
8 // Module Name: top
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////
21
22
23 module top(
24 input clk_osc_50m,
25 input rst,
26 output [3:0] led
27 );
28 endmodule
29

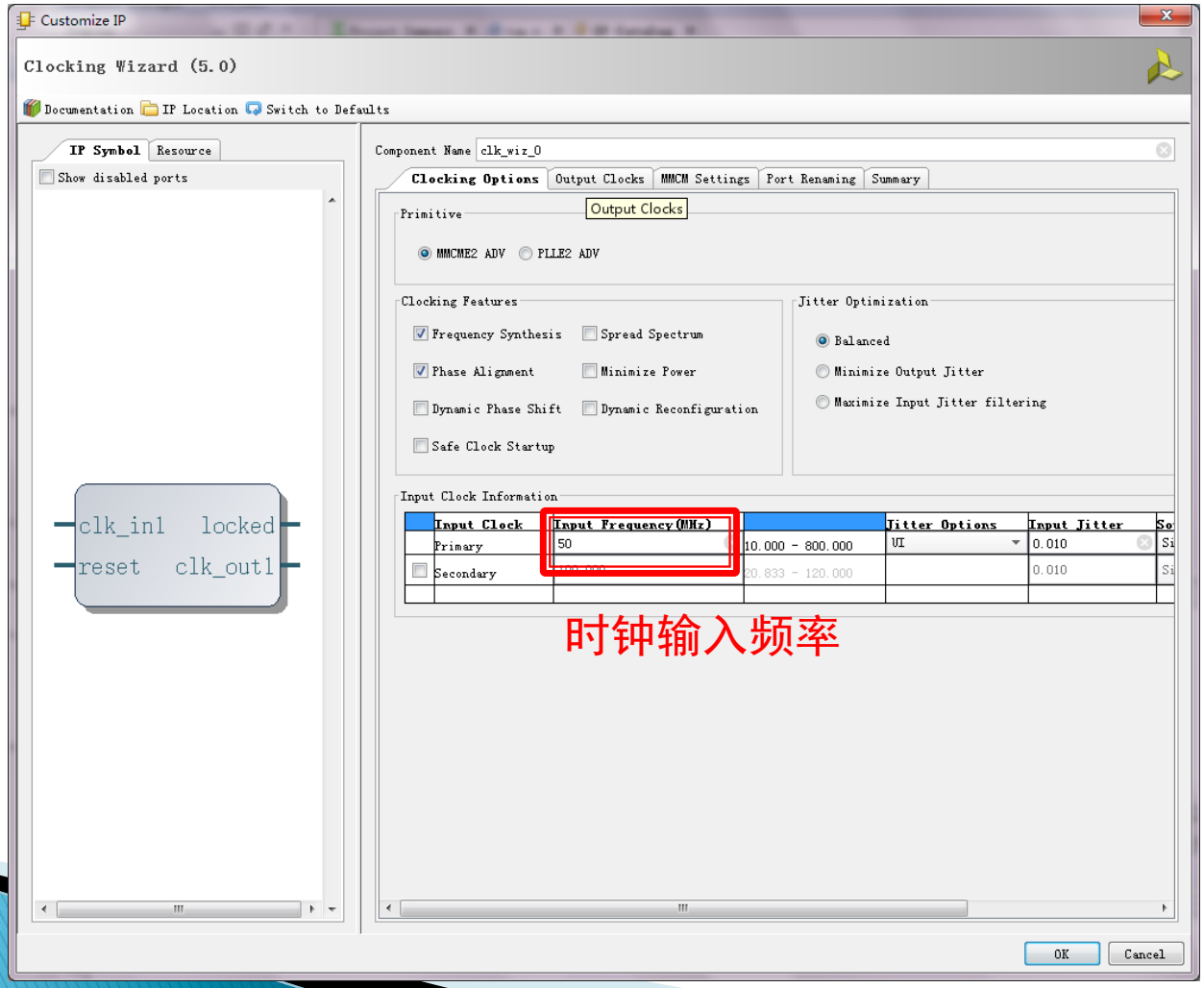
# 示例工程

- 调用IP-PLL时钟



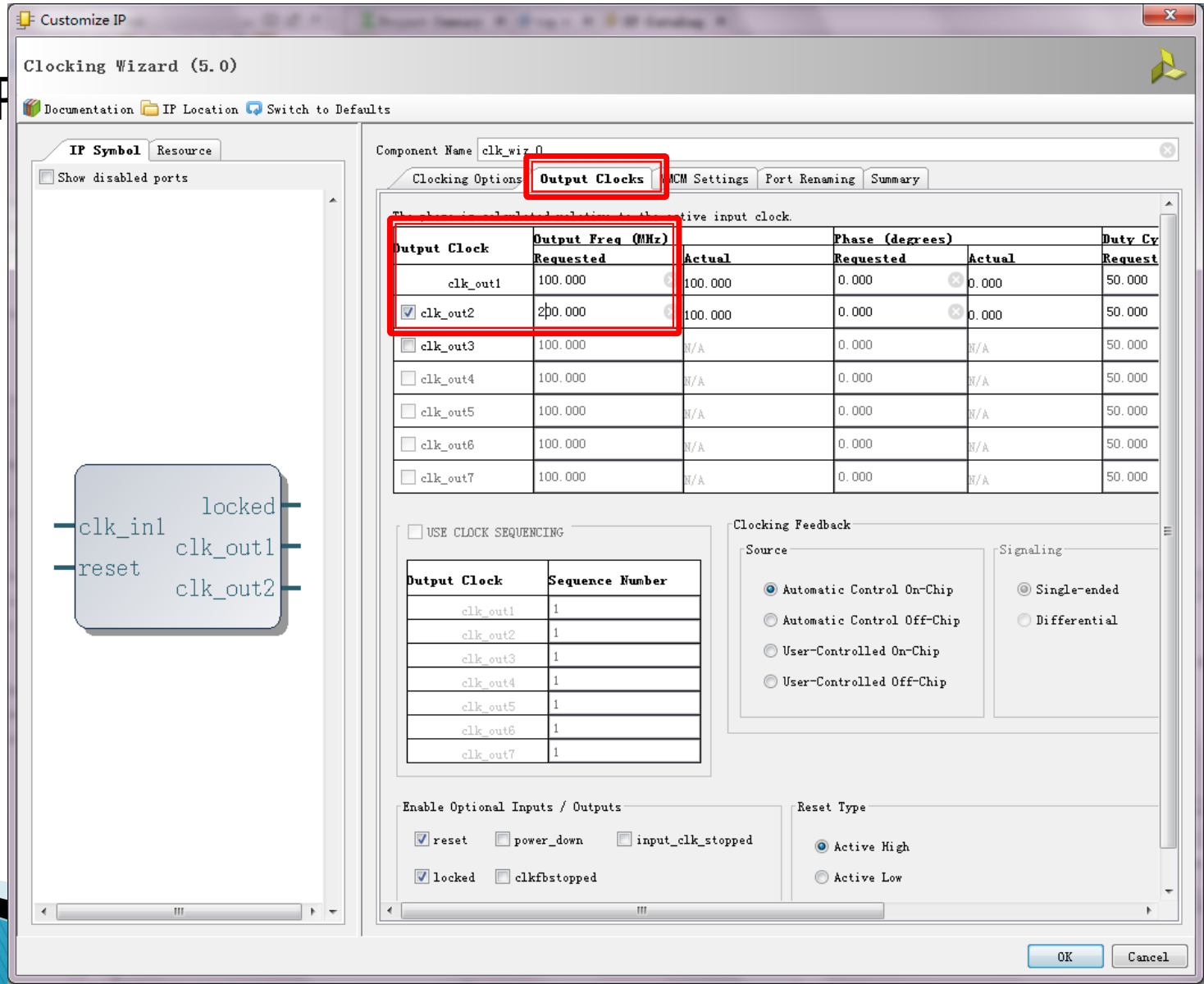
# 示例工程

- 调用IP-PLL时钟

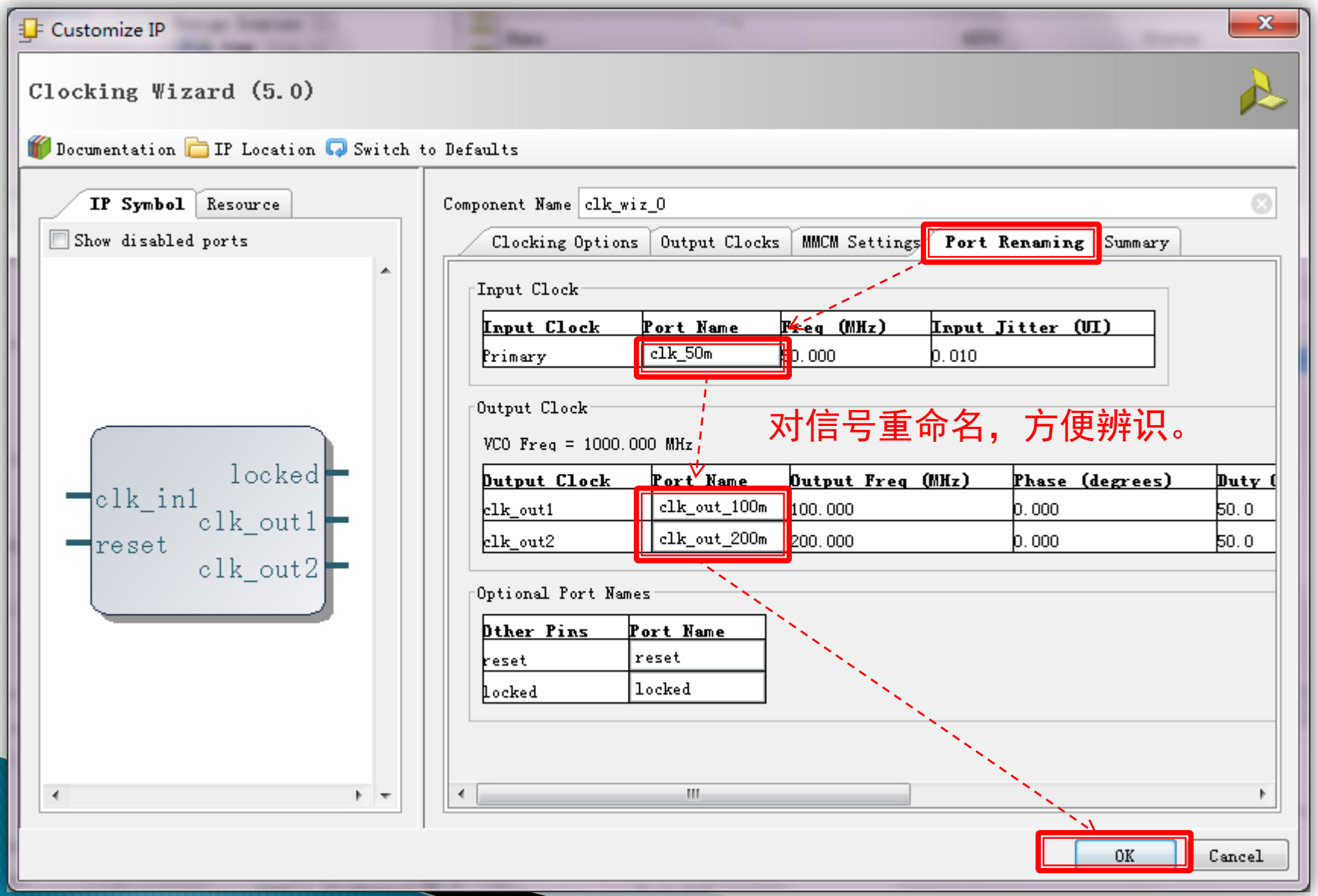


# 示例工程

- 调用IP-R

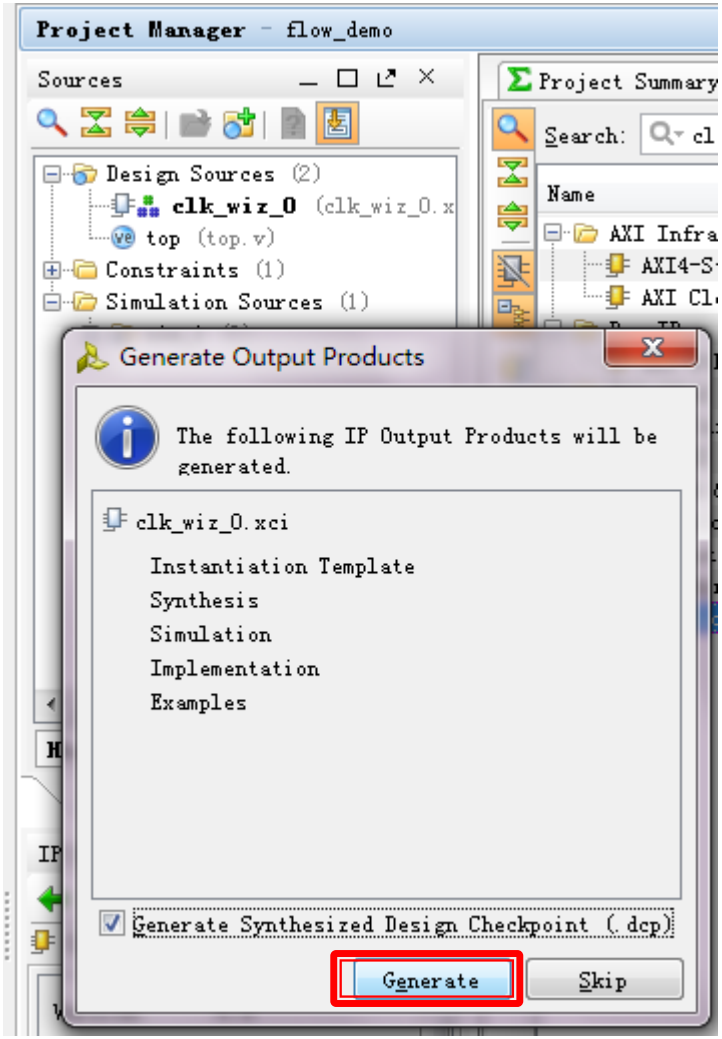

Customize IP
Clocking Wizard (5.0)
Documentation
IP Location
Switch to Defaults
IP Symbol
Resource
Show disabled ports
clk_in1
reset
locked
clk_out1
clk_out2
Component Name
clk_wiz_0
Clocking Options
Output Clocks
MCM Settings
Port Renaming
Summary
Output Clock
Output Freq (MHz)
Requested
Actual
Phase (degrees)
Requested
Actual
Duty Cy
Request
clk_out1 100.000 100.000 0.000 0.000 50.000
clk_out2 200.000 100.000 0.000 0.000 50.000
clk_out3 100.000 N/A 0.000 N/A 50.000
clk_out4 100.000 N/A 0.000 N/A 50.000
clk_out5 100.000 N/A 0.000 N/A 50.000
clk_out6 100.000 N/A 0.000 N/A 50.000
clk_out7 100.000 N/A 0.000 N/A 50.000
USE CLOCK SEQUENCING
Output Clock
Sequence Number
clk_out1 1
clk_out2 1
clk_out3 1
clk_out4 1
clk_out5 1
clk_out6 1
clk_out7 1
Clocking Feedback
Source
Automatic Control On-Chip
Automatic Control Off-Chip
User-Controlled On-Chip
User-Controlled Off-Chip
Signaling
Single-ended
Differential
Enable Optional Inputs / Outputs
reset
power_down
input_clk_stopped
locked
clkfbstopped
Reset Type
Active High
Active Low
OK
Cancel

# 示例工程


Customize IP
Clocking Wizard (5.0)
Documentation
IP Location
Switch to Defaults
IP Symbol
Resource
Show disabled ports
clk_in1
reset
locked
clk_out1
clk_out2
Component Name
clk_wiz_0
Clocking Options
Output Clocks
MMCM Settings
Port Renaming
Summary
Input Clock
Input Clock | Port Name | Freq (MHz) | Input Jitter (UI)
Primary | clk_50m | 50.000 | 0.010
Output Clock
VCO Freq = 1000.000 MHz
Output Clock | Port Name | Output Freq (MHz) | Phase (degrees) | Duty
clk_out1 | clk_out_100m | 100.000 | 0.000 | 50.0
clk_out2 | clk_out_200m | 200.000 | 0.000 | 50.0
对信号重命名，方便辨识。
Optional Port Names
Other Pins | Port Name
reset | reset
locked | locked
OK
Cancel

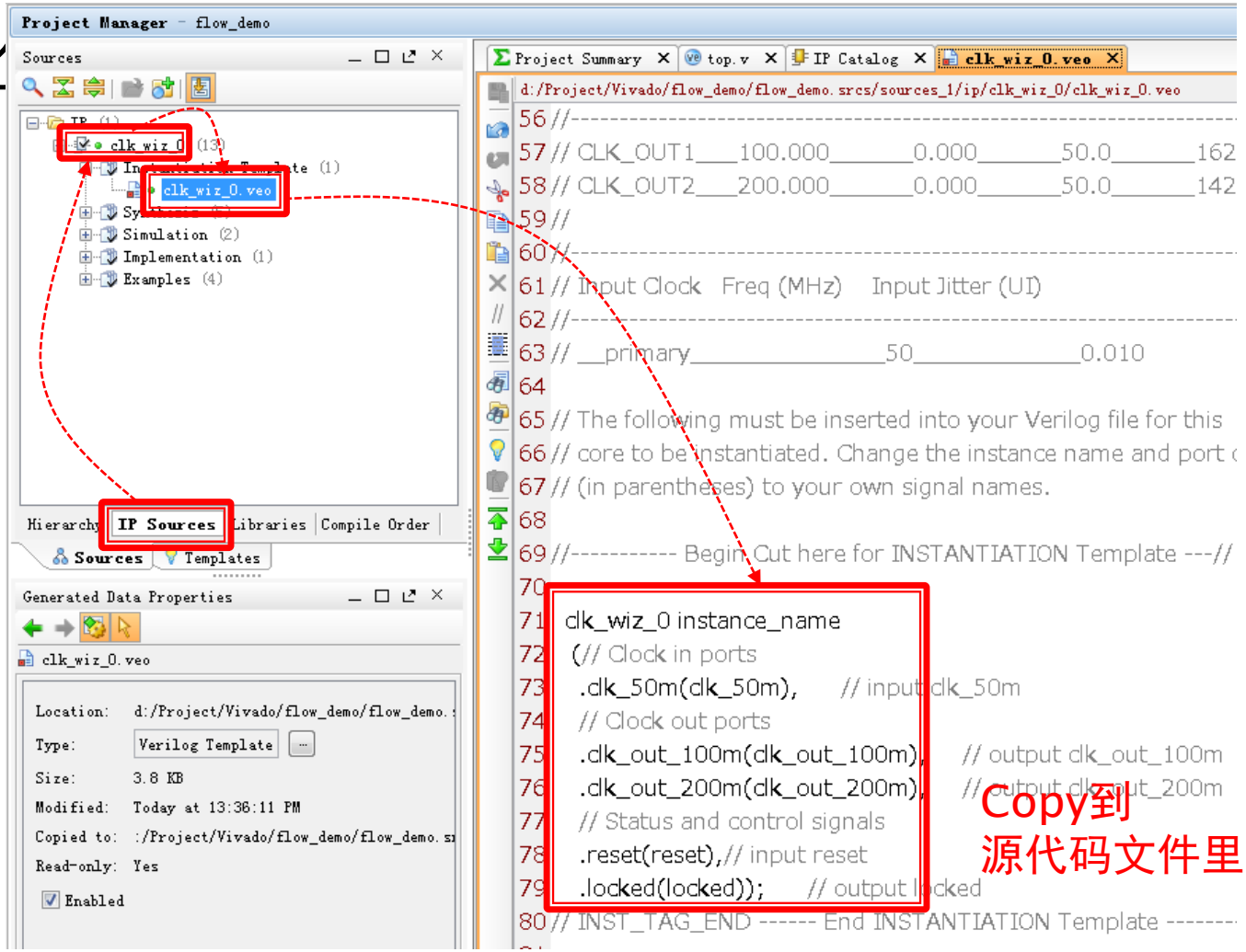
# 示例工程-调用IP

- PLL的生成结果



# 示例工程-调用IP

- 时钟 IP例化



# 示例工程-调用IP

- 时钟 IP例化

Project Summary | top.v * | IP Catalog | clk_wiz_0.veo

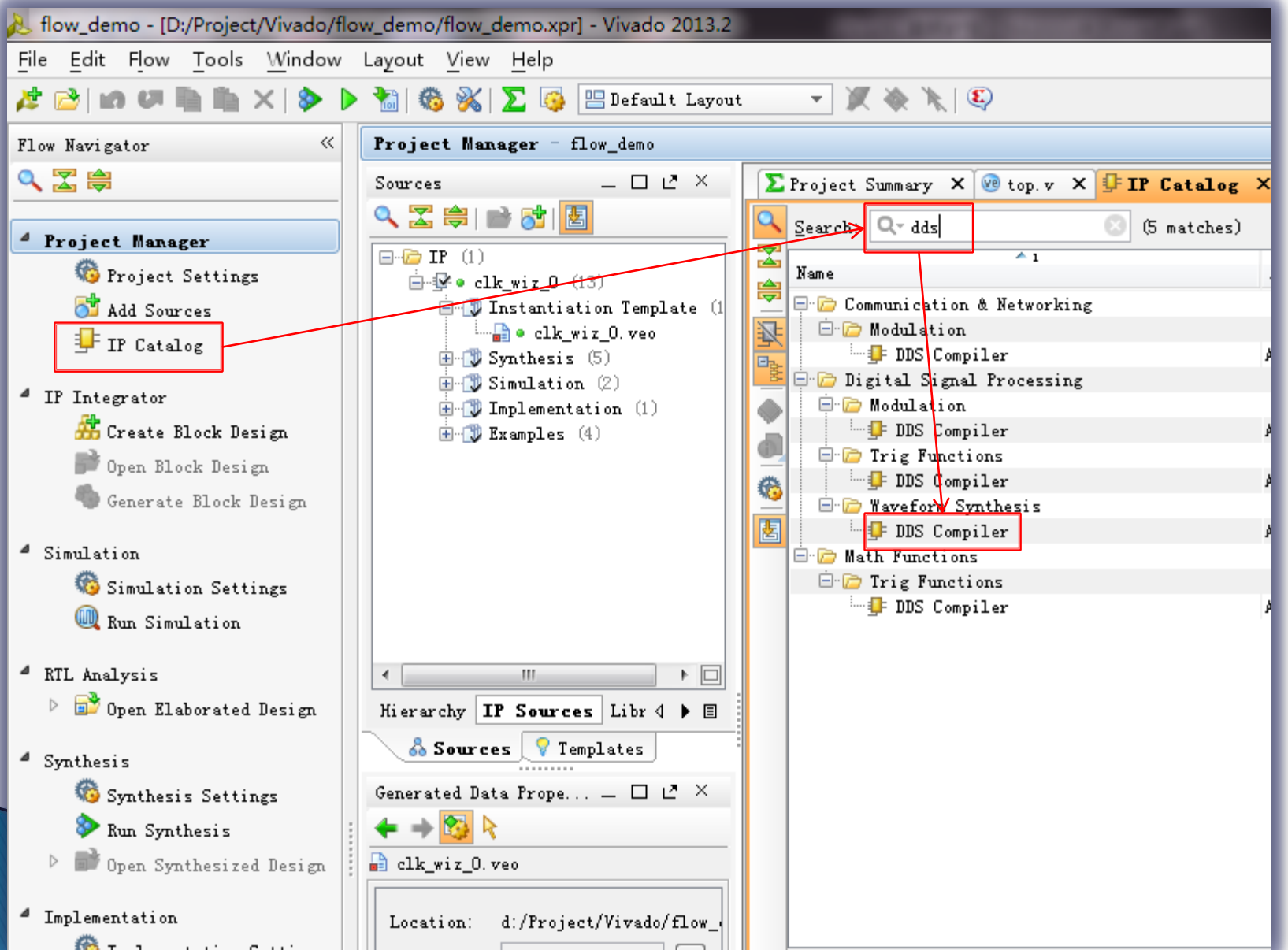
D:/Project/Vivado/flow_demo/flow_demo.srcs/sources_1/new/top.v

```
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////
21
22
23 module top(
24    input clk_osc_50m,
25    input rst,
26    output [3:0] led
27    );
28
29    wire clk_out_100m,clk_out_200m;
30    wire mmcm_locked;
31
32     clk_wiz_0 osc_mmcm_inst0
33    (// Clock in ports
34     .clk_50m(clk_osc_50m),      // input clk_50m
35     // Clock out ports
36     .clk_out_100m(clk_out_100m),     // output clk_out_100m
37     .clk_out_200m(clk_out_200m),     // output clk_out_200m
38     // Status and control signals
39     .reset(rst),// input reset
40     .locked(mmcm_locked));      // output locked
41
42 endmodule
43
```

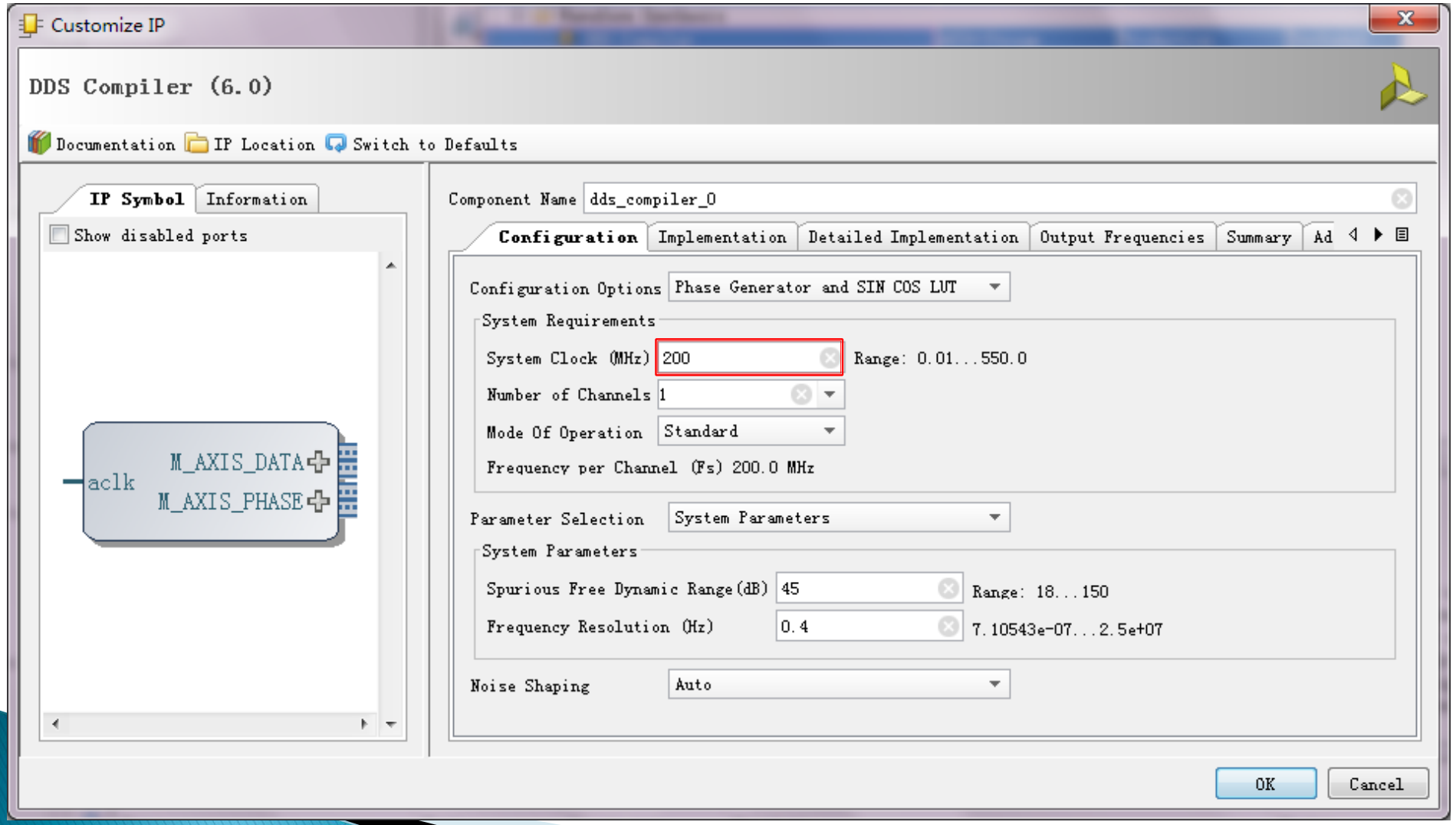
信号声明

IP例化

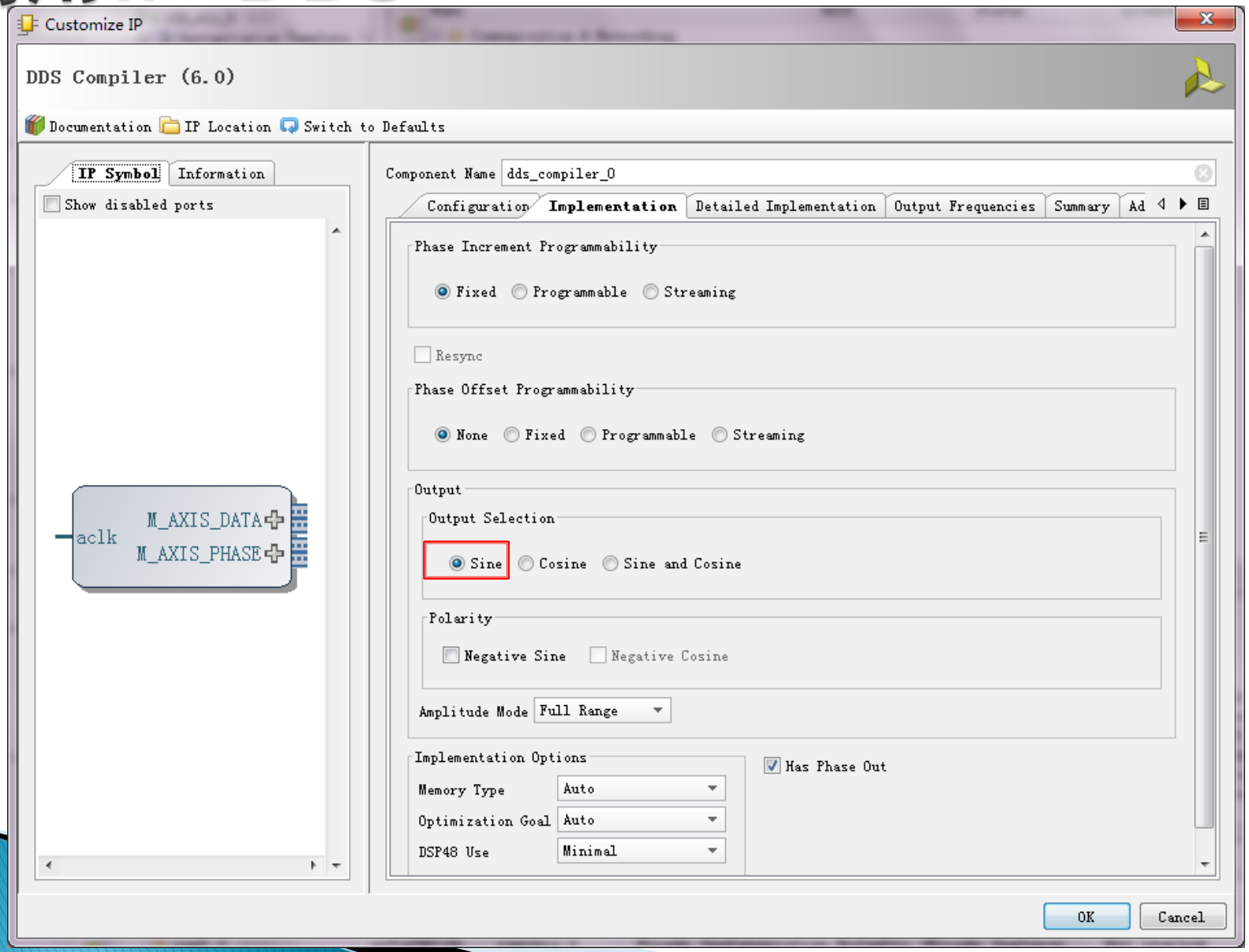
# 调用IP-DDS


flow_demo - [D:/Project/Vivado/flow_demo/flow_demo.xpr] - Vivado 2013.2
File Edit Flow Tools Window Layout View Help
Default Layout
Flow Navigator
Project Manager
Project Settings
Add Sources
IP Catalog
IP Integrator
Create Block Design
Open Block Design
Generate Block Design
Simulation
Simulation Settings
Run Simulation
RTL Analysis
Open Elaborated Design
Synthesis
Synthesis Settings
Run Synthesis
Open Synthesized Design
Implementation
Project Manager - flow_demo
Sources
IP (1)
clk_wiz_0 (13)
Instantiation Template (1
clk_wiz_0.veo
Synthesis (5)
Simulation (2)
Implementation (1)
Examples (4)
Hierarchy IP Sources Libr
Sources Templates
Generated Data Prope...
clk_wiz_0.veo
Location: d:/Project/Vivado/flow_
Project Summary
top.v
IP Catalog
Search: dds
(5 matches)
Name
Communication & Networking
Modulation
DDS Compiler
Digital Signal Processing
Modulation
DDS Compiler
Trig Functions
DDS Compiler
Waveform Synthesis
DDS Compiler
Math Functions
Trig Functions
DDS Compiler

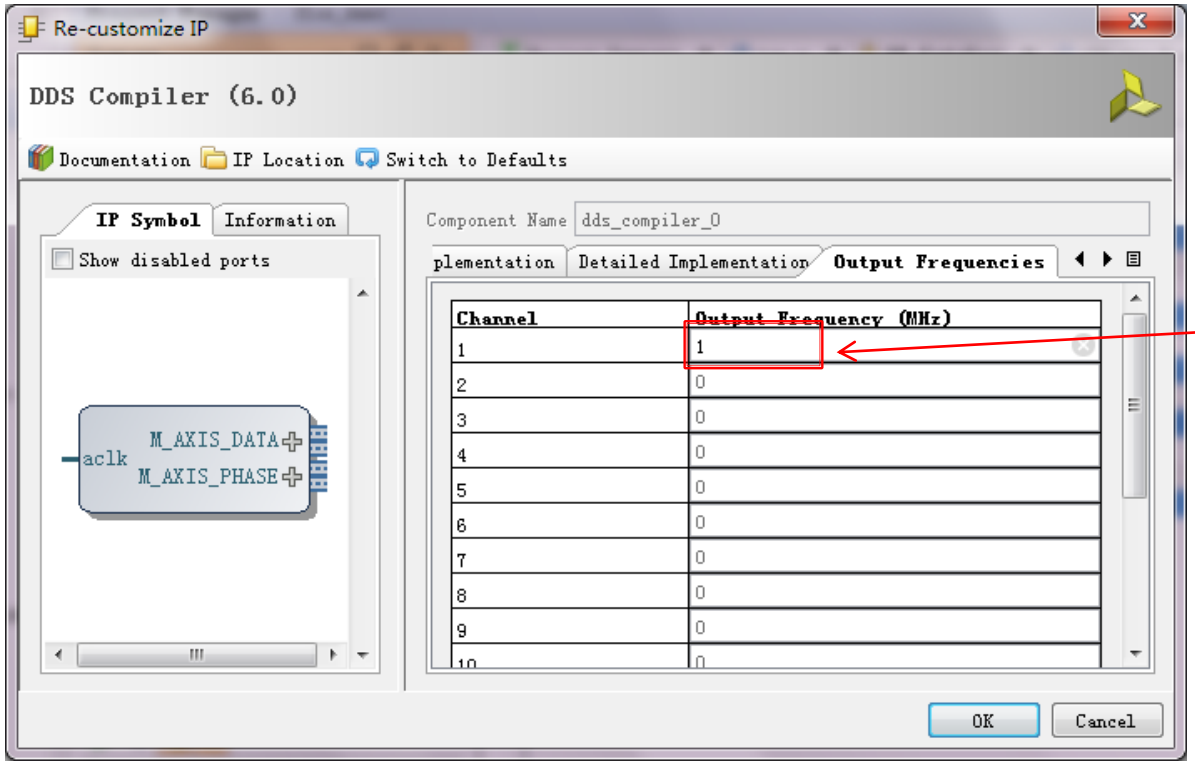
# 调用IP-DDS


Customize IP
DDS Compiler (6.0)
Documentation
IP Location
Switch to Defaults
IP Symbol
Information
Show disabled ports
aclk
M_AXIS_DATA
M_AXIS_PHASE
Component Name
dds_compiler_0
Configuration
Implementation
Detailed Implementation
Output Frequencies
Summary
Configuration Options
Phase Generator and SIN COS LUT
System Requirements
System Clock (MHz)
200
Range: 0.01...550.0
Number of Channels
1
Mode Of Operation
Standard
Frequency per Channel (Fs) 200.0 MHz
Parameter Selection
System Parameters
System Parameters
Spurious Free Dynamic Range(dB)
45
Range: 18...150
Frequency Resolution (Hz)
0.4
7.10543e-07...2.5e+07
Noise Shaping
Auto
OK
Cancel

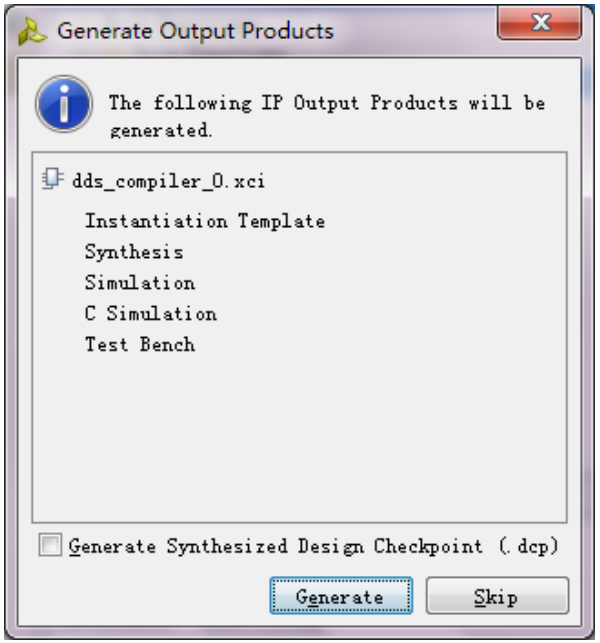
# 调用IP-DDS


Customize IP
DDS Compiler (6.0)
Documentation
IP Location
Switch to Defaults
IP Symbol
Information
Show disabled ports
aclk
M_AXIS_DATA
M_AXIS_PHASE
Component Name
dds_compiler_0
Configuration
Implementation
Detailed Implementation
Output Frequencies
Summary
Phase Increment Programmability
Fixed
Programmable
Streaming
Resync
Phase Offset Programmability
None
Fixed
Programmable
Streaming
Output
Output Selection
Sine
Cosine
Sine and Cosine
Polarity
Negative Sine
Negative Cosine
Amplitude Mode
Full Range
Implementation Options
Memory Type
Auto
Optimization Goal
Auto
DSP48 Use
Minimal
Has Phase Out
OK
Cancel

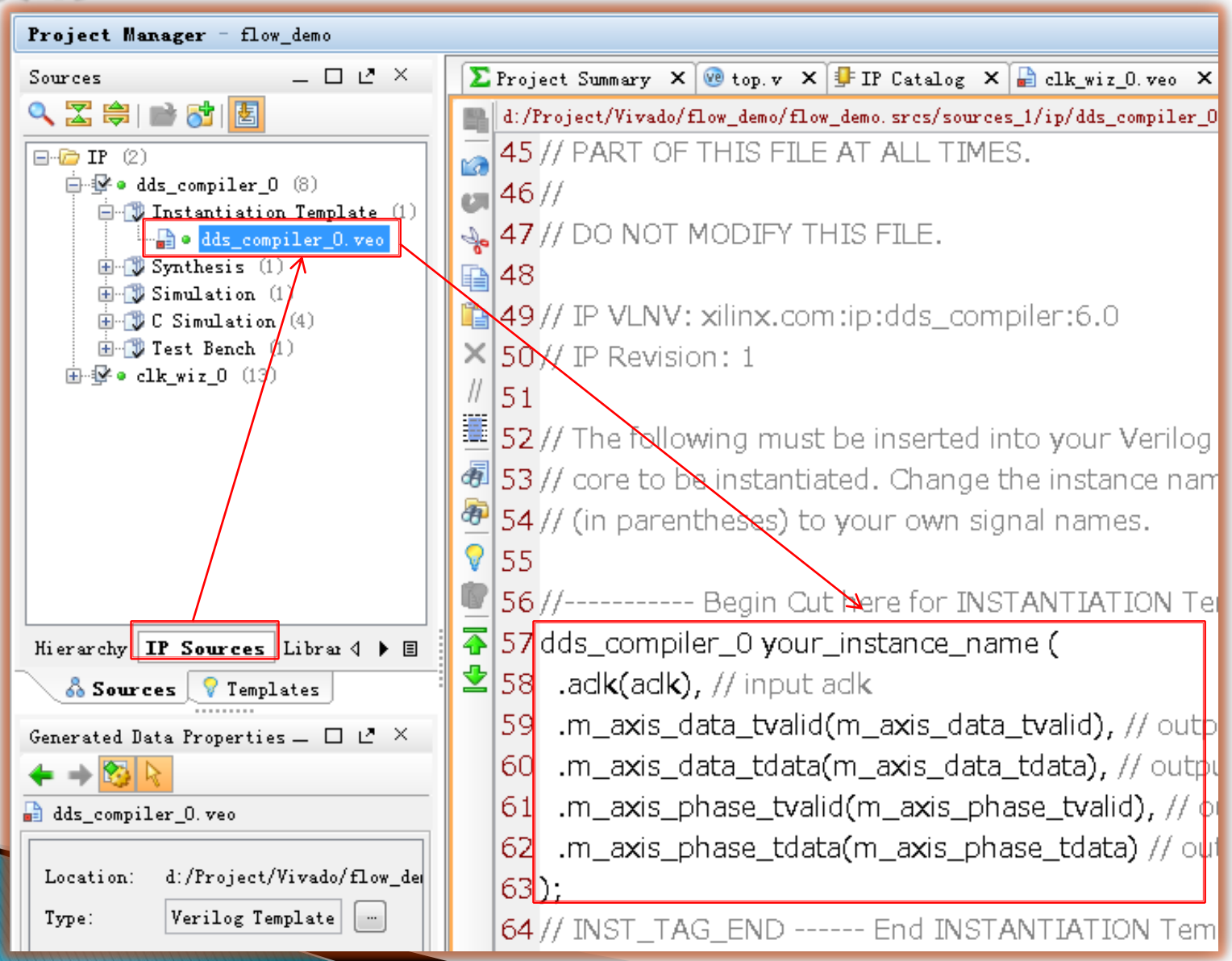
# 调用IP-DDS



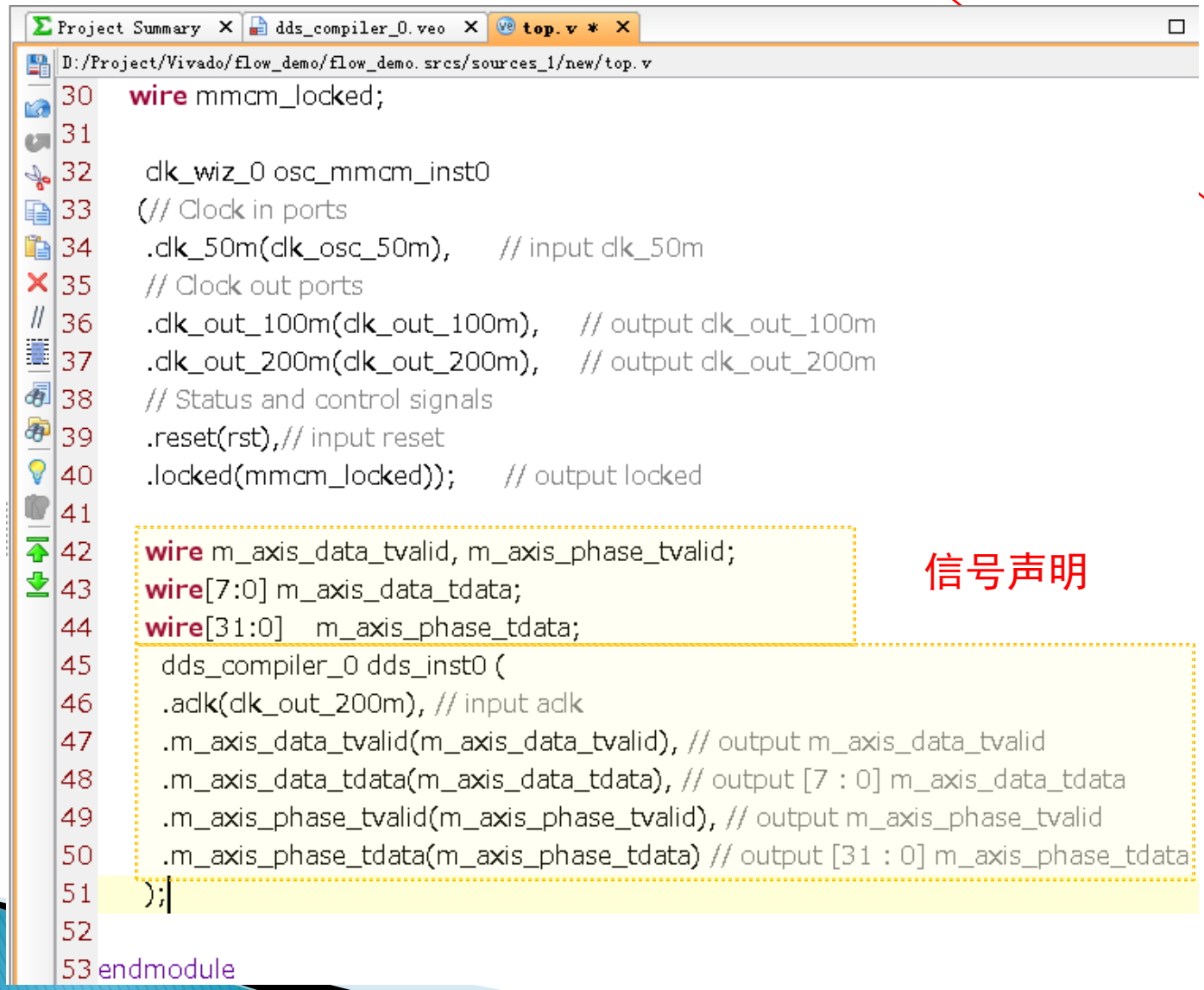

输入需要的频率



# 调用IP-DDS


Project Manager - flow_demo
Sources
IP (2)
dds_compiler_0 (8)
Instantiation Template (1)
dds_compiler_0.veo
Synthesis (1)
Simulation (1)
C Simulation (4)
Test Bench (1)
clk_wiz_0 (13)
Hierarchy
IP Sources
Sources
Templates
Generated Data Properties
dds_compiler_0.veo
Location: d:/Project/Vivado/flow_de
Type: Verilog Template
Project Summary
top.v
IP Catalog
clk_wiz_0.veo
d:/Project/Vivado/flow_demo/flow_demo.srcs/sources_1/ip/dds_compiler_0
45 // PART OF THIS FILE AT ALL TIMES.
46 //
47 // DO NOT MODIFY THIS FILE.
48
49 // IP VLNV: xilinx.com:ip:dds_compiler:6.0
50 // IP Revision: 1
51
52 // The following must be inserted into your Verilog
53 // core to be instantiated. Change the instance nam
54 // (in parentheses) to your own signal names.
55
56 //----------- Begin Cut Here for INSTANTIATION Te
57 dds_compiler_0 your_instance_name (
58 .aclk(aclk), // input aclk
59 .m_axis_data_tvalid(m_axis_data_tvalid), // outp
60 .m_axis_data_tdata(m_axis_data_tdata), // outpu
61 .m_axis_phase_tvalid(m_axis_phase_tvalid), // o
62 .m_axis_phase_tdata(m_axis_phase_tdata) // ou
63 );
64 // INST_TAG_END ------ End INSTANTIATION Tem

# 调用IP-DDS

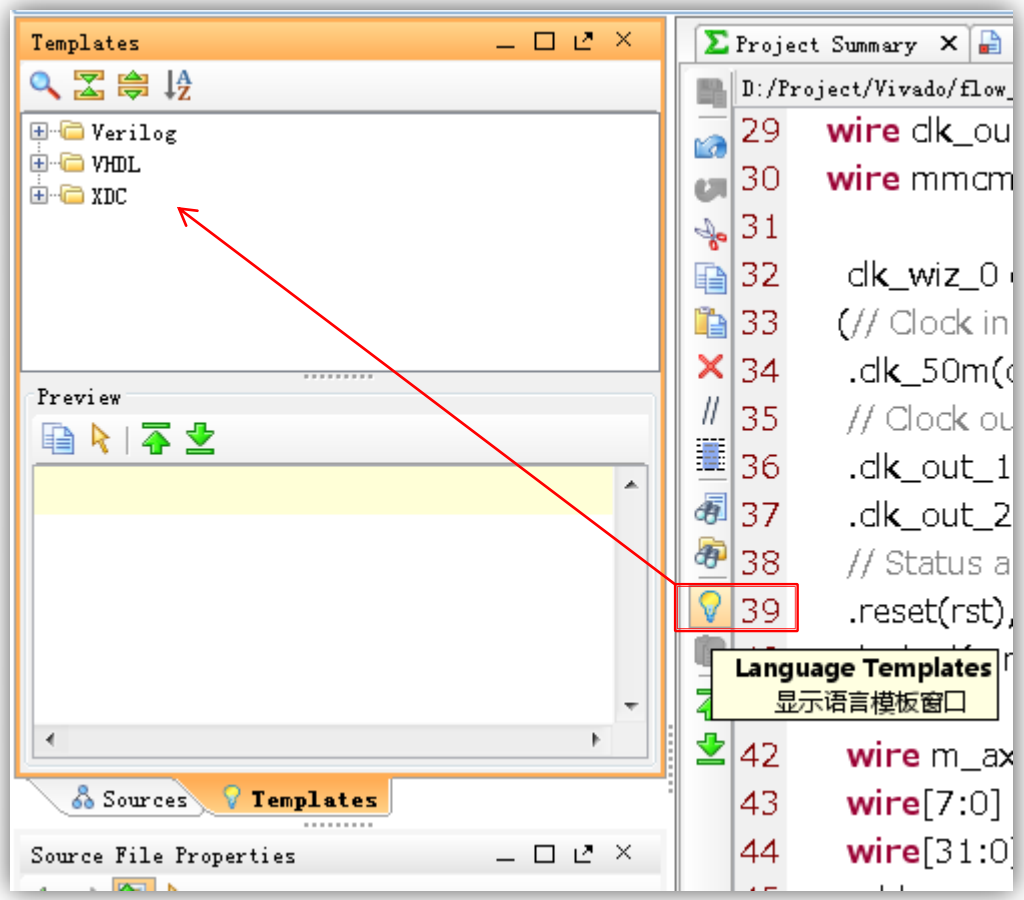

Project Summary
dds_compiler_0.veo
top.v *
D:/Project/Vivado/flow_demo/flow_demo.srcs/sources_1/new/top.v
30 wire mmcm_locked;
31
32 clk_wiz_0 osc_mmcm_inst0
33 (// Clock in ports
34 .clk_50m(clk_osc_50m), // input clk_50m
35 // Clock out ports
36 .clk_out_100m(clk_out_100m), // output clk_out_100m
37 .clk_out_200m(clk_out_200m), // output clk_out_200m
38 // Status and control signals
39 .reset(rst),// input reset
40 .locked(mmcm_locked)); // output locked
41
42 wire m_axis_data_tvalid, m_axis_phase_tvalid;
43 wire[7:0] m_axis_data_tdata;
44 wire[31:0] m_axis_phase_tdata;
45 dds_compiler_0 dds_inst0 (
46 .aclk(clk_out_200m), // input aclk
47 .m_axis_data_tvalid(m_axis_data_tvalid), // output m_axis_data_tvalid
48 .m_axis_data_tdata(m_axis_data_tdata), // output [7 : 0] m_axis_data_tdata
49 .m_axis_phase_tvalid(m_axis_phase_tvalid), // output m_axis_phase_tvalid
50 .m_axis_phase_tdata(m_axis_phase_tdata) // output [31 : 0] m_axis_phase_tdata
51 );
52
53 endmodule
信号声明
IP例化

# Debug变量声明

- 在插入Chipscope ILA模块时，可以直接找到Debug变量。

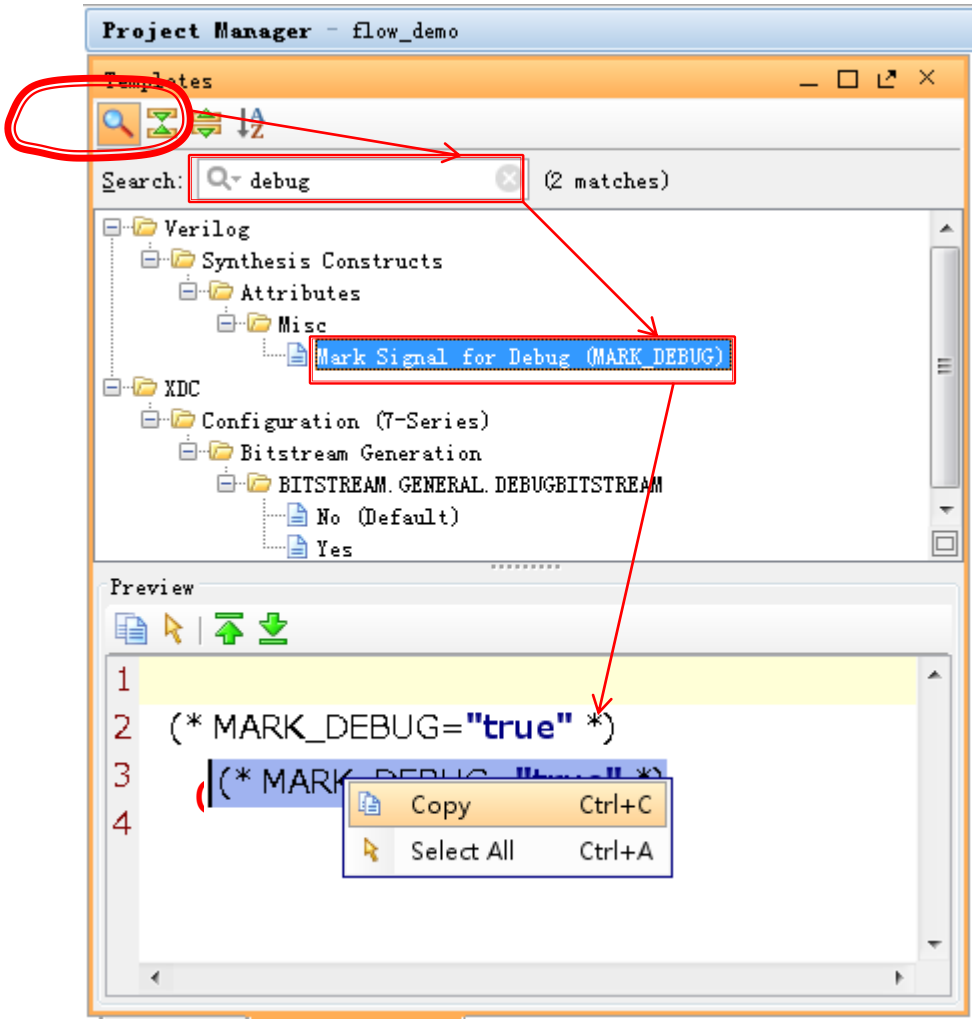
# Debug变量声明

- 启动代码模板



# Debug变量声明

- 搜索”debug”关键字



# Debug变量声明

- 声明为”DEBUG”,即使没有连接到其他模块，也不会被优化掉。

# DDS模块

- DEBUG变量的实现

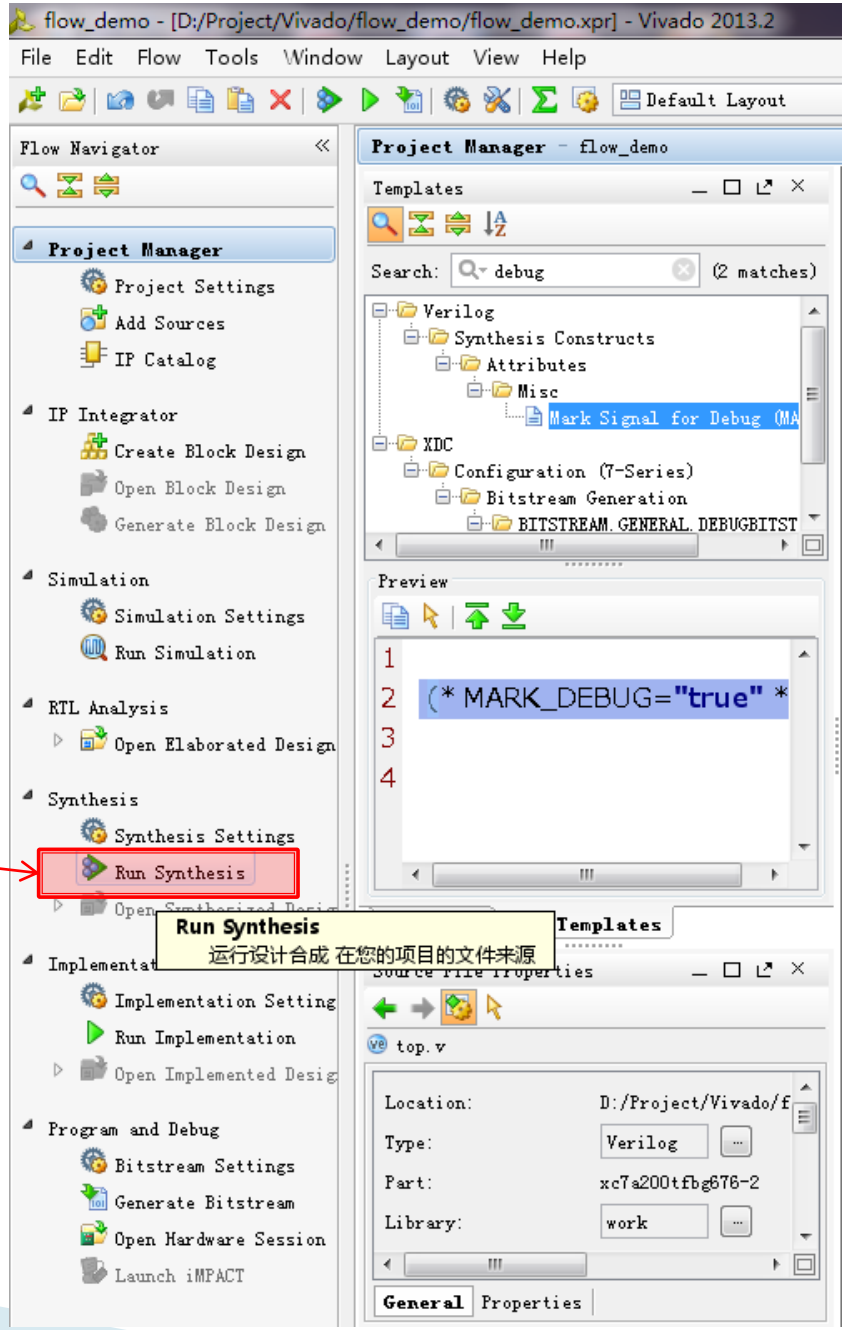
```
52
53      (* MARK_DEBUG="true" *) reg[7:0] m_axis_data_tdata_r;
54      (* MARK_DEBUG="true" *) reg[31:0] m_axis_phase_tdata_r;
55
56      always @(posedge clk_out_200m)
57      begin
58          if(m_axis_data_tvalid)
59              m_axis_data_tdata_r<=m_axis_data_tdata;
60      end
61
62      always @(posedge clk_out_200m)
63      begin
64          if(m_axis_phase_tvalid)
65              m_axis_phase_tdata_r<=m_axis_phase_tdata;
66      end
67
68  endmodule
```

# Counter模块实现

```
68    (* MARK_DEBUG="true" *) reg[23:0] led_r=0;
69
70    always @(posedge clk_out_100m)
71     begin
72         if(mmcm_locked)
73             led_r<=led_r +1;
74     end
75
76     assign led[3:0] =led_r[23:20];
77
78 endmodule
```

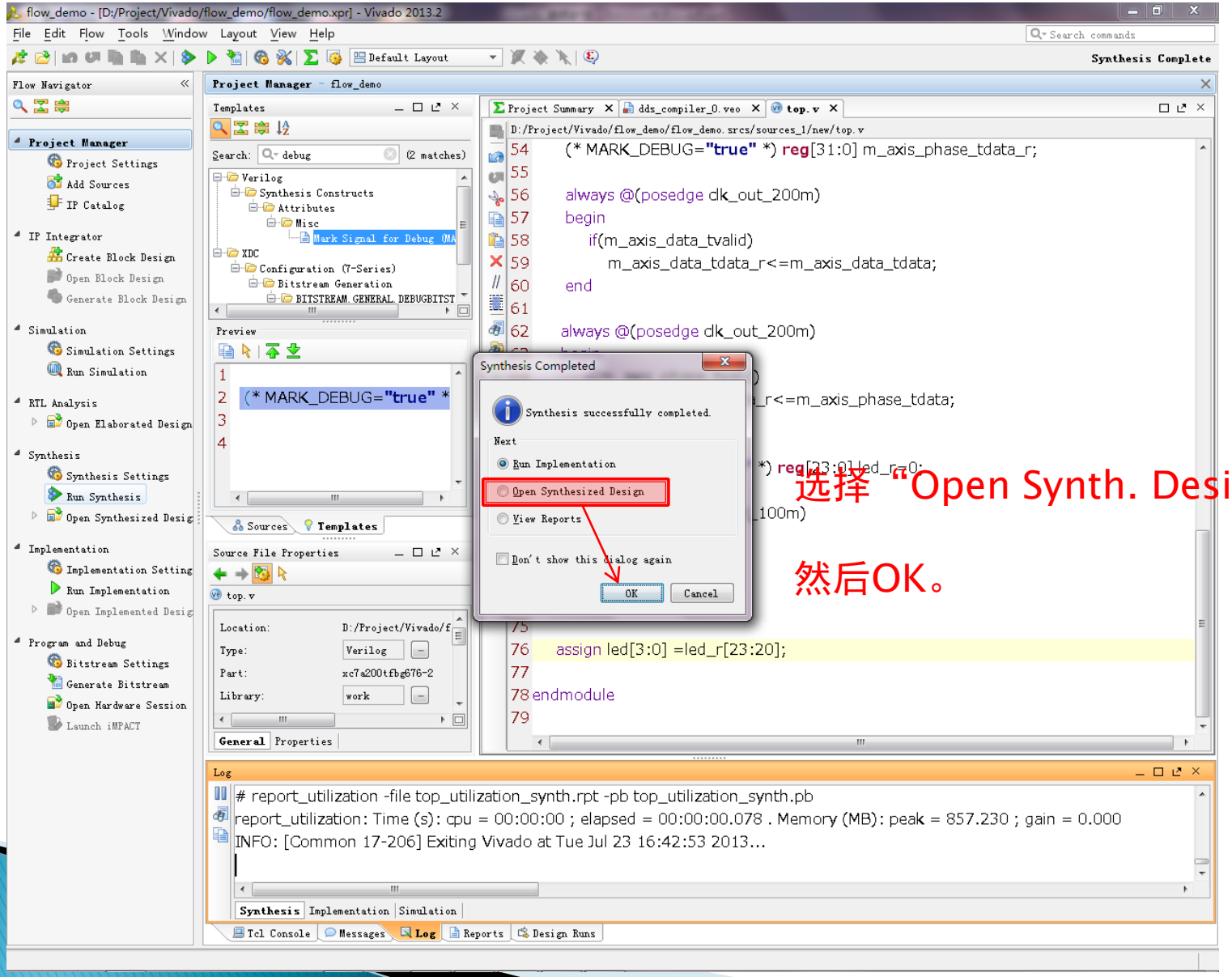
声明时赋初始值；
不要使用reset赋值方式！

# Synthesis !

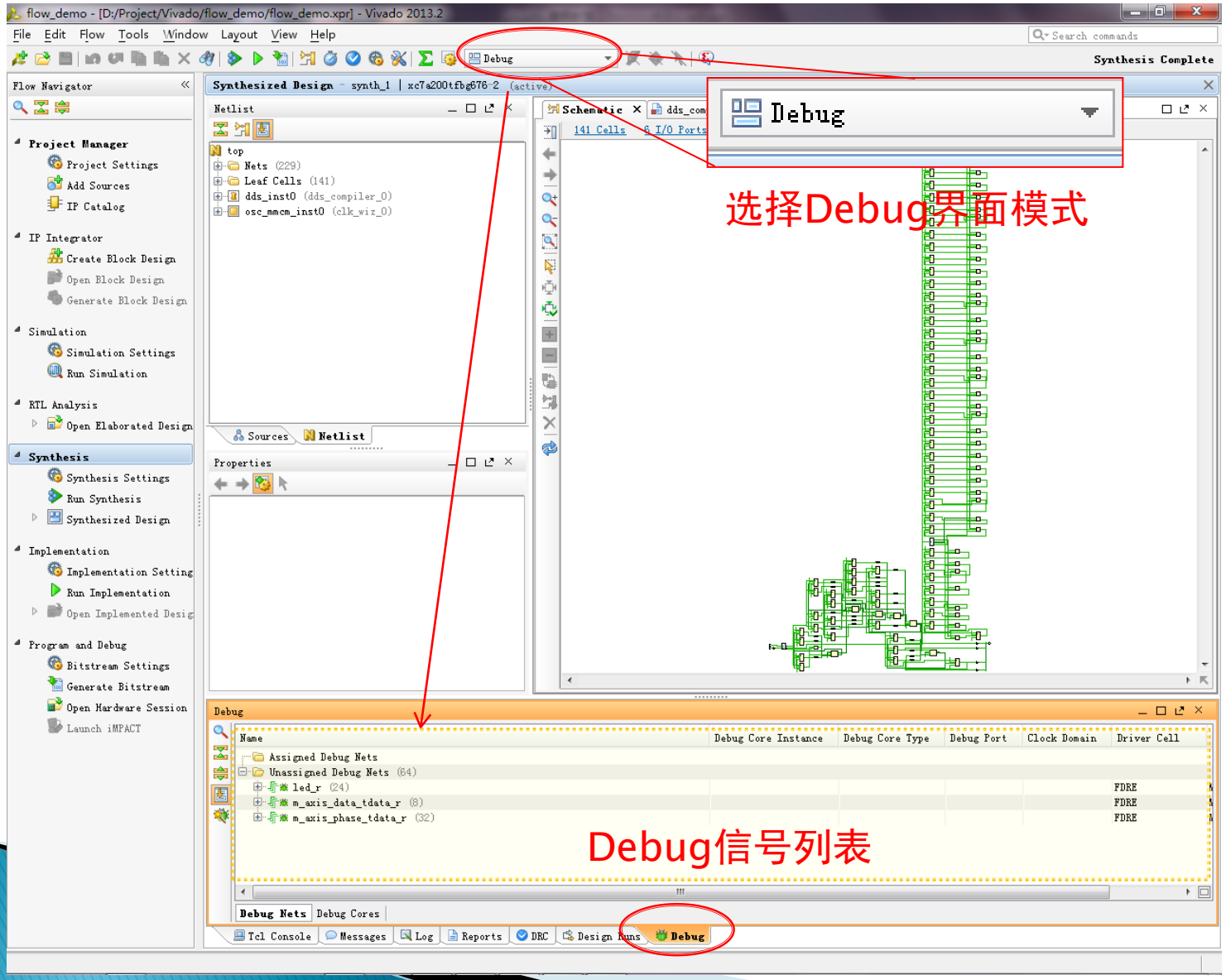



点击它

# Synthesis完毕

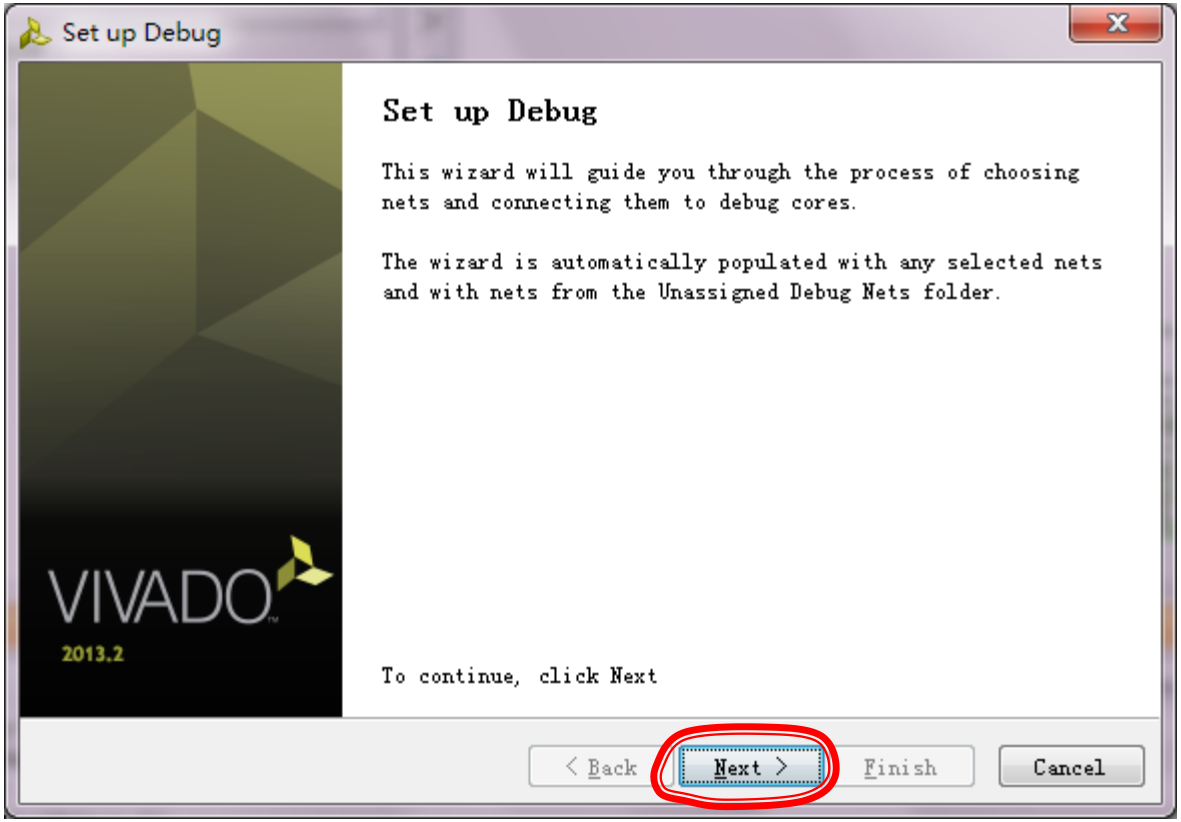


# 设置Chipscope Debug信号

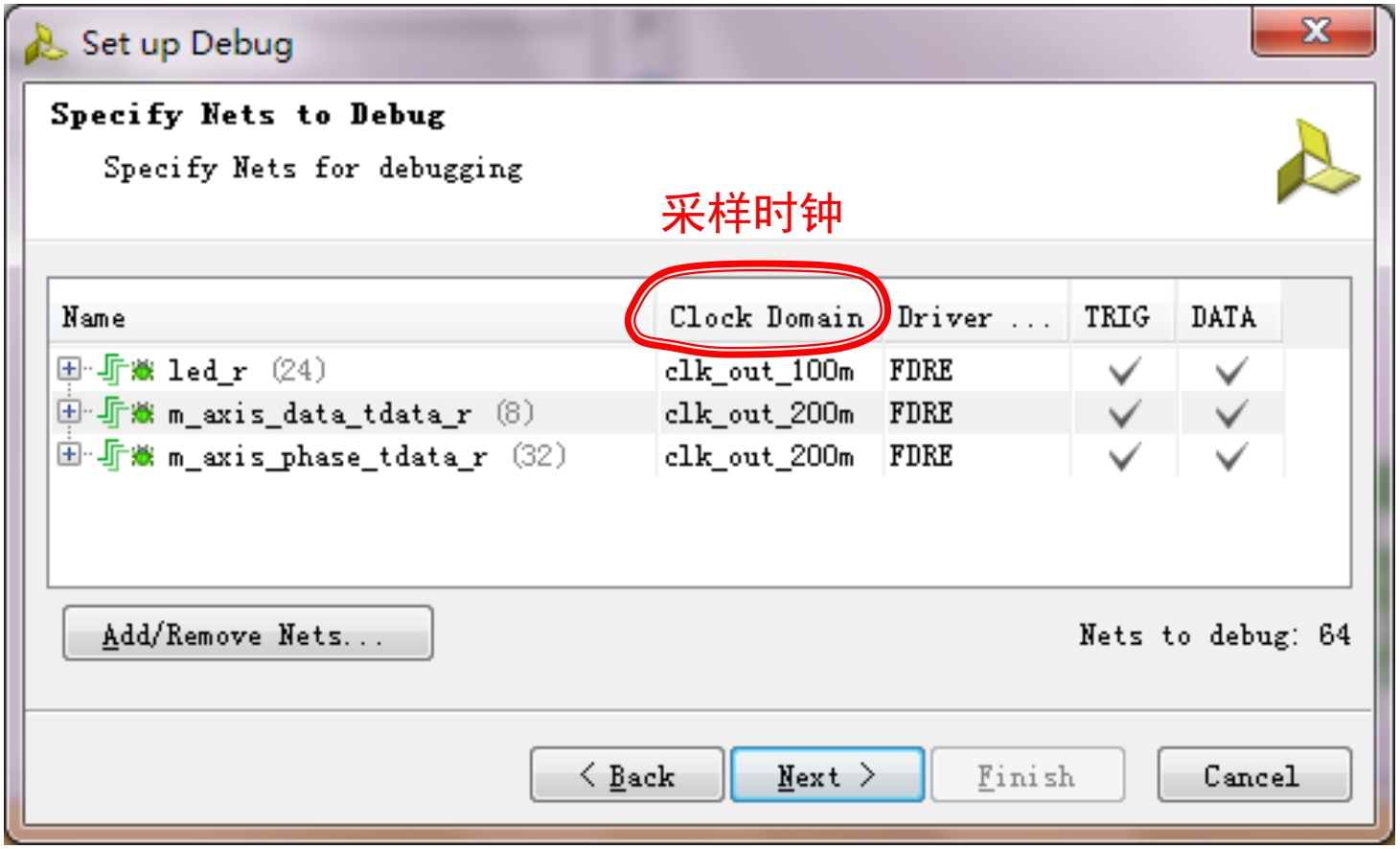


# 设置Chipscope Debug信号

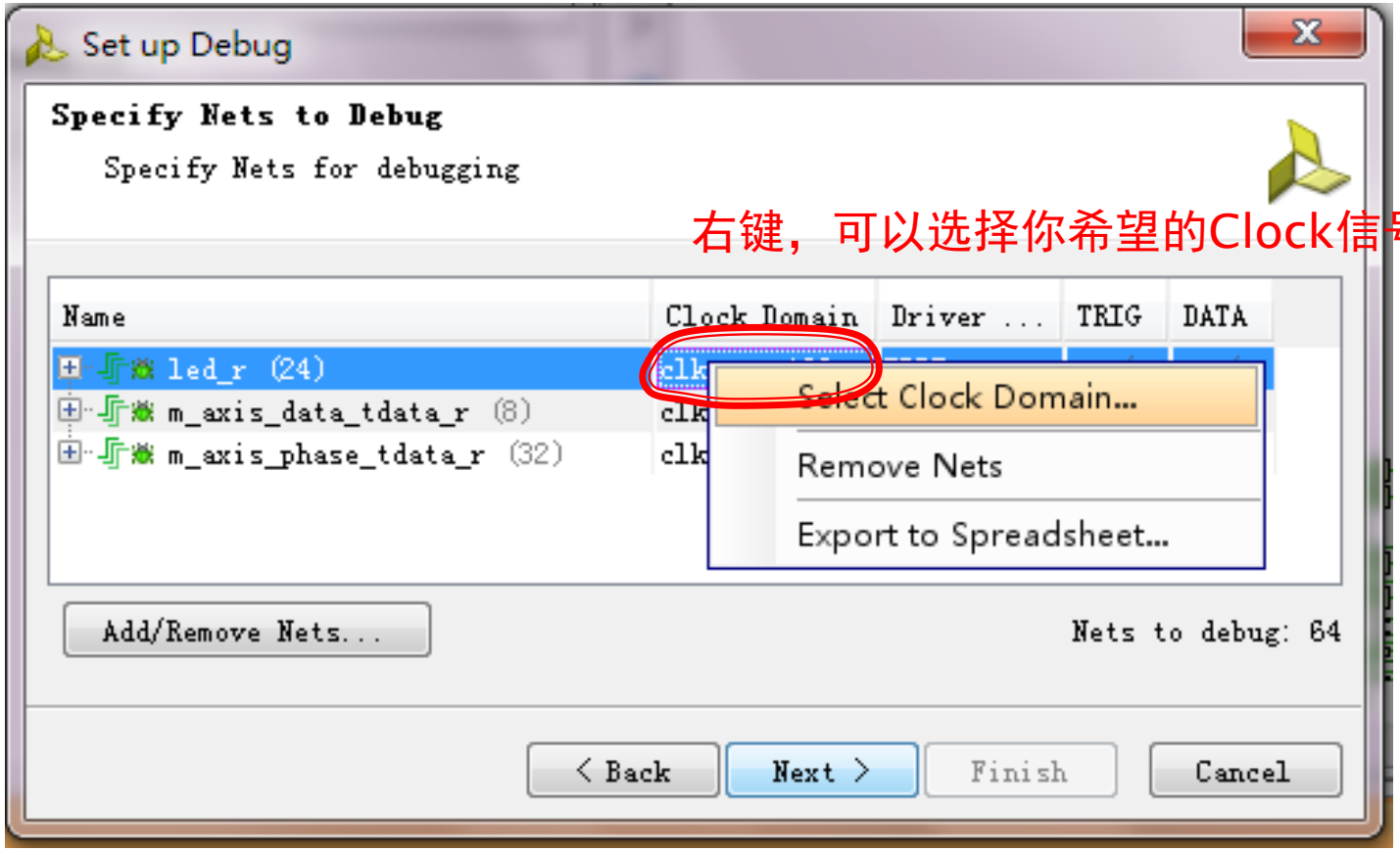
# 设置Chipscope Debug信号



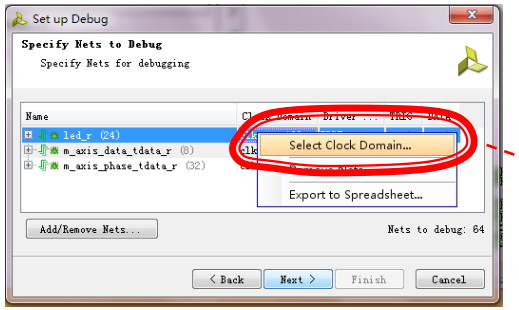
# 设置Chipscope Debug信号

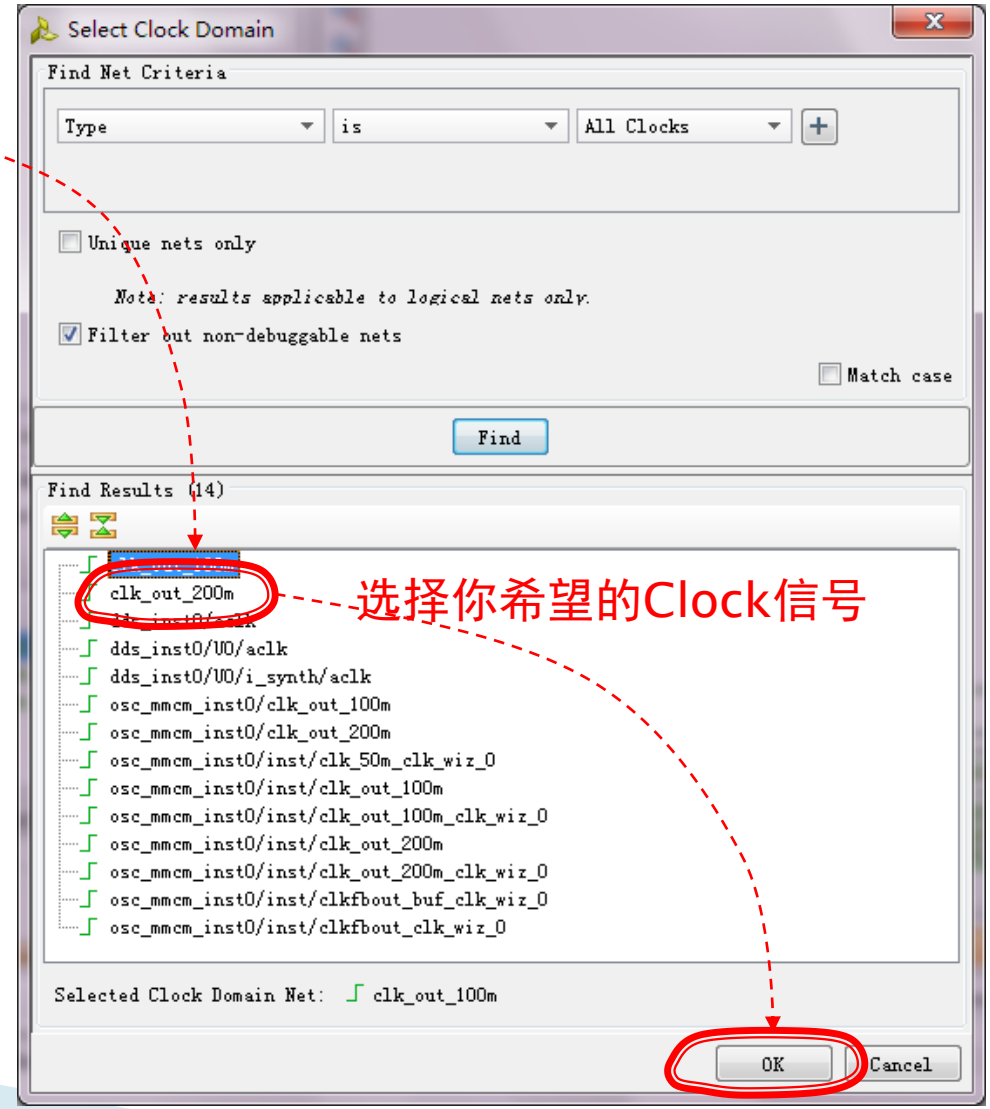


# 设置Chipscope Debug信号

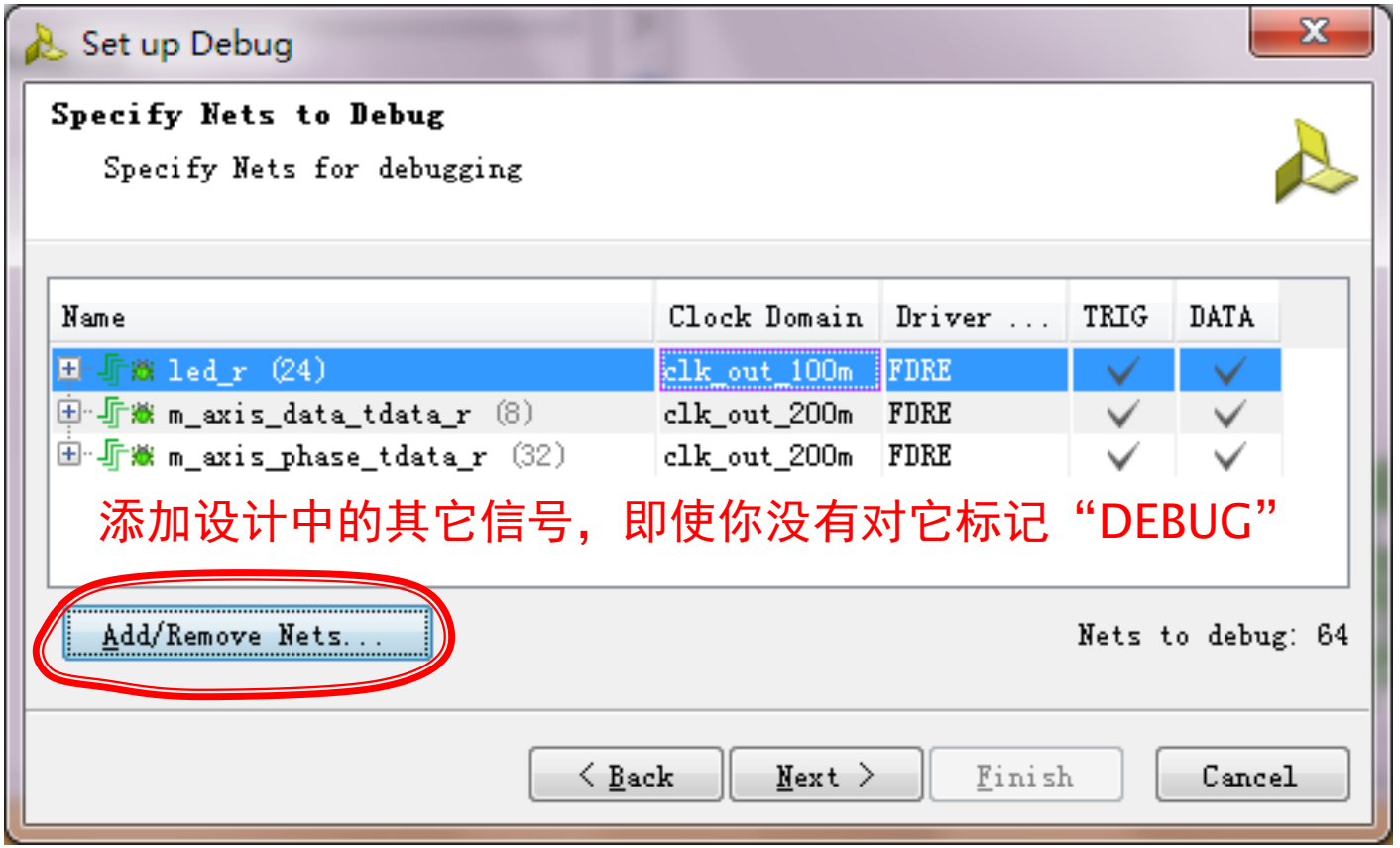


# 设置Chipscope Debug信号

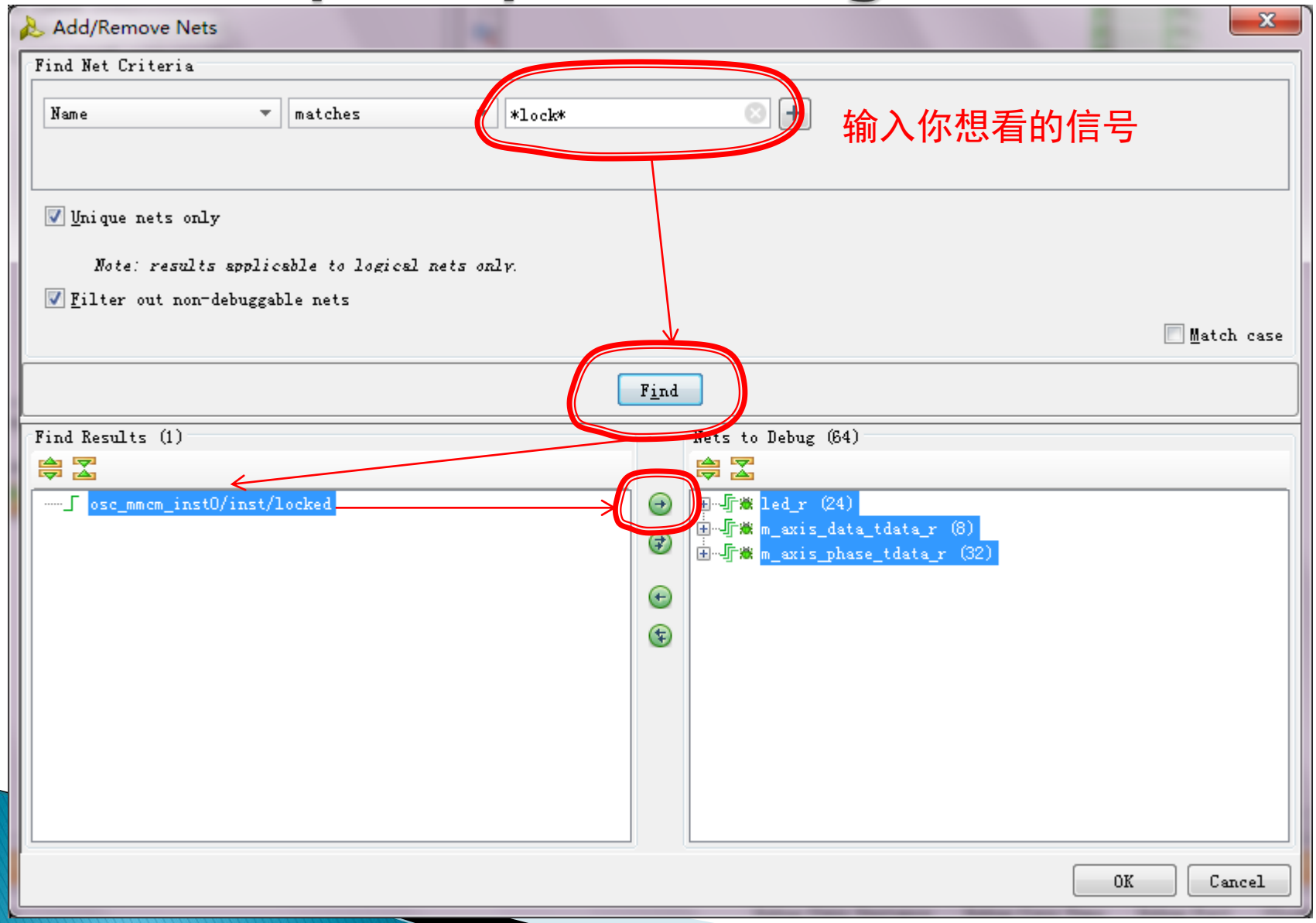


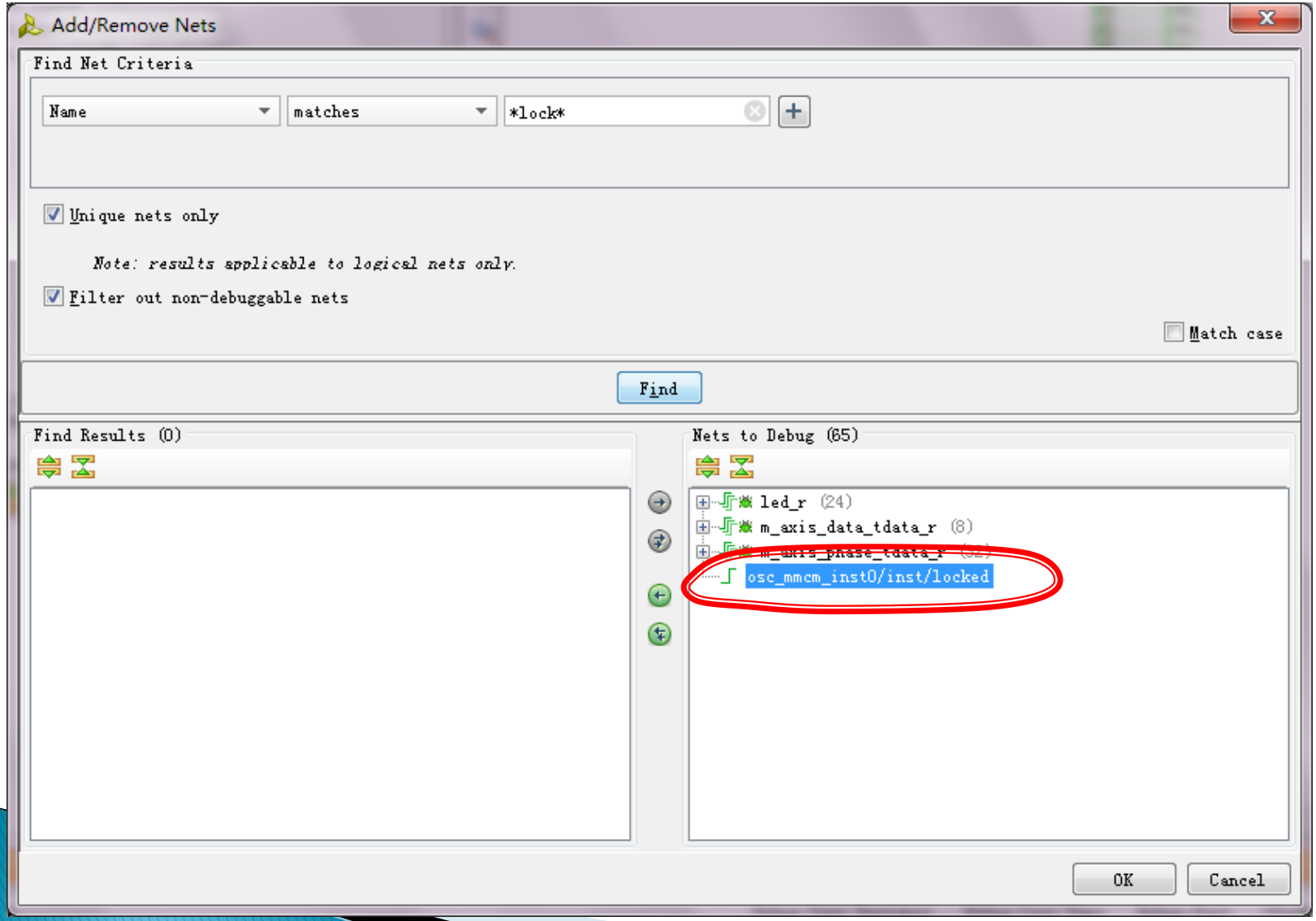
# 设置Chipscope Debug信号



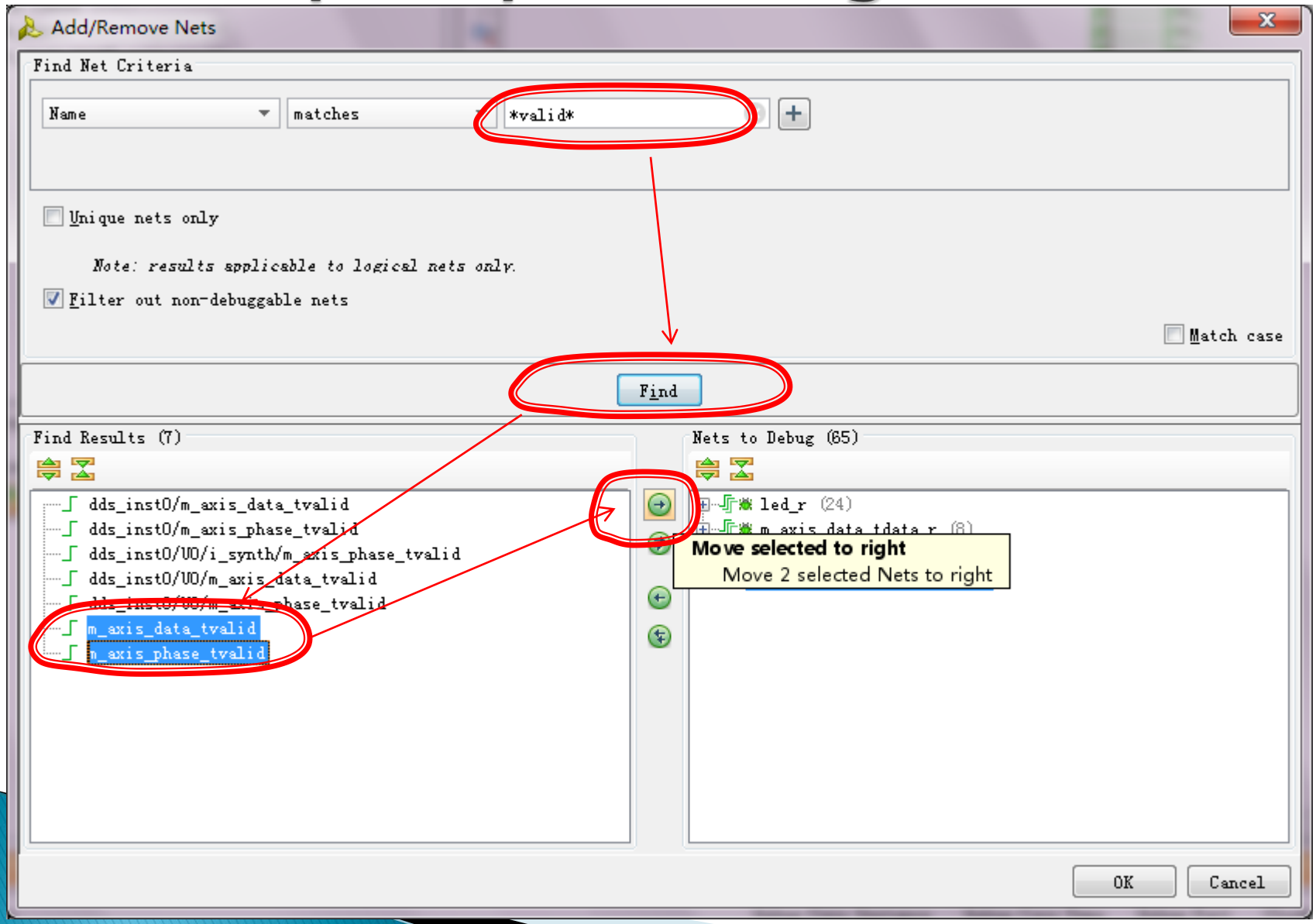
# 设置Chipscope Debug信号



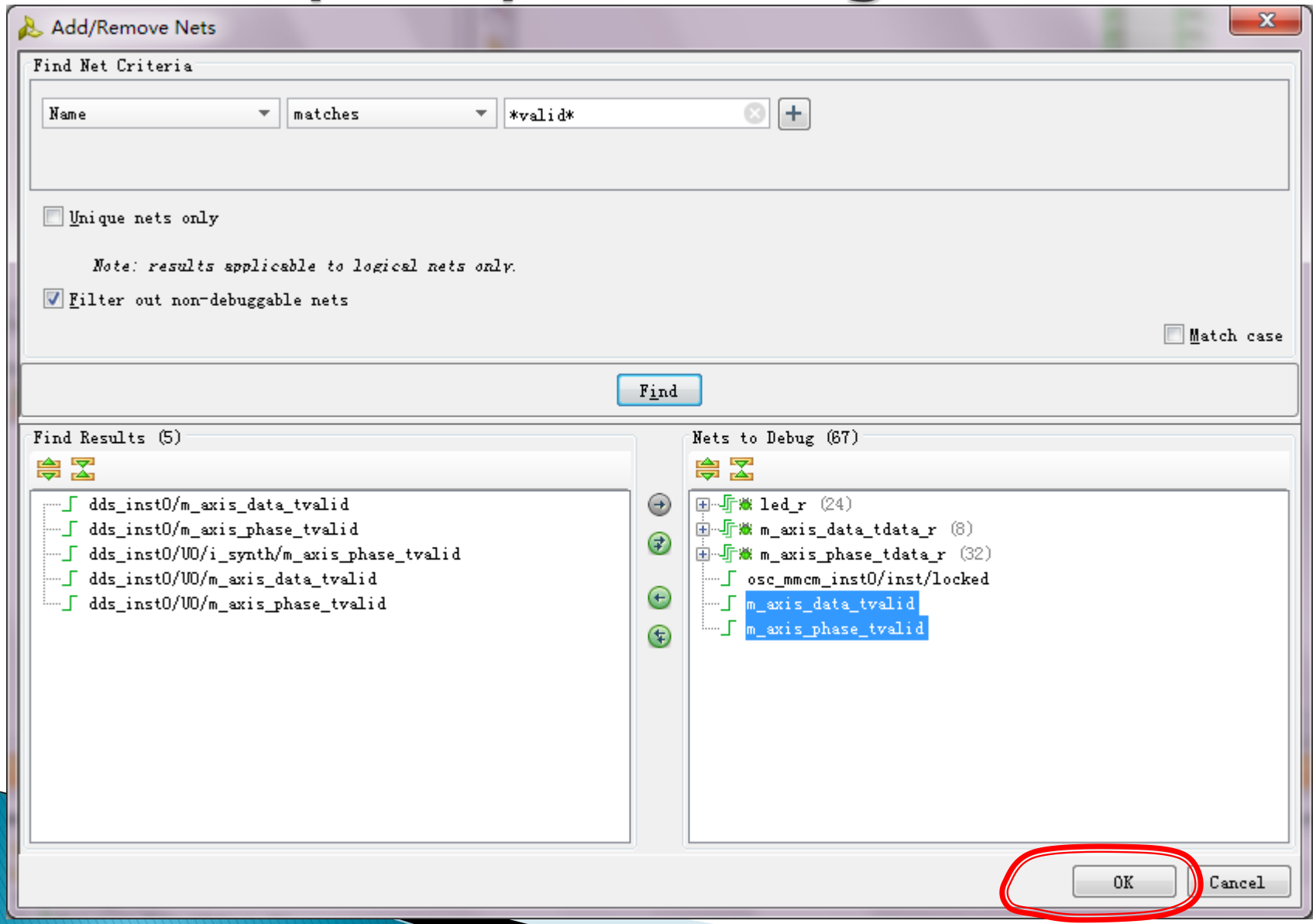
# 设置Chipscope Debug信号



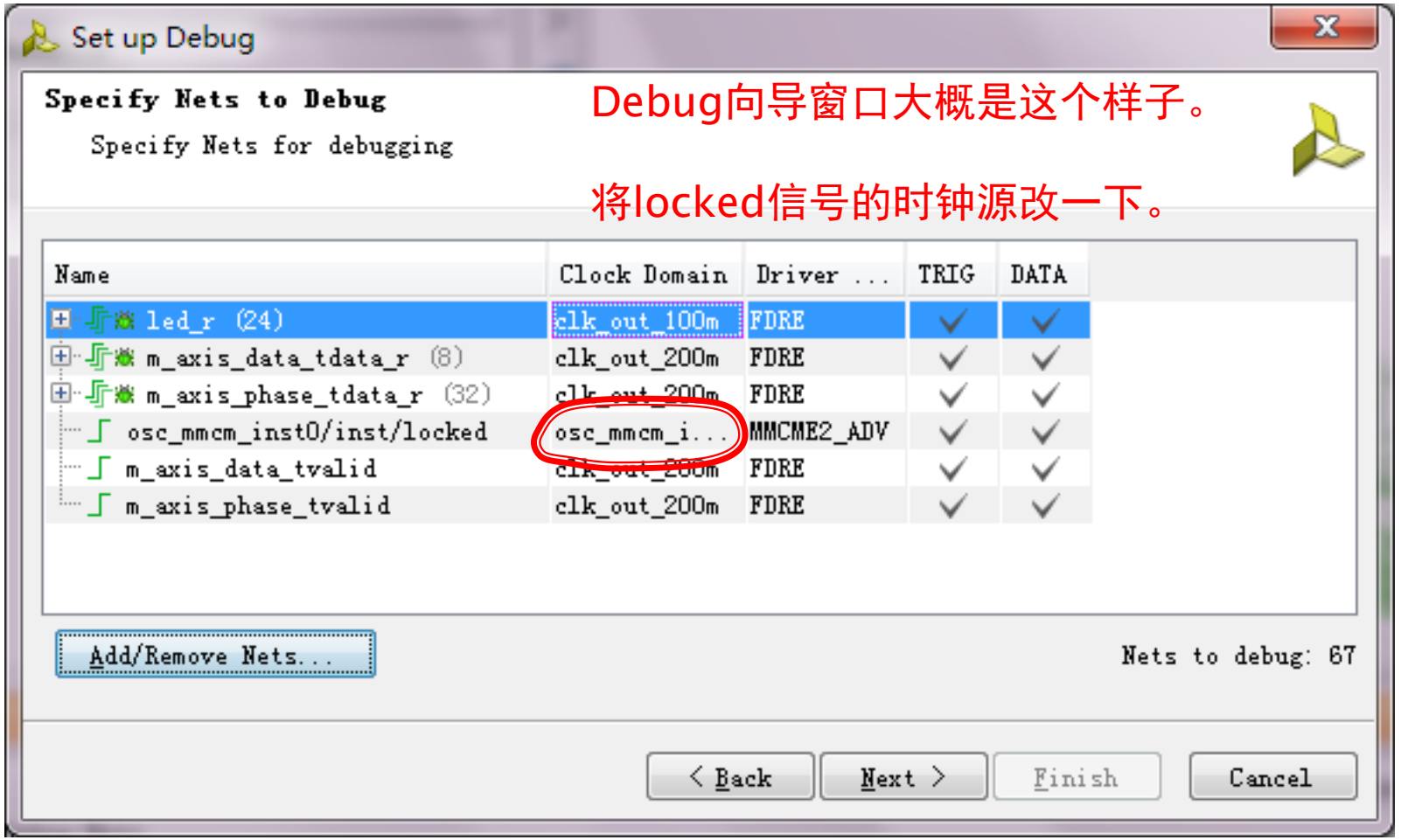
# 设置Chipscope Debug信号



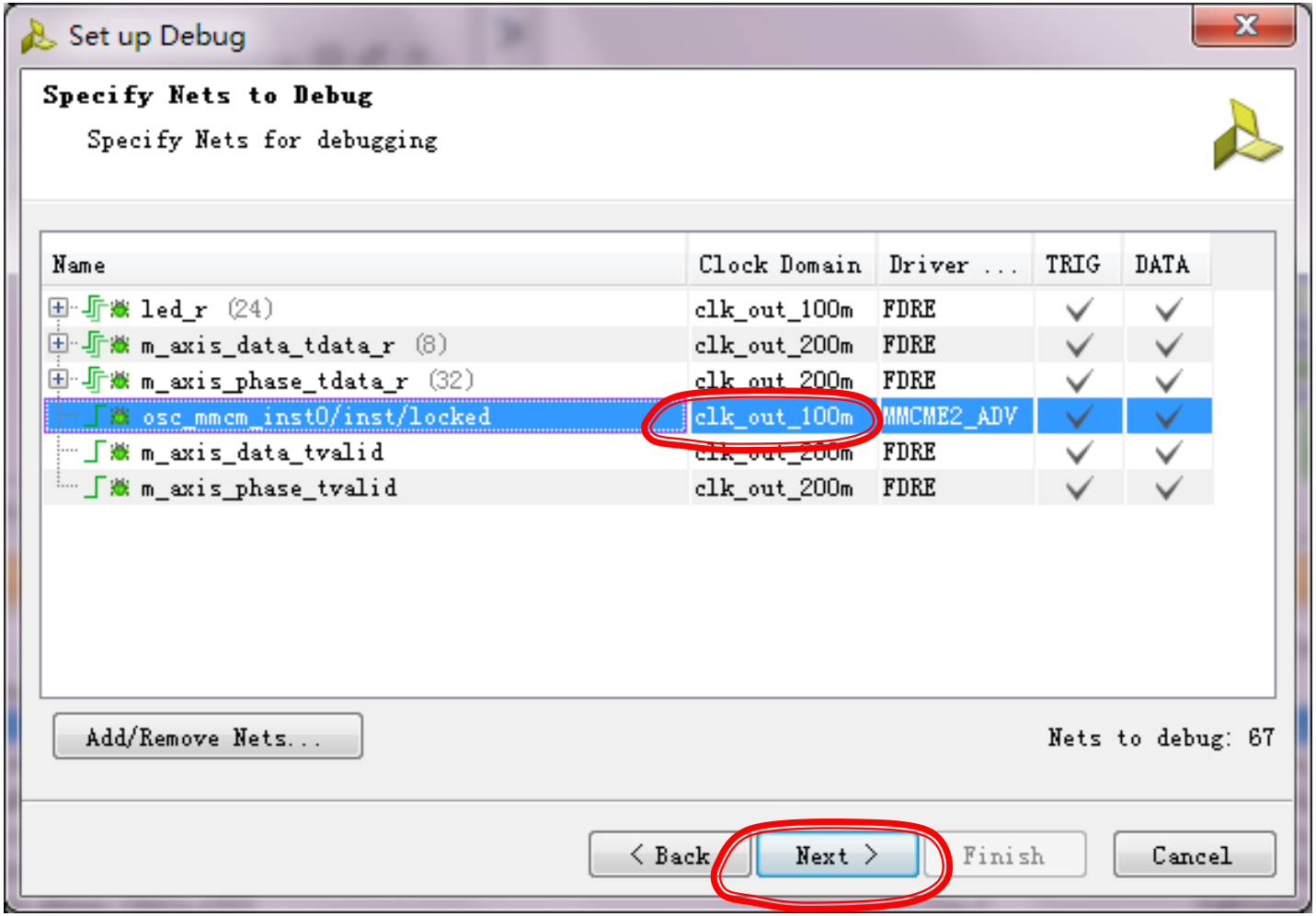
# 设置Chipscope Debug信号



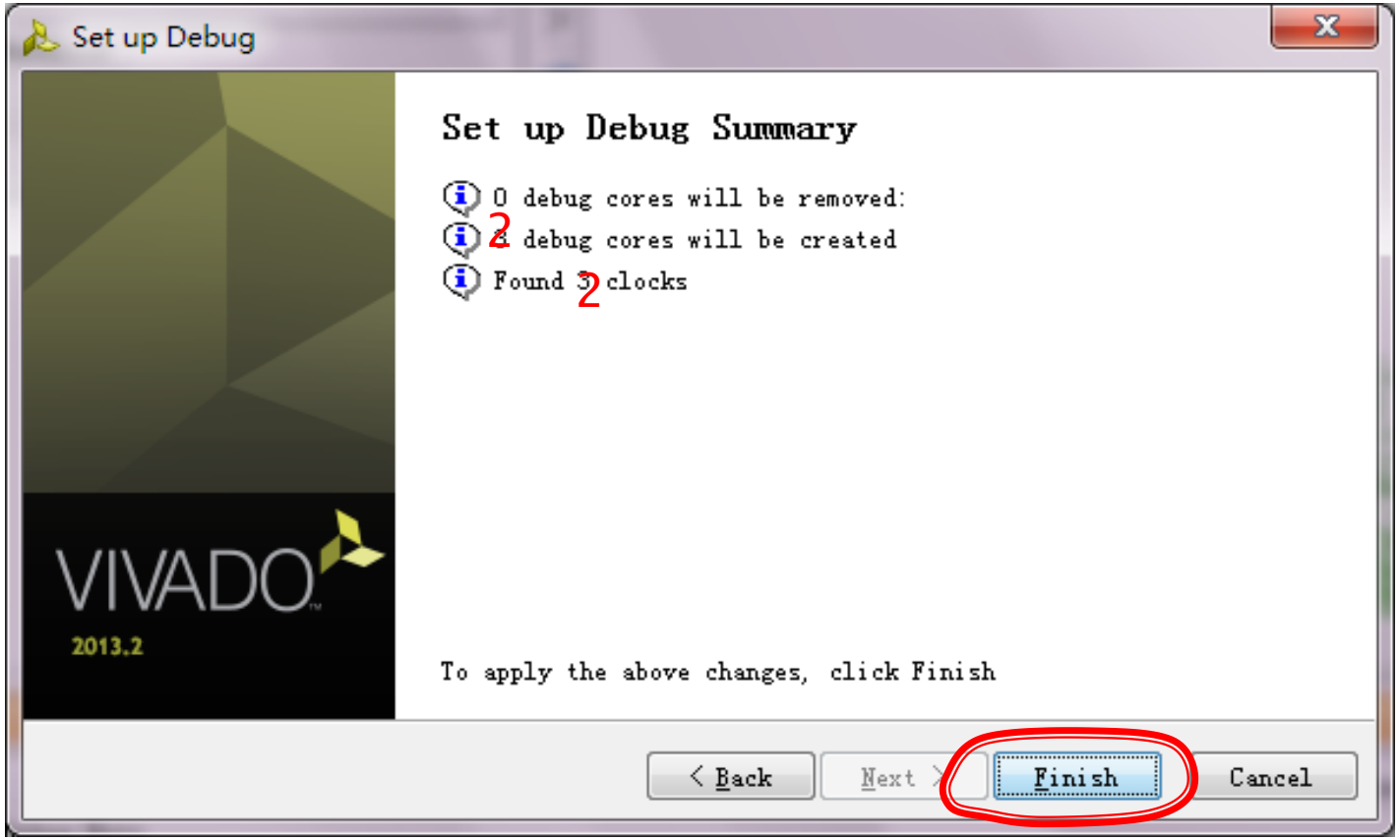
# 设置Chipscope Debug信号



# 设置Chipscope Debug信号

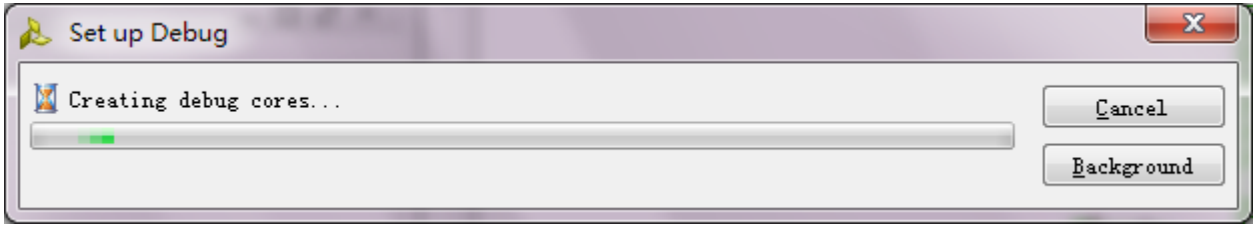


# 设置Chipscope Debug信号

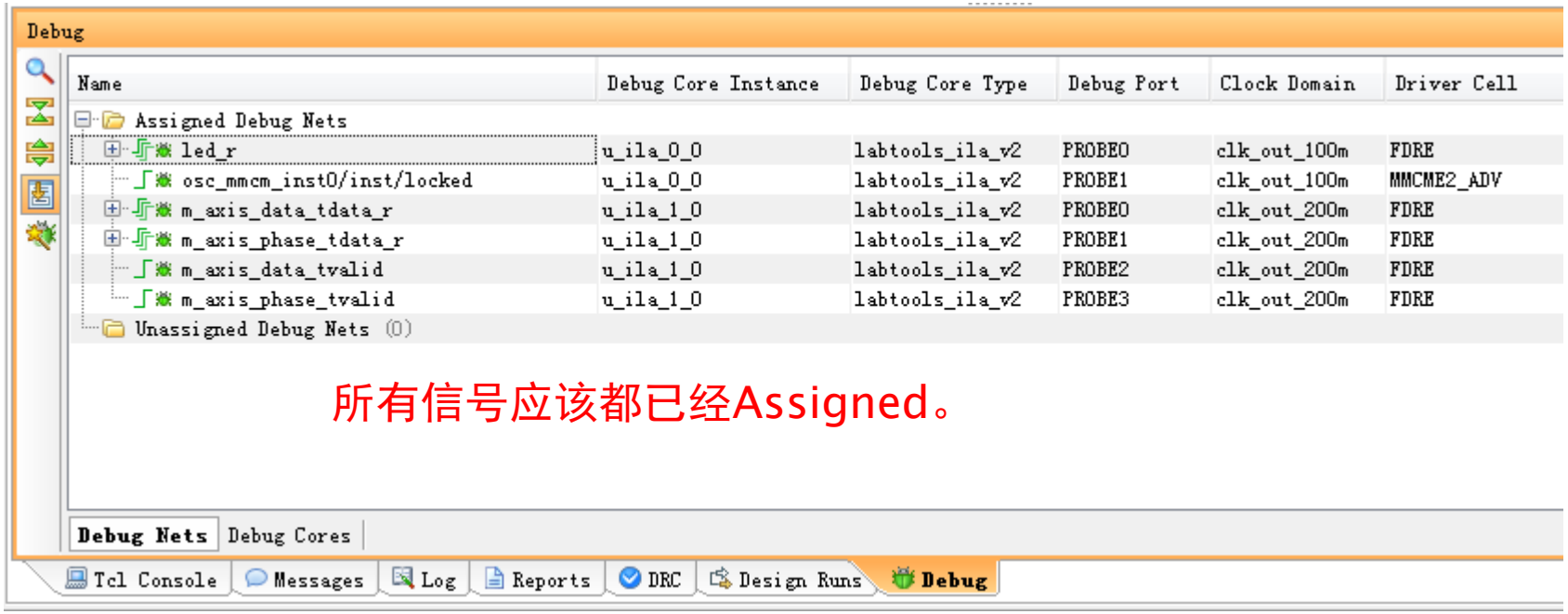


# 设置Chipscope Debug信号

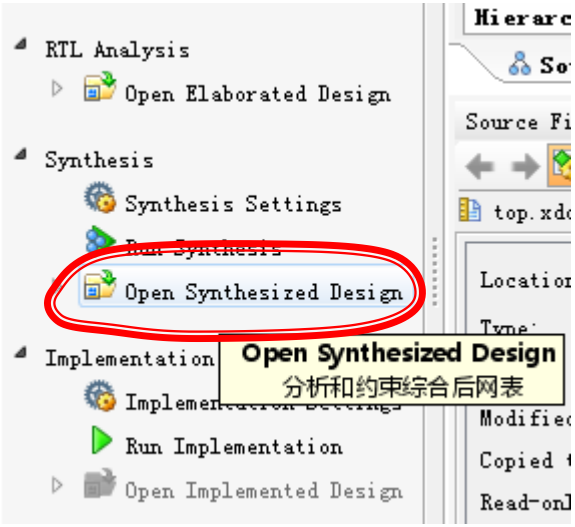
自动设置Debug相关的core。



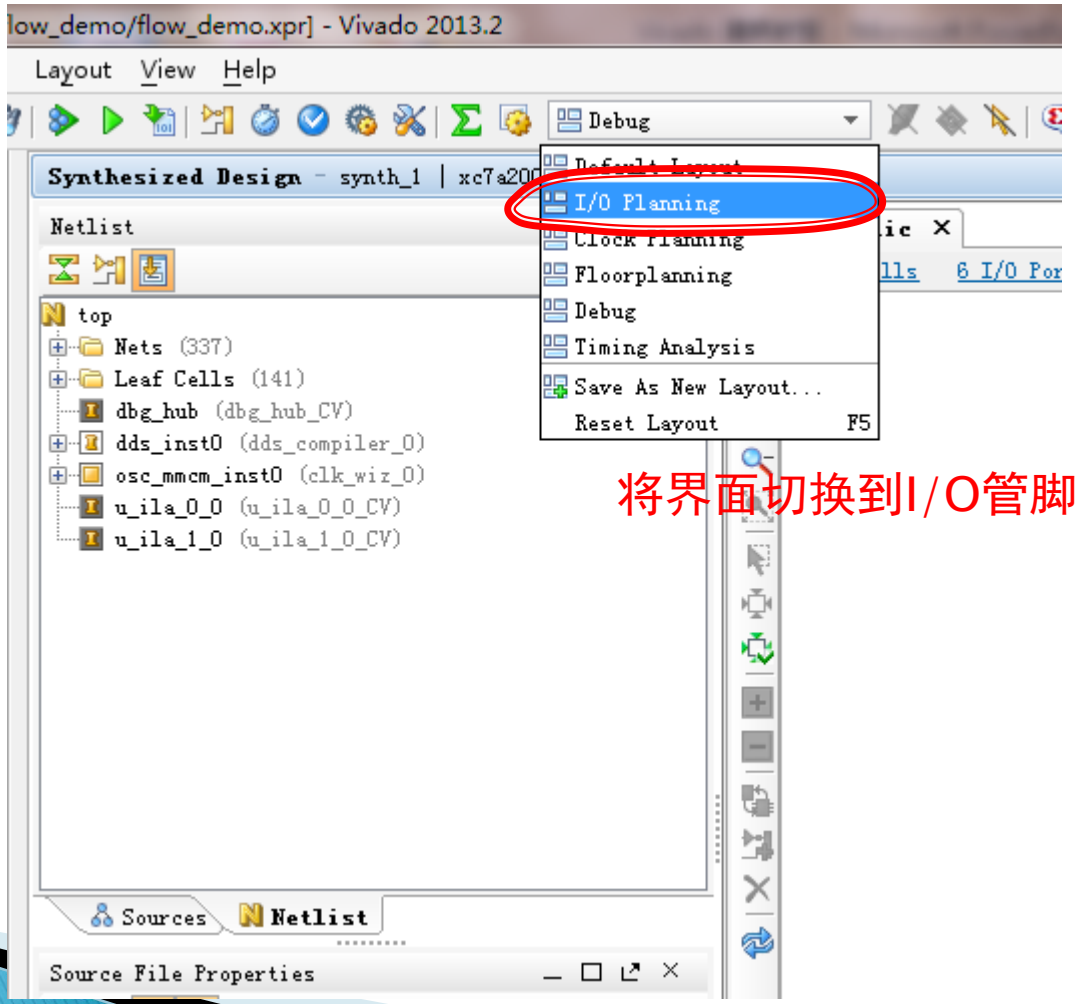
# 设置Chipscope Debug信号



# 锁定管脚


RTL Analysis
Open Elaborated Design
Synthesis
Synthesis Settings
Open Synthesized Design
Implementation
Run Implementation
Open Implemented Design
Open Synthesized Design
分析和约束综合后网表

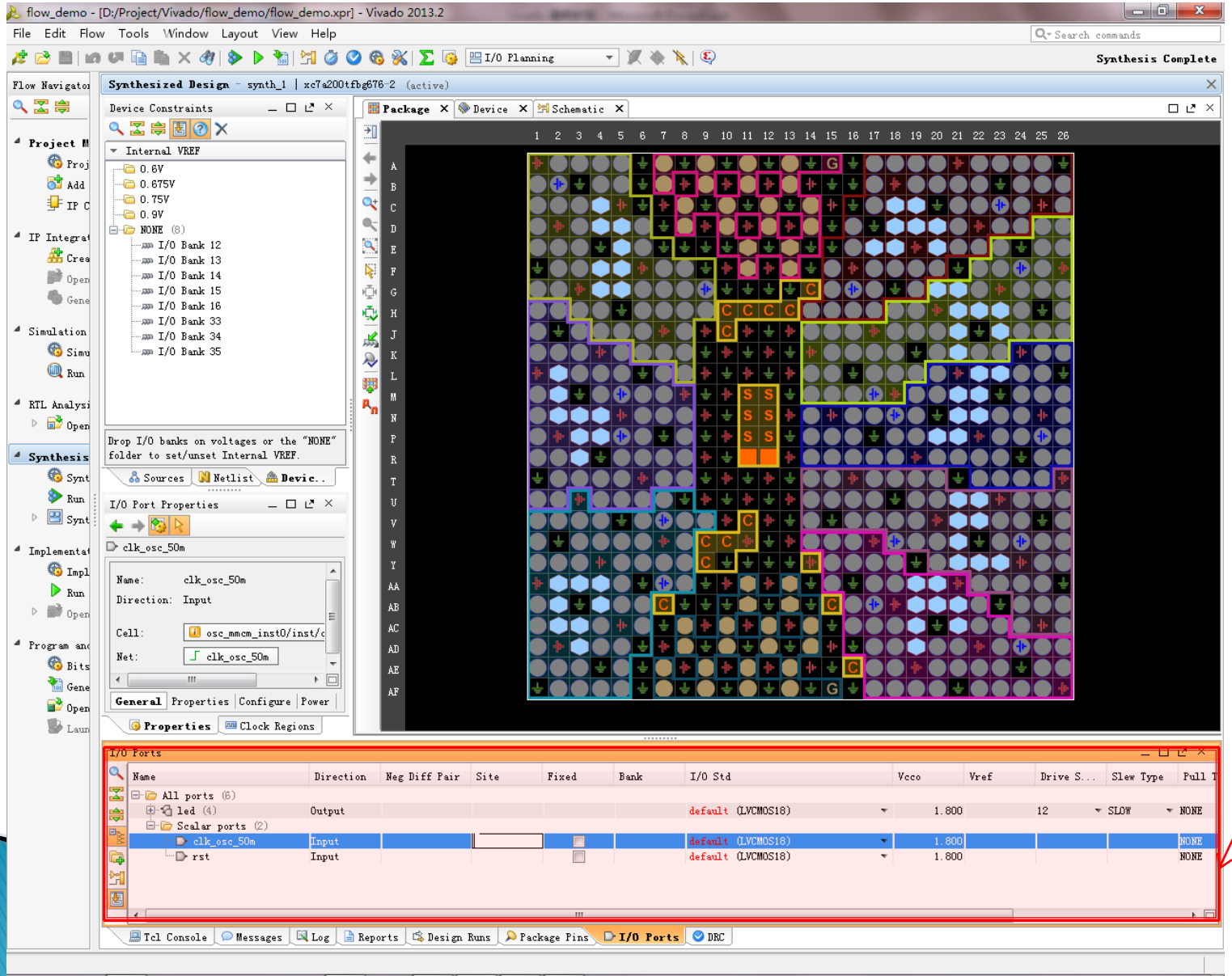
# 锁定管脚




将界面切换到I/O管脚分配模式

# 锁定管脚

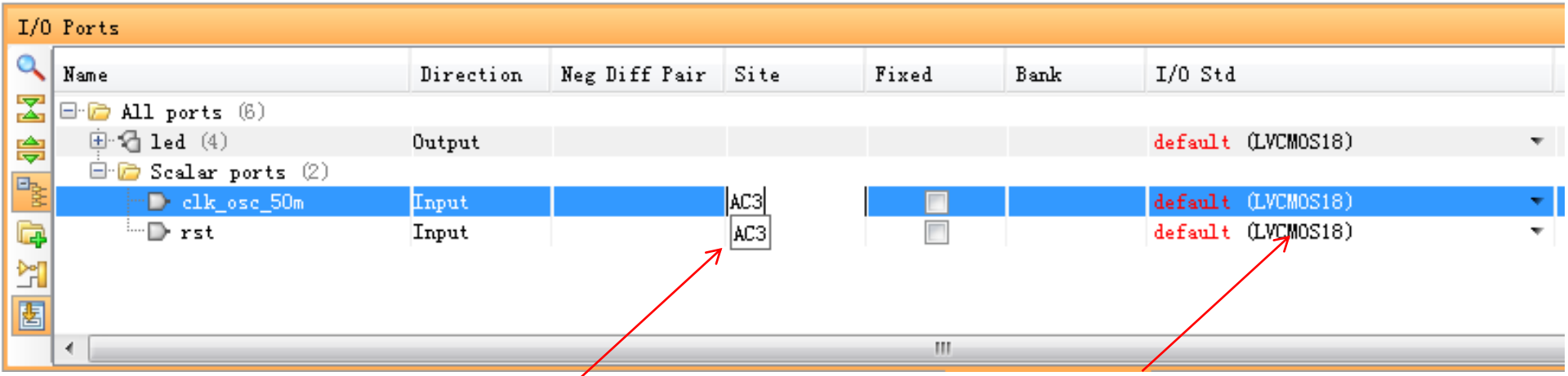
大概是这个样子




主要操作区域

# 锁定管脚 方法一

- 直接在界面里面输入管脚位置；
- 适用于先有硬件，再设计代码；




输入管脚位置
名称，如“AC3”

一定要选择对
应的IO电平

# 锁定管脚 方法二

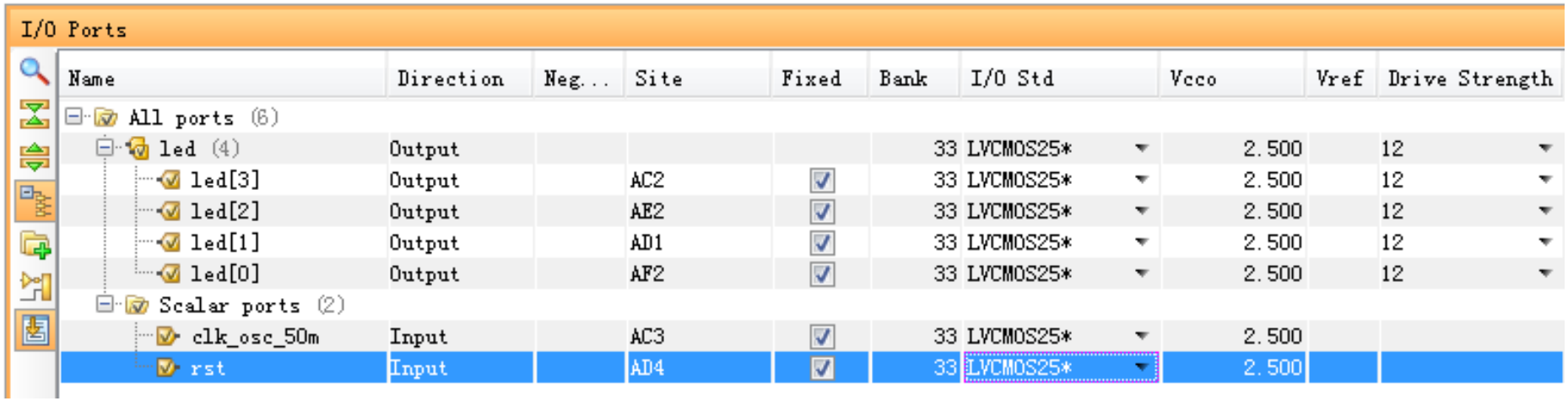
- 将INPUT、OUTPUT信号直接拖放到管脚上；
- 适用于先有代码，再出原理图、PCB；

# 锁定管脚 方法二


op I/O banks on voltages or the "NONE"
older to set/unset Internal VREF.
Sources
Netlist
Devic..
/O Port Properties
clk_osc_50m
Name: clk_osc_50m
Direction: Input
Site: IOB_X1Y72
Fixed
Site type: IO_L14P_T2_SRCC_33
Package pin: AC3
General
Properties
Configure
Power
Properties
Clock Regions
N
P
R
T
U
V
W
Y
AA
AB
AC
AD
AE
AF
/O Ports
Name
Direction
Neg Diff Pair
Site
Fixed
Bank
I/O Std
All ports (6)
led (4)
Output
default (LVCMOS18)
Scalar ports (2)
clk_osc_50m
Input
AC3
33
default (LVCMOS18)
rst
Input
default (LVCMOS18)
点按某个信号，直接拖放到管脚上。
一定要选择对应的IO电平

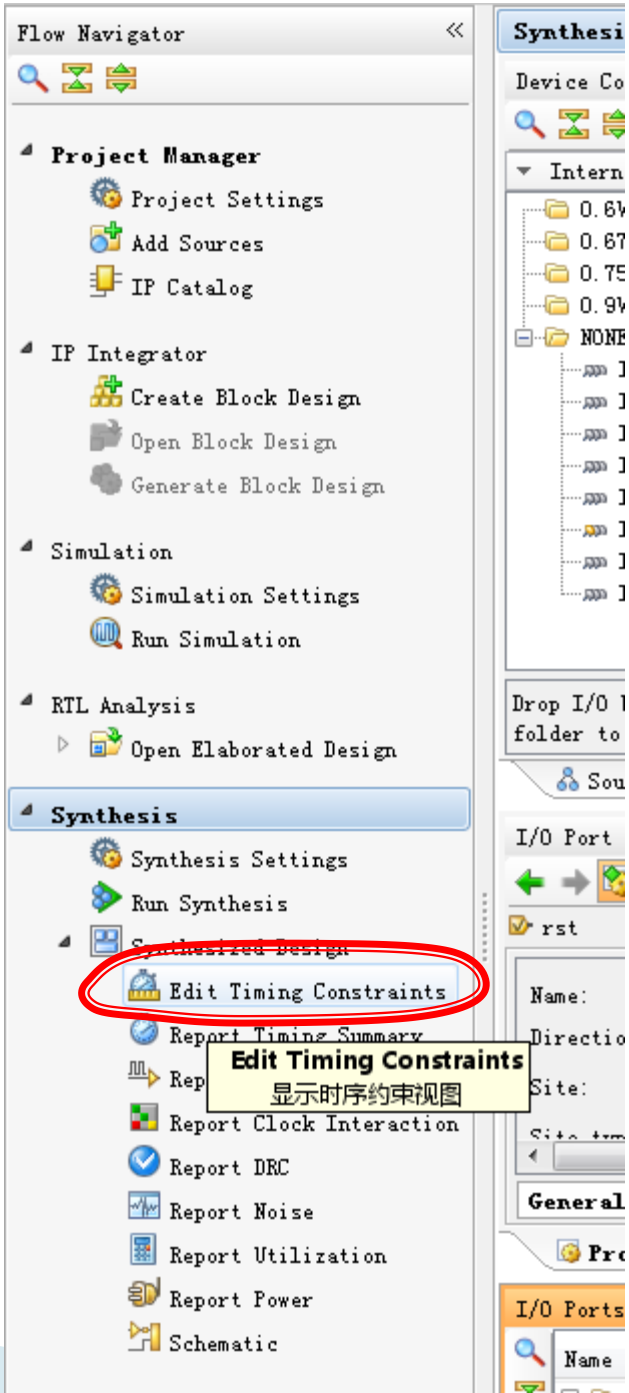
# 锁定管脚

- 将设计里的所有管脚分配好。



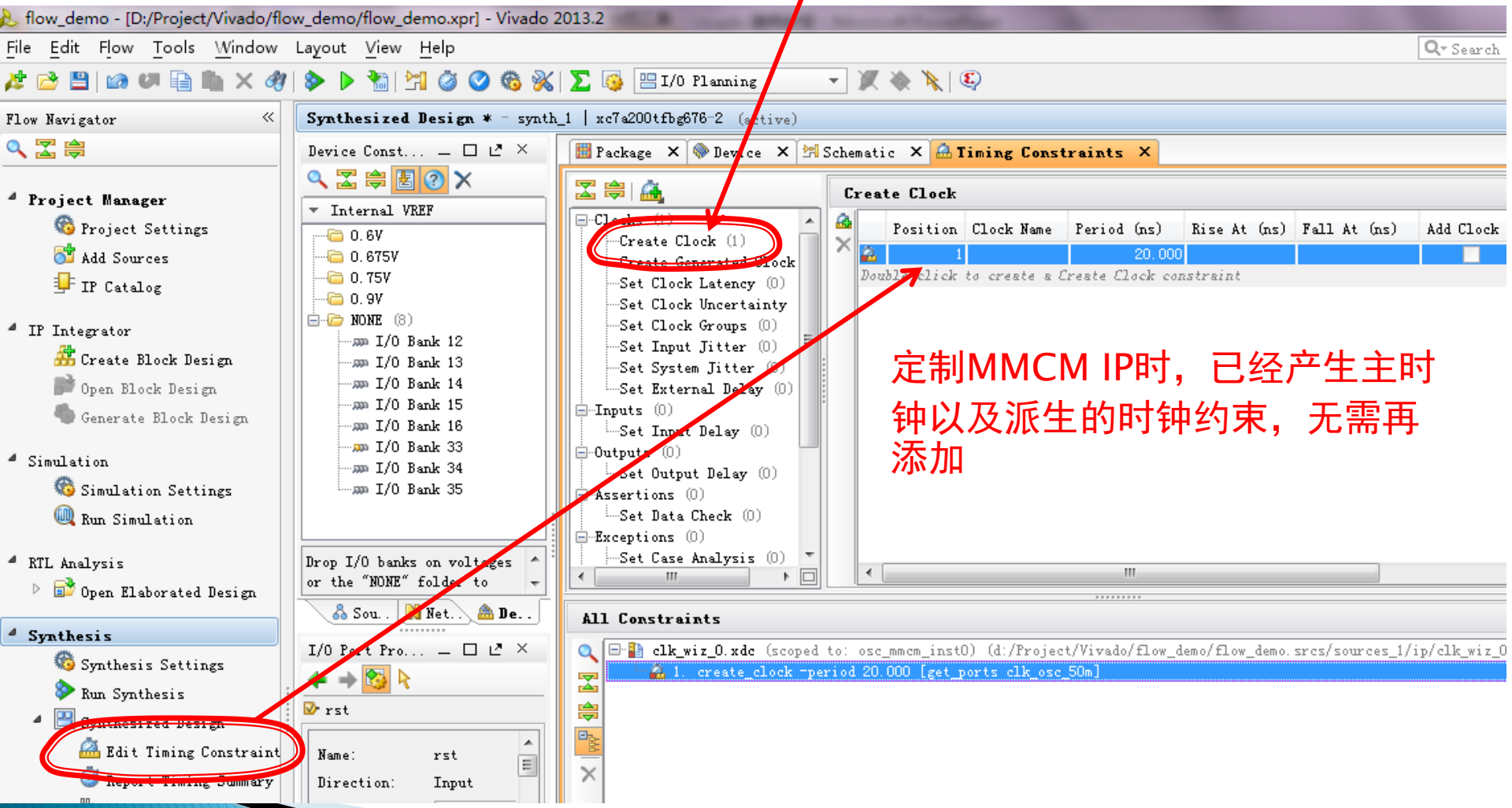
# 主时钟约束

单击，打开


Flow Navigator
Project Manager
Project Settings
Add Sources
IP Catalog
IP Integrator
Create Block Design
Open Block Design
Generate Block Design
Simulation
Simulation Settings
Run Simulation
RTL Analysis
Open Elaborated Design
Synthesis
Synthesis Settings
Run Synthesis
Edit Timing Constraints
Report Timing Summary
Report Clock Interaction
Report DRC
Report Noise
Report Utilization
Report Power
Schematic
Edit Timing Constraints
显示时序约束视图
Synthesi
Device Co
Intern
I/O Port
rst
Name:
Site:
General
I/O Ports
Name

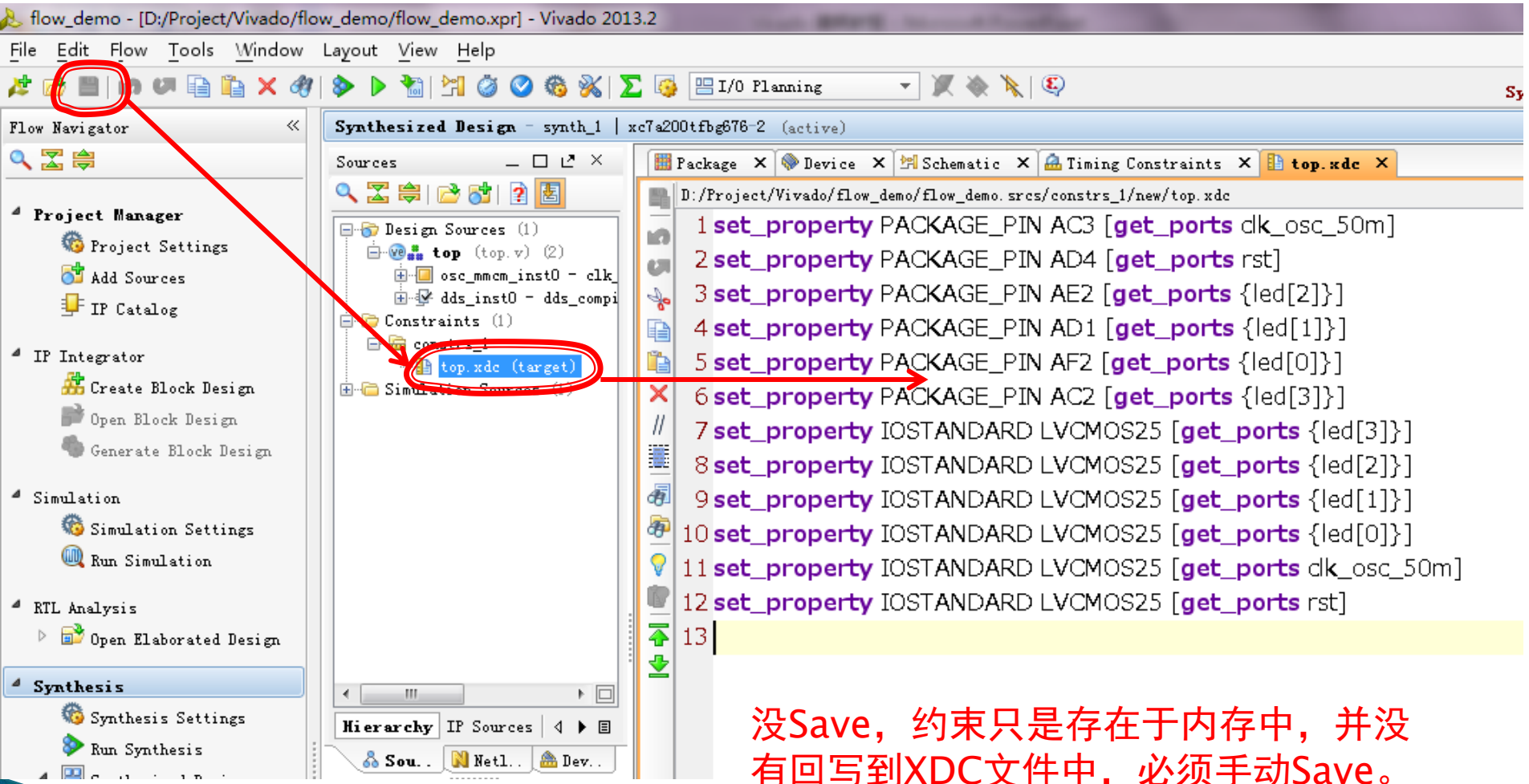
# 主时钟约束

如果有其他时钟需要添加，按照Create Clock向导一步步输入即可。



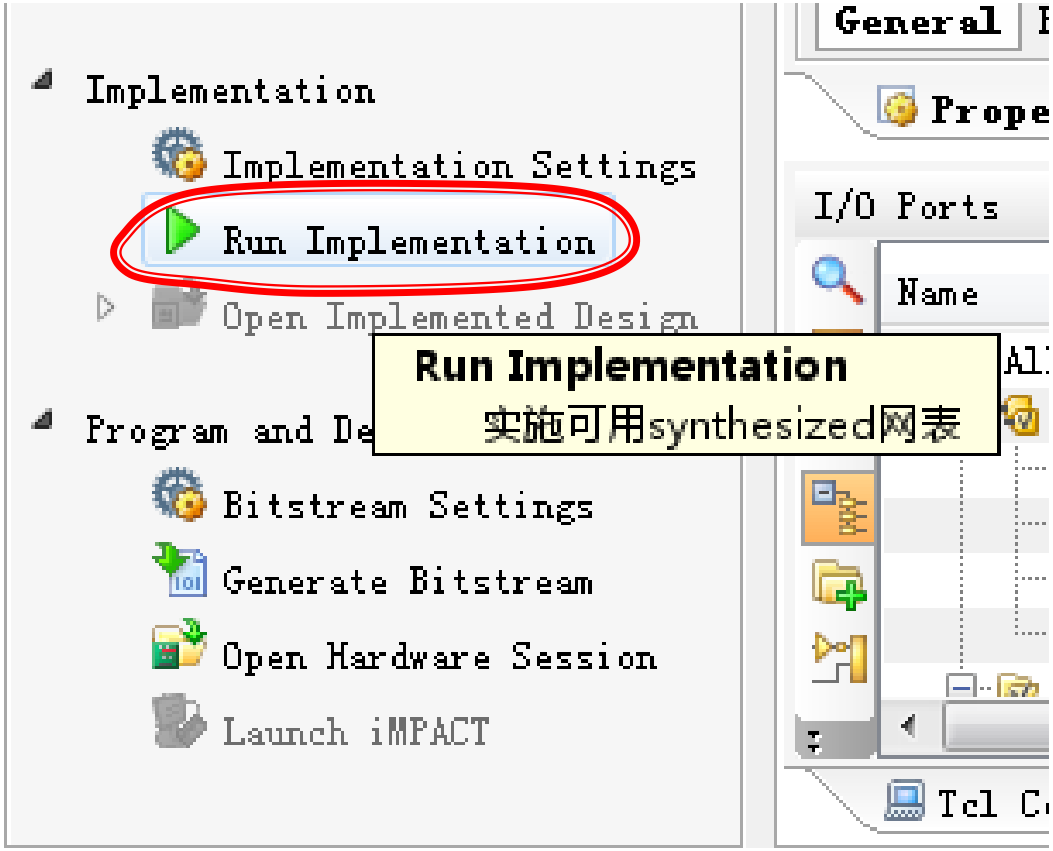

定制MMCM IP时，已经产生主时钟以及派生的时钟约束，无需再添加

# Save并确认约束



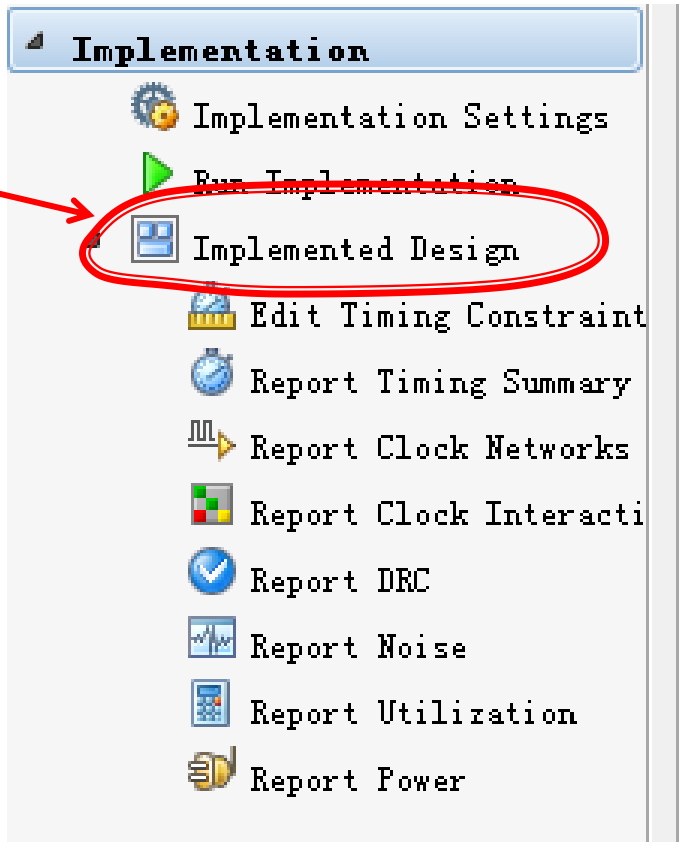

没Save，约束只是存在于内存中，并没有回写到XDC文件中，必须手动Save。

# 实现设计

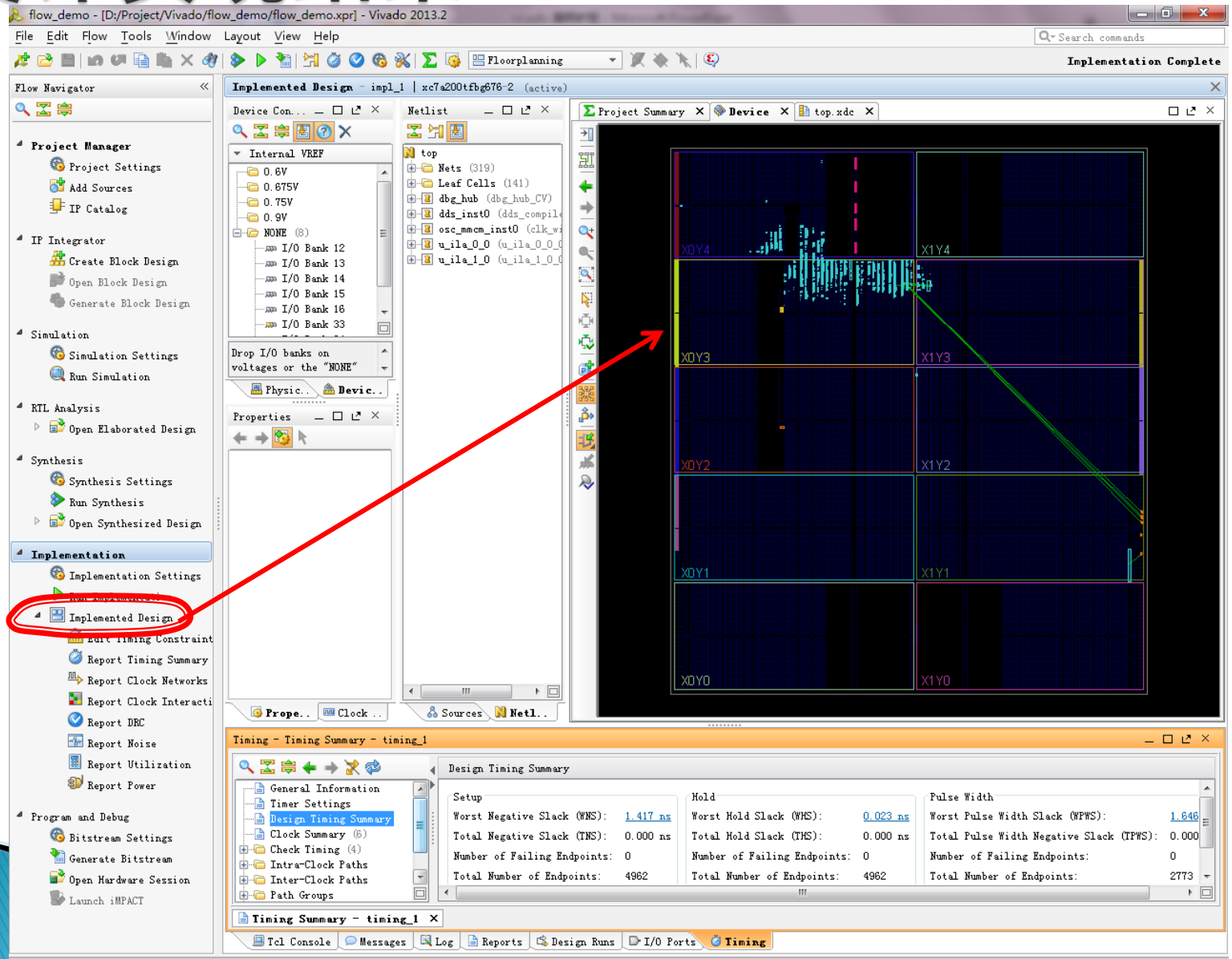

Implementation
Implementation Settings
Run Implementation
Open Implemented Design
Program and De
Bitstream Settings
Generate Bitstream
Open Hardware Session
Launch iMPACT
Run Implementation
实施可用synthesized网表
General
I/O Ports
Name

# 设计实现结果

- Implement完成之后，可以直接打开Implement结果。
- 也可以点击

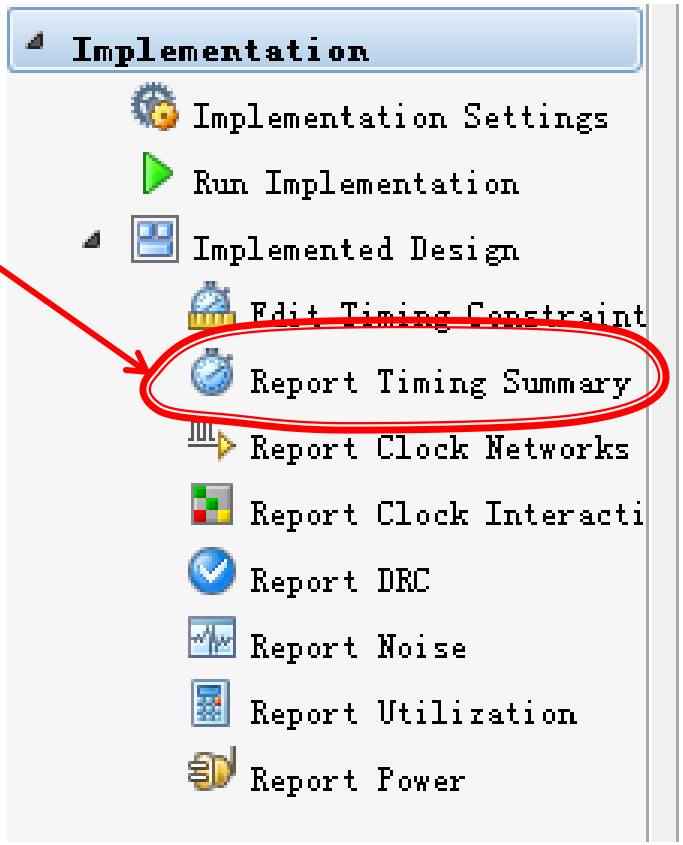


# 设计实现结果


flow_demo - [D:/Project/Vivado/flow_demo/flow_demo.xpr] - Vivado 2013.2
File Edit Flow Tools Window Layout View Help
Floorplanning
Implementation Complete
Flow Navigator
Project Manager
Project Settings
Add Sources
IP Catalog
IP Integrator
Create Block Design
Open Block Design
Generate Block Design
Simulation
Simulation Settings
Run Simulation
RTL Analysis
Open Elaborated Design
Synthesis
Synthesis Settings
Run Synthesis
Open Synthesized Design
Implementation
Implementation Settings
Implemented Design
Report Timing Summary
Report Clock Networks
Report DRC
Report Noise
Report Utilization
Report Power
Program and Debug
Bitstream Settings
Generate Bitstream
Open Hardware Session
Launch iMPACT
Implemented Design - impl_1 | xc7a200tfbg676-2 (active)
Device Con...
Internal VREF
0.6V
0.675V
0.75V
0.9V
NONE (8)
I/O Bank 12
I/O Bank 13
I/O Bank 14
I/O Bank 15
I/O Bank 16
I/O Bank 33
Drop I/O banks on voltages or the "NONE"
Netlist
top
Nets (319)
Leaf Cells (141)
dbg_hub (dbg_hub_CV)
Properties
Project Summary
Device
top.xdc
X0Y4
X1Y4
X0Y3
X1Y3
X0Y2
X1Y2
X0Y1
X1Y1
X0Y0
X1Y0
Timing - Timing Summary - timing_1
Design Timing Summary
General Information
Timer Settings
Design Timing Summary
Clock Summary (6)
Check Timing (4)
Intra-Clock Paths
Inter-Clock Paths
Path Groups
Setup
Worst Negative Slack (WNS): 1.417 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 4962
Hold
Worst Hold Slack (WHS): 0.023 ns
Total Hold Slack (THS): 0.000 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 4962
Pulse Width
Worst Pulse Width Slack (WPWS): 1.648
Total Pulse Width Negative Slack (TPWS): 0.000
Number of Failing Endpoints: 0
Total Number of Endpoints: 2773
Tcl Console
Messages
Log
Reports
Design Runs
I/O Ports
Timing

# 查看时序是否满足

- 点击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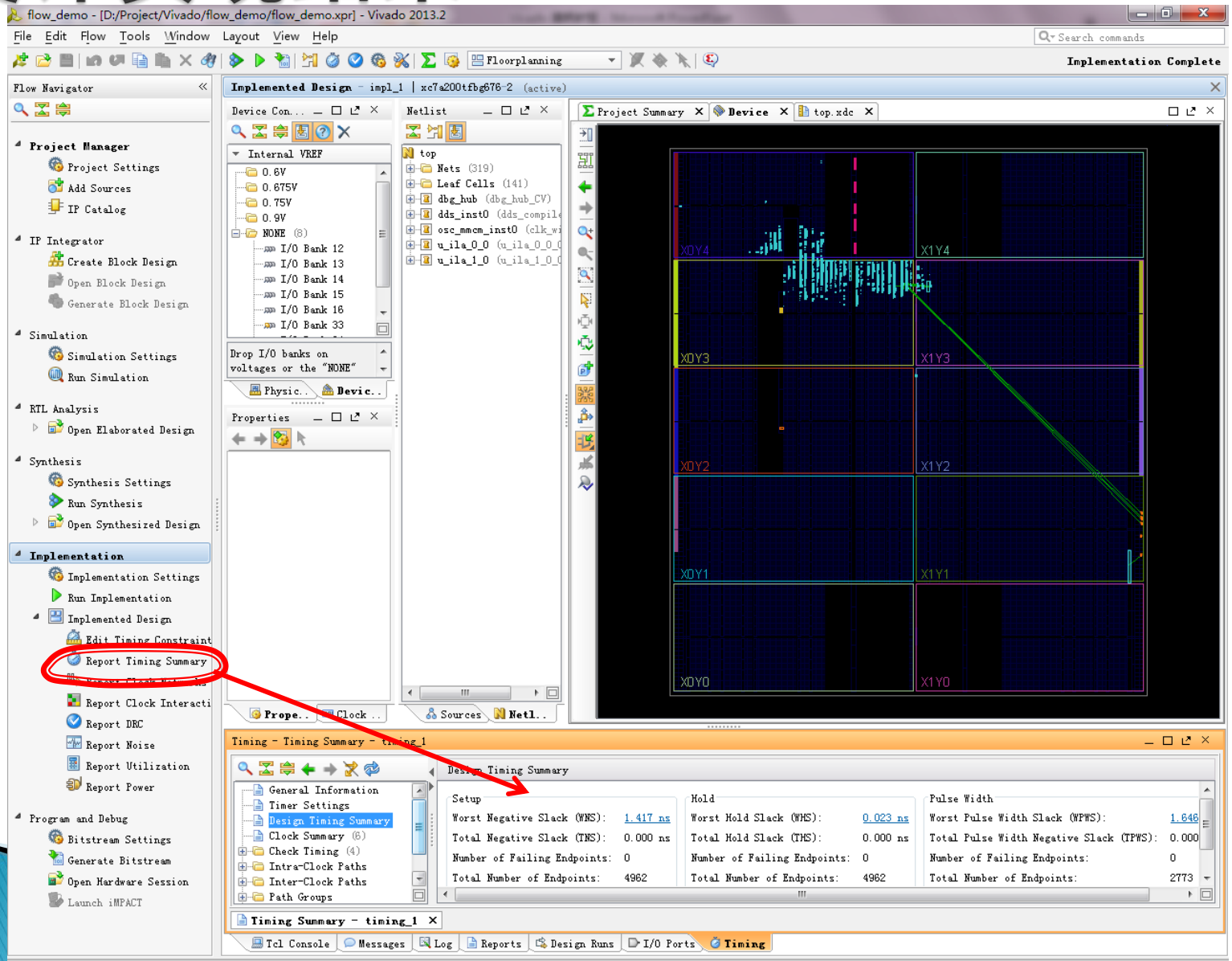
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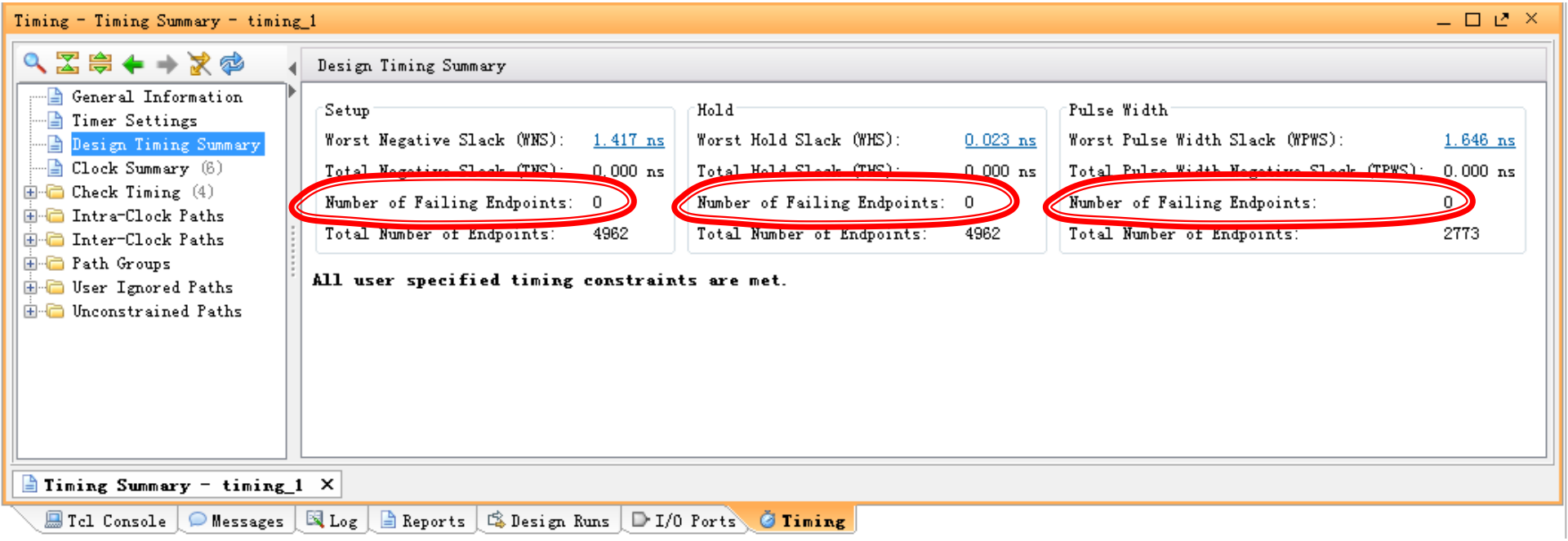
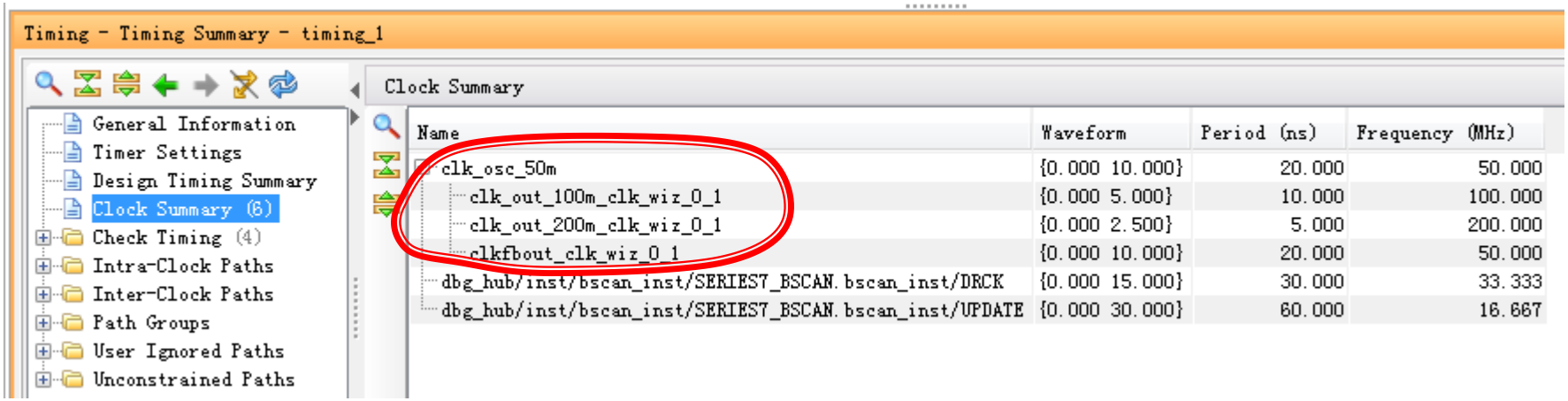
- 不满足的时序会以红色显示

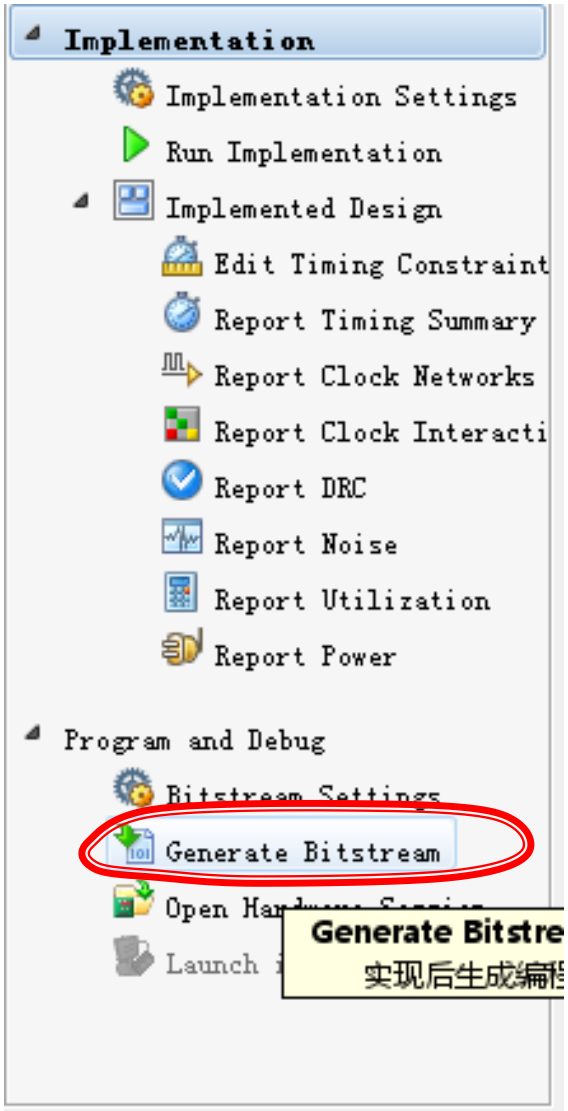


# 设计实现结果

- 定制MMCM时钟IP所隐含的时序



# 生产bit文件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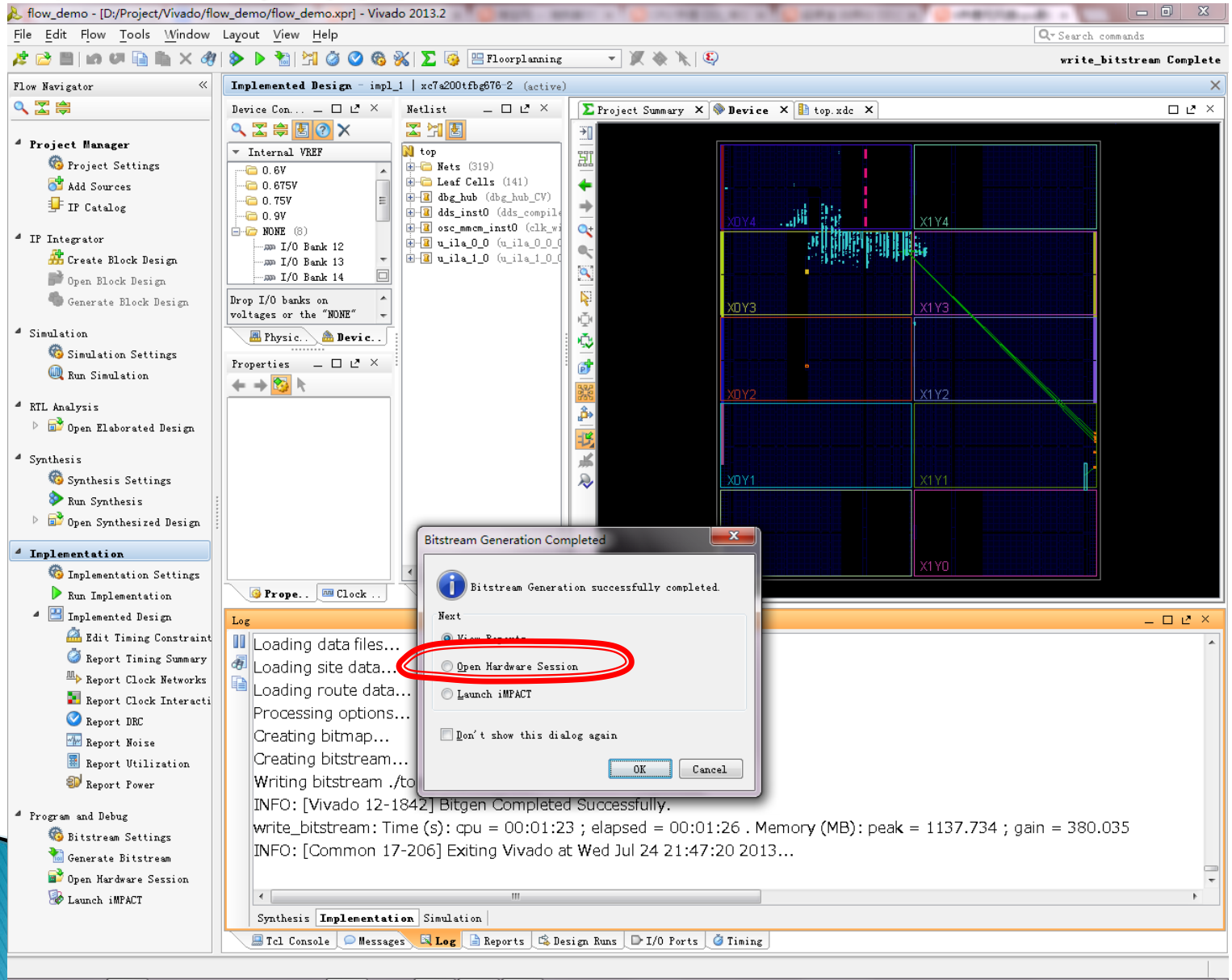
Launch
Generate Bitstre
实现后生成编

# 生产bit文件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Generate Bitstream
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Implemented Design - impl_1 | xc7a200tfbg676-2 (active)
Bitstream Generation Completed
Bitstream Generation successfully completed.
Next
Open Hardware Session
Launch iMPACT
Don't show this dialog again
OK
Cancel
Loading data files...
Loading site data...
Loading route data...
Processing options...
Creating bitmap...
Creating bitstream...
INFO: [Vivado 12-1842] Bitgen Completed Successfully.
write_bitstream: Time (s): cpu = 00:01:23 ; elapsed = 00:01:26 . Memory (MB): peak = 1137.734 ; gain = 380.035
INFO: [Common 17-206] Exiting Vivado at Wed Jul 24 21:47:20 2013...